PDE Ledger

Archive Ledger Skeleton

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July 23, 2026

Purpose and Scope

TODO. B3 will fill the rebuilt ledger purpose, source hierarchy, and scope boundaries.

0.1 The Role of This Ledger

Placeholder for the rebuilt ledger's role and audience.

0.2 Primary Source Hierarchy

Placeholder for the source hierarchy used by the rebuilt ledger.

0.3 Document Contract

Placeholder for the provenance, status, verification, and downstream-use contract.

0.4 Scope Boundaries

Placeholder for the limits of the rebuilt ledger.

0.5 Completeness Criterion

Placeholder for the completion standard used by later build phases.

How to Read This Ledger

TODO. B3 will fill the reader guide after the rebuilt stage and Part structure is set.

0.6 Layer Structure

Placeholder for the relationship between theorem-block prose, stage cards, notes, and audits.

0.7 Reading Paths

Placeholder for operational, verification, provenance, and authoring reading paths.

0.8 Citation Discipline

Placeholder for result anchors, theorem labels, equation labels, and stage provenance.

0.9 Stage Entries

Placeholder for the expected fields in future stage cards.

0.10 Gap Handling

Placeholder for demotion, splitting, open-gate, and failed-route handling.

Claim-Status Firewall

TODO. B3 will fill the rebuilt ledger's status rules and examples.

0.11 Allowed Status Tags

Status tag	Meaning
Exact	Placeholder.
Exact Within Closure	Placeholder.
Reduced / Controlled Reduction	Placeholder.
Effective Closure	Placeholder.
Numerically Located	Placeholder.
Open	Placeholder.

0.12 Audit-Status Vocabulary

Placeholder for audit evidence categories.

0.13 Status Inheritance

Placeholder for weakest-essential-input and status propagation rules.

0.14 Citation Language

Placeholder for safe status-aware citation wording.

Notation Firewall

TODO. B3 will fill notation rules after the rebuilt sectors and stage packets are fixed.

0.15 Global Anti-Ambiguity Rules

Placeholder for global notation safeguards.

0.16 Coordinate, Index, and Operator Conventions

Placeholder for coordinate, index, and operator conventions.

0.17 Field and Sector Variables

Placeholder for sector-specific variable conventions.

0.18 High-Risk Symbol Collision Table

Placeholder for symbol collision accounting.

0.19 Notation Checklist

Placeholder for future stage write-up checks.

Result Anchor Map

TODO. B3 will assign rebuilt result anchors after the Part and stage plan is settled.

0.20 Anchor Use Rules

Placeholder for stable citation-anchor rules.

0.21 Anchor Naming Convention

Placeholder for rebuilt anchor naming.

0.22 Top-Level Result Anchors

Anchor	Stage range	Citation target
No anchors assigned yet.		

0.23 Maintenance Checklist

Placeholder for anchor maintenance rules.

Dependency Map

TODO. B3 will fill the rebuilt dependency graph after stages are authored.

0.24 Arrow Types

Placeholder for dependency-arrow types.

0.25 Global Dependency Spine

Placeholder for the rebuilt ledger's dependency spine.

0.26 Top-Level Dependency Table

Block	Inputs	Outputs
No dependency blocks assigned yet.		

0.27 Revision Rules

Placeholder for dependency-map revision rules.

Contents

Part I

Part I – The medium

Chapter 1

Medium Ledger Stages

Stage cards for Part I are collected in Appendix A.

Part II

Part II – Gravity

Chapter 2

Gravity Ledger Stages

Stage cards for Part II are collected in Appendix B.

Part III

Part III – Light

Chapter 3

Light Ledger Stages

Stage cards for Part III are collected in Appendix C.

Appendix A

Part I Stage Cards

A.1 Stage 004: GNLS Action and Dimensional Foundation

Part / anchor. Part I – The medium

Claim status. Exact Within Closure

Purpose. Fix the field content and $\{L, T, M\}$ dimensional dictionary of the GNLS parent action, derive the pin null-relation and healing scale, and land the base-system verdict `RETAIN_L_T_M`.

Status. Exact Within Closure. The dimensional dictionary, sound-speed and healing anchors, pin null-relation, and base-system verdict are exact symbolic results; the honest landing carries named residuals (mass non-emergence and the $\{L, T\}$-conversion gaps), deferred to pathA_21/22 and not repaired here.

Parent action and field content. The 4D-bulk medium is a GNLS condensate with order parameter ψ, number density $\rho = |\psi|^2$, and EOS $P = K\rho^5$ (from $U(\rho) = \frac{K}{4}\rho^5$):

$$\mathcal{L} \sim i\hbar\,\psi^*\partial_t\psi - \frac{\hbar^2}{2m_{\mathrm{GNLS}}}\,|\nabla\psi|^2 - U(\rho).$$

Dimensional dictionary (two tiers). **Primitive inputs.** $[\hbar] = ML^2T^{-1}$, $[m_{\mathrm{GNLS}}] = M$, $[\rho_0] = L^{-4}$ (state datum) — posted and consistency-checked against action usage. **Derived by composition.** $[\psi] = L^{-2}$, $[\rho] = L^{-4}$, $[K] = ML^{18}T^{-2}$, forced by action homogeneity across all 17 harness checks (14 foundation, including the bulk continuity equation, plus 3 $\{L, T\}$-representation gates).

Derived anchors. **Sound speed and healing length.**

$$c_{s0}^2 = \frac{5K\rho_0^4}{m_{\mathrm{GNLS}}}, \quad [c_{s0}] = LT^{-1}; \qquad h_0 = \frac{5K}{4}\rho_0^4 = \frac{m_{\mathrm{GNLS}}c_{s0}^2}{4}, \quad \xi_h = \frac{\sqrt{2}\,\hbar}{m_{\mathrm{GNLS}}c_{s0}}.$$

Pin null-relation. Four pins $\{a, c_{s0}, \hbar, m_{\mathrm{GNLS}}\}$ on three base dimensions give rank 3, nullity 1:

$$a = \frac{\hbar}{m_{\mathrm{GNLS}}c_{s0}}, \qquad a/\xi_h = 1/\sqrt{2}.$$

Verdict. **RETAIN_L_T_M.** Computed from two conjuncts: the $\{L, T\}$-rejection gate (dropping M leaves ≥ 1 non-dimensionless target, needing M-bearing conversion factors LT^2M, L^2T^2M, L^2T^2) *and* the mass fork (m_{GNLS} explicit of dimension M; m_{defect} not derived, `INFLOW_MASS_SOURCE_MISSING`). The $\{L, T\}$ projection is a natural-unit representation only, not a base-system change.

Carried residuals (first-class, not repaired). `INFLOW_MASS_SOURCE_MISSING` (m_{defect} emergence deferred to pathA_21; $\hbar J/c_\gamma^2 = M$ is dimensional-only), `LT_R_norm_gate_fails_without_new_conversion_factor` (`REJECTS_TRUE_LT_BASE`) and the two 4D/3D R-norm conversion gaps (pathA_22), plus `A_PIN_IS_BRANCH_MOMENT_NO` `EOS_FROM_GNLS_FACTOR`, `NO_NET_ACCRETION_BC_UNDERIVED`, `M_TO_G_UNIFICATION`, `SCALE_MAP_INPUTS`.

A.2 Stage 005: EOS Sound Speed and Light/Sound Ratio as a Free Knob

Part / anchor. Part I – The medium

Claim status. Reduced / Controlled Reduction

Purpose. Derive the phonon sound speed from the imposed EOS, pin the three velocity scales and the $c = c_\gamma$ wave-sector ceiling, and land the light/sound ratio $\lambda_\gamma = c_\gamma/c_s$ as a free calibration input (`C_GAMMA_RATIO_UNDERDETERMINED`).

Status. Reduced / Controlled Reduction. The sound-speed law and the $c = c_\gamma$ ceiling are exact symbolic results *within the imposed EOS closure*; the honest headline is the CALIBRATED landing: $\lambda_\gamma = c_\gamma/c_s$ is *unpinned by the parent action* – a free calibration input, with the tail $(c/c_s)^3 = \lambda_\gamma^3$ carried symbolic (not set to 1). Consuming stage: it cites `ledger_stage004` (I-1)'s dictionary and `EOS_FROM_GNLS_FACTOR` handoff (h_0, ξ_h) and owns the sound-speed derivation I-1 deferred here.

Sound speed from the EOS. From the imposed $P = K\rho^5$ (`EOS_CLOSURE_IMPOSED`) and the barotropic definition, by symbolic differentiation:

$$c_s^2(\rho) = \frac{1}{m_{\text{GNLS}}}\frac{dP}{d\rho} = \frac{5K\rho^4}{m_{\text{GNLS}}}, \quad [c_s] = LT^{-1}, \qquad \frac{d\ln c_s}{d\ln\rho} = 2.$$

The factor 5 falls out of the derivative. EARNED relative to the imposed EOS.

Three speeds and the $c = c_\gamma$ ceiling. **Velocity scales.** $v_b = (\hbar/m_{\text{GNLS}})\nabla\theta$, c_s, c_γ, each of dimension LT^{-1} (v_b's pin consumed from I-1). **Ceiling (derived from the wave sector).** For the trapped mode $\omega = c_\gamma\sqrt{k^2 + k_\perp^2}$,

$$\frac{d\omega}{dk} = \frac{c_\gamma k}{\sqrt{k^2 + k_\perp^2}}, \qquad c_\gamma^2 - \left(\tfrac{d\omega}{dk}\right)^2 = \frac{c_\gamma^2 k_\perp^2}{k^2 + k_\perp^2} \geq 0,$$

strictly positive for $k_\perp \neq 0$: the group speed is bounded by c_γ and approaches it. The bound-mode clock $\omega_0 = c_\gamma k_\perp$ gives the boosted internal rate ω_0/γ (time dilation) with no $E = mc^2$ premise.

Coupled principal symbol (pathA_20b support). On the neutralized homogeneous background the coupled symbol block-diagonalizes: the Madelung phonon determinant $\det M = \hbar(\omega^2 - c_{s0}^2 k^2)$ with $c_{s0}^2 = 5K\rho_0^4/m_{\rm GNLS}$ falling out via $h' = 5K\rho_0^3$, the gauge block $C_E\omega^2 - C_B k^2 = C_E(\omega^2 - c_{\rm bulk}^2 k^2)$ ($c_{\rm bulk}^2 = C_B/C_E$), and $\det P = \det M \cdot P_T^2$. The off-diagonal current/London terms are lower order and do not set the cone (`BULK_PRINCIPAL_TRANSVERSE_BRANCH_ESTABLISHED`); $[c_s] = [c_\gamma]$ is labeled *non-evidentiary* for equality.

The free-knob landing and its negative control. C_GAMMA_RATIO_UNDERDETERMINED. Computed, not asserted. With $C_E, C_B, K, \rho_0, m_{\rm GNLS}$ independent, the residual $C_B/C_E - 5K\rho_0^4/m_{\rm GNLS}$ is not identically zero and no source equation is present, so `forced_equals_valid = False` $\Rightarrow$ `FORCED_EQUALITY_REJECTED_WITHOUT_SOURCE_EQUATION`. It is *reversible*: inserting the identity $C_B \to 5K\rho_0^4 C_E/m_{\rm GNLS}$ flips it to `C_GAMMA_EQUALS_C_S` – so `EQUALS` is reachable only via an actually-inserted source equation the action does not supply. Two-layer sharpening: bulk `C_GAMMA_BULK_UNDERDETERMINED`, brane `C_GAMMA_RATIO_STILL_UNDERDETERMINED`; if $c_{\rm bulk}$ is ρ_0-independent then $\lambda_\gamma \propto \rho_0^{-2}$ ($d\ln\lambda_\gamma/d\ln\rho_0 = -2$).

Carried residuals (first-class, not repaired). `EOS_CLOSURE_IMPOSED`; `C_GAMMA_RATIO_UNDERDETERMINED`; `STATIONARY_PROFILE_UNDERDETERMINED_BY_BRANCH_DATA`; `NO_NET_ACCRETION_BC_UNDERIVED`; `HBAR_PROVENANCE_U` `HBAR_FREE_SUBSTRATE_RELATION_MISSING`; `H_2PI_RATE_CLASSIFICATION_UNDERDETERMINED`; `BULK_METRIC_SPEED_` `PARENT_METRIC_ACOUSTIC_IDENTIFICATION_MISSING`; `BRANE_ZERO_MODE_REDUCTION_UNDERIVED`; `BRANE_PHOTON_CONE_REQUIRES_PROFILE`. The mass bridge $\hbar J/c_\gamma^2 = M$ is a labeled dimensional candidate only, not a $m_{\rm defect}$ derivation.

A.3 Stage 006: Two-Phase Material-State Ontology (Order Field χ_B)

Part / anchor. Part I – The medium

Claim status. Exact Within Closure

Purpose. Formalize the brane/bulk ontology as a POSTULATED two-phase action – one conserved medium, order field $\chi_B \in [0, 1]$; brane = ordered shear-supporting phase, bulk = the same medium destructured, throat = phase-conversion site – dimensionally classified (`ACTION_SPECIFIED_CLASSIFIED`), with the recovery reduction to the frozen old-ledger $S_{\rm leak}$ verified assert-zero and the θ-as-Maxwell-φ no-go carried able-to-fail.

Status. Exact Within Closure. **The wall is made explicit as a postulated field – it is NOT derived from the one medium.** Every term of the free-energy functional and order-state balance is POSTULATED and labeled (13 pinned modeling choices P1–P13); the EARNED content is exact *within that imposed closure*: structural-closure identities, single-kink wall admission, and the recovery reduction against the frozen firewall. Cross-engine agreement proves consistent encoding of the same postulated F, not derivation. Consuming stage: cites `ledger_stage004` (dictionary, single-well U), `ledger_stage005` (c_{s0}^2), and `ledger_stage003` ($c_\gamma^2 = \mu_R/\rho_{br}$, $C_J = -J\rho_{B0}$, $B_{\rm eff} = \rho_{B0}^2/\chi_c$, the second-class$\Rightarrow$stray-DOF classification rule) with exact-value citation-integrity checks. *Decision 16 (2026-07-21)* retires the sole P-dependent piece — the $\alpha_{\rm aniso}$ anisotropy — with the polar field P (operative `DRIFT(5)`; analysis in the failures-paper backlog); SymPy 121 / Mathematica 119 PASS post-amendment.

The postulated action and balance.

$$F=\int d^4X\Big[\tfrac{1}{2}m_{\mathrm{GNLS}}n|\mathbf{u}|^2+U(n)+f_B(\chi_B)+\tfrac{\kappa_B}{2}|\nabla_4\chi_B|^2+\chi_B f_{\mathrm{shear}}+f_{\mathrm{throat}}+f_{\mathrm{mix}}\Big],$$

$$\partial_t n+\nabla_4\!\cdot\!(n\mathbf{u})=0,\qquad \partial_t(\chi_B n)+\nabla_4\!\cdot\!(\chi_B n\mathbf{u}+\mathbf{J}_\chi)=n\Gamma_B,\quad [\Gamma_B]=T^{-1}.$$

Key pins: kinetic term carries the constituent mass (`KINETIC_MASS_FACTOR_PINNED`; $[n]=L^{-4}$); double-well $f_B=a_B\chi_B^2(1-\chi_B)^2$ with minima $\{0,1\}$, n-independent (`SAME_DENSITY_DEGENERACY_POSTULATED`) – required because the cited $U(\rho)=(K/4)\rho^5$ is *single*-well; MacCullagh-form shear gate on the displacement curl; χ_B is *the* postulated wall order parameter (the $\alpha_{\mathrm{aniso}}\chi_B(P\cdot\hat{w})^2$ anisotropy is **retired by Decision 16**); $\mathbf{J}_\chi=0$ default; global returns $R_0=-M_0$, $R_1=-D_1$ printed as postulates, not asserted locally; throat = phase-conversion ontology, not a solve. Every integrand term audited to $ML^{-2}T^{-2}$; adjunct rows $(\mu_\chi,M_\chi,P_{\mathrm{order}},M_n)$ dimension-pinned, with the energy ledger fixed to the variationally consistent $P_{\mathrm{order}}=\int\mu_\chi D_t\chi_B\,d^4X$ (`HANDOFF_P_ORDER_N_PLACEMENT_CORRECTED`).

Structural closure and wall admission (EARNED within closure). The advective form $D_t\chi_B=\Gamma_B-(1/n)\nabla_4\!\cdot\!\mathbf{J}_\chi$, the disorder complement for $n_D=(1-\chi_B)n$, and the Γ_B-cancelling sum reproducing total conservation are derived by real derivative expansion in both engines. The static EL equation $\kappa_B\chi_B''=f_B'$ admits the kink $\chi_B=\frac{1}{2}(1+\tanh(w/2\delta))$ with solved width $\delta=\sqrt{\kappa_B/2a_B}$ and surface tension $\sigma_{\mathrm{wall}}=\sqrt{2a_B\kappa_B}/6$, $[\sigma_{\mathrm{wall}}]=ML^{-1}T^{-2}$. **Slab caveat (carried).** Bulk on both sides makes the brane a finite kink–antikink *slab*; the double-well selects no slab width – W_{slab} is an un-earned input mapping onto the old ledger's L/a self-selection item (sim-deferred). Wall admission $\neq$ slab stability.

Recovery reduction (EARNED within closure; the frozen firewall). Projection with unit-normalized $W(w)$ and in-engine IBP gives the two-source law $\partial_t\rho_B+\nabla_3\!\cdot\!\mathbf{J}_B=S_{\mathrm{flux}}+S_{\mathrm{convert}}$ with the general $\chi_B(w),\Gamma_B(w),J_\chi^w(w)$ computationally live; the in-engine limit $\chi_B\to 1,\Gamma_B\to 0,\mathbf{J}_\chi\to 0$ reduces it assert-zero to the *independently constructed* frozen target $S_{\mathrm{leak}}=-[Wj^w]_\partial+\int W'j^w\,dw$, $j^w=nu^w$ (stage_243), including the stage_244 Gaussian one-mode closed form $S_{\mathrm{leak}}=\sqrt{2}\,\mu_w\rho_0\mathcal{E}_0/(2\sqrt{\pi}\lambda^3)$ re-derived by direct integration (`RECOVERY_REDUCTION_VERIFIED`). Conditionality computed: $\chi_B=c\neq 1$, $\Gamma_B\neq 0$, $J_\chi^w\neq 0$ each leave nonzero computed residuals – the limit does real work.

The θ-as-Maxwell-φ no-go (carried FATAL_FLAW). A stable order-parameter phase has positive gradient energy ($\kappa_{\mathrm{phase}}>0$) $\Rightarrow$ $K_\theta=-\kappa_{\mathrm{phase}}<0$ in the Lagrangian; Maxwell's electric square needs $K_\theta=C_J^2/\rho_{br}>0$ – opposite signs (exact predicate). The constraint bracket $\{\Phi_1,\Phi_2\}=k^2(J^2\rho_{B0}^2+\kappa_{\mathrm{phase}}\rho_{br})/\rho_{br}$ has its zero locus solved in-engine at exactly $K_\theta=+J^2\rho_{B0}^2/\rho_{br}$: via the consumed stage_003 rule the provenance branch lands `FAIL_CAUCHY_STRAY_LONGITUDINAL` (finite $B_{\mathrm{eff}}=\rho_{B0}^2/\chi_c$), the $B_{\mathrm{eff}}=0$ branch `FAIL_C5_LONGITUDINAL_ZERO_MODE`, and the tuned fixture ($K_\theta=J^2\rho_{B0}^2/\rho_{br},B_{\mathrm{eff}}=0,m_\theta^2=0$) `C5_RESOLVED_MAXWELL_BY_TUNING` – flagged `BY_TUNING`, never `WITH_PROVENANCE` (flag computed from provenance booleans; detunings re-fire the pathA_36 control tokens). The only provenance sign-flip ($\kappa_{\mathrm{phase}}<0$) is a Lifshitz instability, $k_*^2=-\kappa_{\mathrm{phase}}/2\kappa_4>0\iff\kappa_{\mathrm{phase}}<0$ – the already-killed pathA_25 wall (`FAIL_LIGHT_STARVED` lineage). The transverse sector is untouched: the ε-parametrized dispersion derived in-engine gives $\omega^2=(\mu_R/\varepsilon)k^2$; $\varepsilon=\rho_{br}$ reproduces the consumed c_γ^2 (`PASS_TRANSVERSE_UNDISTURBED`), $\varepsilon\neq\rho_{br}$ fires `FAIL_TRANSVERSE_DISTURBED`. **This stage does not earn light** – the no-go is the point.

Drift and carried items (first-class, not repaired). Operative DRIFT(5) computed: $\{\chi_B$; a_B; κ_B; Γ_B; gating str (α_{aniso} retired with P, Decision 16 $\to$ failures-paper backlog; rung-W reconciliation printed; THETA_BRANCH_DEAD_NOT_ADMITTED; ρ_{B0}, χ_c cross-referenced to the pathA_41 Part-VI drift trio, not double-counted). Carried: SECOND_MEDIUM_DRIFT lineage, the slab width W_{slab}, Gate-L $\delta w = u_w$ translation Goldstone (deferred), $\mathbf{J}_\chi \neq 0$, $f_{\text{throat}}/f_{\text{mix}}$ placeholders, dynamics/energy-ledger adjunct. Two source-prose dimensional defects pinned and ablation-documented (kinetic mass factor; P_{order} n-placement).

A.4 Stage 007: Shear-Surface G0 Freeze (Frozen Medium Action, Drift Ledger, DOF)

Part / anchor. Part I – The medium (closing Part-I stage)

Claim status. Exact Within Closure

Purpose. Formal home of the pathA_35 G0 shear-surface *constitutive* freeze, amended by Decision 16 (P-retirement). Operative frozen action $\{S_{\text{GNLS}}, L_{\text{Mac}}, L_{u_w}\}$ under $g_\ell(w)$, hash-guarded (T0_SHEAR_FROZEN(d9520d3819c3)), POST_D16_DRIFT(7), DOF$=4$, $c_\gamma^2 = \mu_R/\rho_{br}$; the retired P-complement $\{L_{\text{pol}}, L_{Pu}, \lambda_{Pu}\}$ is byte-embedded in the frozen block as a hash-verified historical record only.

Status. Exact Within Closure. **The operative POST_D16_DRIFT(7) inputs are POSTULATED/CALIBRATED – the freeze freezes *terms*, not gate answers.** The EARNED content is exact within that declared closure: (i) *freeze fidelity* – both engines re-extract the canonical freeze-action blocks by byte-exact fence-parsing (never fixed offsets) and assert SHA-256 equality (G0 d9520d3819c3f7..., T0 8fa41ac51e88... byte-embedded); (ii) the *operative flat-brane linear DOF* $= 4$ computed from rank/nullity bookkeeping (in-plane u^a curl-transverse 2 + kinetic–curl 1; u_w 1; C5 φ **0** – the absent longitudinal partner the later C5 crux attacks); (iii) the *dimensional firewall* over every operative frozen term. This stage computes **no** gate verdict (Gate-L excluded; exposure names carried as prose provenance only) and does **not** earn light (Stage 003 derives the two transverse photons at $c_\gamma^2 = \mu_R/\rho_{br}$ from this stage's frozen L_{Mac}). *(The audit script additionally computes the historical freeze-as-run tier — DOF=8, SECOND_MEDIUM_DRIFT_AT_FREEZE(11) — as verification provenance; not reproduced here.)*

Decision 16 amendment (P-retirement, 2026-07-21). Decision 16 retires the brane polar field P and its couplings — L_{pol}, L_{Pu}, λ_{Pu}, and structural postulates 3/4/5 — as a light-stiffness source that was tried and failed (analysis in the failures-paper backlog). The freeze block is unchanged on disk and both engines still SHA-256-verify it (the retirement is a symbolic action-summand set partition over the immutable hash anchor, no byte surgery; token POST_D16_ACTION{S_GNLS,gL_Mac,gL_uw}_OF(d9520d3819c3)), so the historical freeze-as-run record stands. Operative action $\{S_{\text{GNLS}}, L_{\text{Mac}}, L_{u_w}\}$ under g_ℓ, DOF$=4$, POST_D16_DRIFT(7) (3 constants $\{\rho_{br}, \mu_R, \Omega_w\}$ + g_ℓ + survivor postulates $\{1, 2, 6\}$; postulate 1's annotation softened to "intrinsic wall normal"). $c_\gamma^2 = \mu_R/\rho_{br}$, L_{Mac}, L_{u_w}, g_ℓ, and the R10 reduction are untouched; $\chi_B = |P_\parallel|^2$ (route (c)) stays a named high-risk future gate needing a new T0 freeze, not foreclosed.

The frozen action (operative; $\{L_{\rm pol}, L_{Pu}, \lambda_{Pu}\}$ retired by Decision 16). Kept operative (0 new): the GNLS parent action $S_{\rm GNLS}$. The retired P-complement ($L_{\rm pol}$, L_{Pu}) stays byte-embedded in the frozen block as a hash-verified historical record only (analysis in the failures-paper backlog). Operative brane blocks under $S_{\rm brane} = \int dt\, d^4X\, g_\ell(w)\,[L_{\rm Mac} + L_{u_w}]$:

$$L_{\rm Mac} = \tfrac{1}{2}\rho_{br}(\partial_t u^a)^2 - \tfrac{1}{2}\mu_R\,\Omega^a_u\Omega^a_u, \qquad L_{u_w} = \tfrac{1}{2}\rho_{br}(\partial_t u_w)^2 - \tfrac{1}{2}\rho_{br}\Omega^2_w u^2_w,$$

$$g_\ell(w) = \frac{e^{-(w/\ell_g)^2}}{\sqrt{\pi}\,\ell_g}, \quad \int g_\ell\, dw = 1 \text{ (derived)}.$$

The operative $\hat{w}$ is the intrinsic wall normal (postulate 1, softened; no longer a concession *for* the retired P–u operator).

The drift ledger (computed). Both engines *compute* the operative subcounts and the verdict string (the source gate's hardcoded literals are retired): `POST_D16_DRIFT(7)` = **3 constants** $\rho_{br}\,[ML^{-3}]$, $\mu_R\,[ML^{-1}T^{-2}]$, $\Omega_w\,[T^{-1}]$ (table dims asserted against the firewall targets) + **1 function** $g_\ell(w;\ell_g)$ (fixed Gaussian, one width, target-blind admission) + **3 survivor structural postulates** $\{1, 2, 6\}$ (conceded $\hat{w}$+wall; free-slip u^a; no C5 φ) — derived as the set partition historical$\setminus\{\lambda_{Pu}$, postulates 3/4/5$\}$ (not a recount; a same-cardinality Ω_w/λ_{Pu} swap fails the partition assert). New-field content (u^a 3 + u_w 1 = 4 DOF) is counted *separately*. **Anti-absorption (2026-07-04 erratum residue):** $\{\rho_{br}, \mu_R\}$ are genuine postulated shear-surface inertia/modulus carrying the registered-pending Route-A reduction (`ROUTE_A_UNDERDETERMINED_MISSING_NONLINEAR_THROAT`, register R10); the cross-sector drift $\{\rho_{B0}, \chi_c, C_{hu}\}$ is a Part-VI (pathA_41) item, guarded in-engine against absorption here.

Supersession and firewalls. Both facts asserted: the Stage 006 χ_B wall superseded the fixed-shape $g_\ell(w)$ as the *material-state* closure, while G0 *remains* the light-sector *constitutive* freeze (Stage 003 consumes $L_{\rm Mac}$ as-is) – a scope split (register R21), not a reduction. Notational firewall (register R22): μ_R (3D brane, $ML^{-1}T^{-2}$) and Stage 006's $\mu^{(4)}_R$ (4D density, $ML^{-2}T^{-2}$) are asserted dimensionally distinct, related only by the *pending* R17 projection $\mu_R = \int \chi_B\, \mu^{(4)}_R dw$ (dim-consistency asserted).

Verification. Dual-engine, independent routes. After the Decision 16 amendment (2026-07-21): **SymPy 142 PASS / Mathematica 140 PASS**, both exit 0, CWD-independent (the amendment directive cleared the Codex→Grok→Codex bookend — `DIRECTIVE_SOUND` + `GROK_COMPUTE_CLEAN`). Able-to-fail teeth: the hash fence-parse teeth, the action-summand partition teeth, operative-DOF teeth (one P^i mode → 5, full block → 8; survivor ablations → 3), the operative-drift set-partition and same-cardinality-swap teeth, and the μ_R firewall tooth. *Fresh-agent tri-review of the amended scripts+docs is the next gate.* (Original pre-amendment build: SymPy 96 / Mathematica 94, tri-review `FIDELITY_CLEAN`+`ADVERSARIAL_CLEAN`, 26-run matrix.)

Appendix B

Part II Stage Cards

B.1 Stage 001: Solid-Angle and Second-Moment Primitives

Part / anchor. Part II – Gravity

Claim status. EXACT

Purpose. Derive the unit-sphere solid angles and normalized second moments used by the pathA_21c matter-stress force assembly.

Status. EXACT. This stage is closed geometry: no profile, normalization, or field-equation residual remains inside the stated claims.

Derivation. For $S^2 \subset \mathbb{R}^3$, take

$$n(\theta,\phi) = (\cos\theta, \sin\theta\cos\phi, \sin\theta\sin\phi), \qquad dS = \sin\theta\, d\phi\, d\theta.$$

Then **Omega2 solid angle.**

$$\Omega_2 = \int_0^\pi \int_0^{2\pi} \sin\theta\, d\phi\, d\theta = 4\pi.$$

With $n_1 = \cos\theta$ and $n_2 = \sin\theta\cos\phi$, **S2 second moments.**

$$\langle n_1^2 \rangle_{S^2} = \frac{1}{4\pi} \int_0^\pi \int_0^{2\pi} \cos^2\theta\, \sin\theta\, d\phi\, d\theta = \frac{1}{3}, \qquad \langle n_1 n_2 \rangle_{S^2} = 0.$$

The cross moment vanishes because $\int_0^{2\pi} \cos\phi\, d\phi = 0$.

For $S^3 \subset \mathbb{R}^4$, take

$$n(\chi,\theta,\phi) = (\cos\chi, \sin\chi\cos\theta, \sin\chi\sin\theta\cos\phi, \sin\chi\sin\theta\sin\phi), \qquad dS = \sin^2\chi\, \sin\theta\, d\phi\, d\theta\, d\chi.$$

Then **Omega3 solid angle.**

$$\Omega_3 = \int_0^\pi \int_0^\pi \int_0^{2\pi} \sin^2\chi\, \sin\theta\, d\phi\, d\theta\, d\chi = 2\pi^2.$$

With $n_1 = \cos\chi$ and $n_2 = \sin\chi\cos\theta$, **S3 second moments.**

$$\langle n_1^2 \rangle_{S^3} = \frac{1}{2\pi^2} \int_0^\pi \int_0^\pi \int_0^{2\pi} \cos^2\chi\, \sin^2\chi\, \sin\theta\, d\phi\, d\theta\, d\chi = \frac{1}{4}, \qquad \langle n_1 n_2 \rangle_{S^3} = 0.$$

The cross moment vanishes because $\int_0^\pi \cos\theta \sin\theta\, d\theta = 0$.

Physical use. $\Omega_2 = 4\pi$ and $\Omega_3 = 2\pi^2$ set the reduced r^{-2} and bulk R^{-3} Gauss normalizations. The second moments give the δ_{ij}/d angular average used in the convective factor $-(1+1/d)$ and the pressure factor $1/d$.

B.2 Stage 002: Matter-Stress Inter-Defect Force Assembly

Part / anchor. Part II – Gravity

Claim status. Reduced / Controlled Reduction

Purpose. Assemble the inter-defect force between two drain defects from the matter (Noether) stress, earning the bilinear Q_1Q_2 structure, the reduced-3D r^{-2} and bulk-4D R^{-3} power laws, and the attractive like-drain sign, consuming the Stage 001 geometry primitives.

Status. Reduced / Controlled Reduction. Three verdict layers are carried from the earned source: the leading matter-stress sign `FORCE_ATTRACTIVE_DERIVED` (target-blind); the full sign `SIGN_RESIDUAL_QUANTUM_VCONF_MAXWELL_PROFILE` (honest residual); acceptance `PASS_WITH_NAMED_RESIDUALS`. The bilinear structure, the power laws, and the attractive sign are earned (form + sign); the overall magnitude is a calibration knob; the full sign carries named, profile/sim-deferred residuals.

Derivation. The medium obeys $P = K\rho^5$, $h = (5K/4)\rho^4$, with the barotropic identity $dP/d\rho = \rho\,(dh/d\rho)$. The matter (Noether) stress in Madelung form is

$$\Pi_{ij} = m\rho\, v_i v_j + \delta_{ij} P(\rho) + \sigma^Q_{ij}, \qquad \sigma^Q_{ij} = \frac{\hbar^2}{4m}\Big[\frac{(\partial_i\rho)(\partial_j\rho)}{\rho} - \partial_i\partial_j\rho\Big],$$

with the quantum-stress divergence contributing no net far-field force at this order. A defect draining flux Q sets up, by Gauss's law, the inflow

$$v = -\frac{Q\,\hat{n}}{\Omega_{d-1}\, r^{\,d-1}},$$

where $\Omega_2 = 4\pi$ ($d = 3$) and $\Omega_3 = 2\pi^2$ ($d = 4$) are the Stage 001 solid angles. The Bernoulli cross-pressure of the two overlapping inflows is $\delta P = -m\rho\,(v_1\!\cdot\! v_2)$. Integrating the traction $\Pi_{ij}n_j$ over a control sphere around defect 2 and using the Stage 001 second moment $\langle n_i n_j\rangle = \delta_{ij}/d$ gives the convective factor $-(1+1/d)$ and the pressure factor $+1/d$, whose sum is the total flux $-\,mNQ_2v_1$ (the $1/d$ pieces cancel). The stationary force $F_{12} = -\oint \Pi_{ij}n_j\, dS$, with defect 1's own inflow substituted, yields the two lanes.

Reduced-3D force law.

$$F_{12} = -\frac{m\,N_3\,Q_1Q_2}{4\pi\, r_{12}^2}\,\hat{r}_{12}.$$

Bulk-4D force law.

$$F_{12} = -\frac{m\,\rho_4\,Q_1Q_2}{2\pi^2\, R_{12}^3}\,\hat{R}_{12}.$$

Attractive sign. Since $F \propto -Q_1Q_2$, like drains ($Q_1Q_2 > 0$) attract and opposite signs repel; the sign is derived from the Gauss-drain / Bernoulli / surface-integral chain, not assumed.

Physical use and named residuals. Target-blind, the inter-defect force is bilinear in the drain strengths, falls as r^{-2} (reduced 3D) and R^{-3} (bulk 4D), and is attractive between like drains. Honest scope: the overall normalization (I_F/Θ_Q/branch-profile) is a *calibration* knob, not derived; and the full sign carries the `QUANTUM_STRESS_FAR_FIELD`, `VCONF_BODY_FORCE`, and `MAXWELL_Z_GAUGE_JEXT` residuals, which are profile/sim-deferred and first-class.

B.3 Stage 008: Monopole/Dipole Return Constraint Specification

Part / anchor. Part II – Gravity (first return/radiation stage)

Claim status. OPEN

Purpose. Render the old ledger's open-item #9 as an exact constraint specification: the raw $\ell = 0/1/2$ outgoing c_s-wave amplitudes and radiation orders, and the precise source moments any brane↔bulk return must cancel (`MONOPOLE_DIPOLE_RETURN_CONDITIONAL`). The formal home of the targets $R_0 = -M_0$, $R_{1,i} = -D_{1,i}$ consumed by Stages 009/010 (pathA_29) and 022/023 (pathA_34); register edge R23.

Status. OPEN – deliberately. **This is a verified CONSTRAINT-SPECIFICATION, not a falsifiable test of no-monopole-radiation** (the source gate's own scope caveat, carried verbatim-class). GR forbids monopole/dipole gravitational radiation; the drain breaks brane mass conservation, so suppression is conditional on a return that cancels the moments below. The cancellation condition is algebraic bookkeeping (an $x - x$ identity, labeled as such in both engines); `cancellation_possible` is a literal parameter flag printed as *SCOPE: parameter, not computed – track-3 decides.* The falsification lives downstream (Stages 009/010; the Gate-6 $Z_{0,\mathrm{ret}}/Z_{1,\mathrm{ret}}$ selector).

The DtN ladder (earned; scanned, not typed). From spherical Hankel closed forms $h_\ell = j_\ell + iy_\ell$, both engines form $\Lambda_\ell = k\, h'_\ell/h_\ell|_{z=ka}$, normalize the admittance by its static value, and *scan* for the first purely-imaginary k-coefficient:

$$p_{\mathrm{raw}} = 1, 3, 5; \qquad \text{kernels } i\frac{\omega}{c_S}a, \quad i\frac{\omega^3}{c_S^3}\frac{a^3}{2}, \quad i\frac{\omega^5}{c_S^5}\frac{a^5}{27},$$

$$A^{(\ell 0)}_{\mathrm{raw}} = \frac{iM_0 a\omega}{c_S}, \qquad A^{(\ell 1)}_{\mathrm{raw}} = \frac{iD_1 a^3\omega^3}{2c_S^3}, \qquad A^{(\ell 2)}_{\mathrm{raw}} = \frac{iQ_2 a^5\omega^5}{27c_S^5},$$

all symbolically nonzero; steady control $\lim_{\omega\to 0} A^{(\ell 0)} = 0$; dominance $p(\ell{=}0) < p(\ell{=}2)$ from the scanned powers.

Cancellation targets (bookkeeping, labeled). $R_0(\omega) = -M_0(\omega)$ with $M_0 = \int_{\mathrm{brane}} S_{\mathrm{leak}}\, d^3x$ (cancels $O(\omega^1)$); $R_{1,i}(\omega) = -D_{1,i}(\omega)$ with $D_{1,i} = \int x_i S_{\mathrm{leak}}\, d^3x + \int j_i\, d^3x$ including the carried odd wake (cancels $O(\omega^3)$). Conditions, not closure. The derivative outlet vertex $g_{W0} = \eta\omega$ is recorded as a `branch_assumption`, not verdict-bearing. Live-source consistency is anchored by direct integration of the frozen Stage-243/244 closed forms ($\int W' j^w dw = -\sqrt{2}\,\ell_w j_0/4$; $S_{\mathrm{leak}} = \sqrt{2}\,\mu_w \rho_0 E_0/(2\sqrt{\pi}\lambda^3)$) with the strict-recovery limits computed but *not* taken as a basis; Stage 006 owns the recovery-reduction derivation (boundary printed).

Verdict machinery and guards. The verdict is computed from computed booleans plus the literal flag; the synthetic control $(\mathrm{True}, \mathrm{False}, \mathrm{False}) \to$ `MONOPOLE_RADIATION_UNAVOIDABLE` proves the rung reachable. The nine source controls are carried honestly: four computed, five printed as declarations (not counted checks), with the source's framing – guards against obvious tautologies, not evidence of suppression-vs-unavoidable. Consumed from Stage 005: $c_s^2 = 5K\rho^4/m_{\mathrm{GNLS}}$ (cited, exact-value integrity check $5 \cdot \frac{1}{500} \cdot (\sqrt{10})^4 = 1$); frozen slice $G = c = c_s = 1$, $K_{\mathrm{eos}} = 1/500$, (a^*, L^*) exact rationals, cited. Q_2 is exported as a **free anchor** (not derived $\ell = 2$ physics) for Stage 026.

Verification. Dual-engine, independent routes, zero file I/O (the source pair's JSON/YAML engine bridge is retired): SymPy 54 PASS / Mathematica 57 PASS (3 = arity self-checks), both exit 0, CWD-independent. Eight able-to-fail teeth fire (Hankel-form corruption trips the scan; kernel-coefficient, moment-kill, bookkeeping-integrity, verdict-machinery, anchor-prefactor, steady-limit, consumed-law corruptions). Full tri-review on fresh agents: `FIDELITY_CLEAN` (coverage-diff, math re-derived by hand) and `ADVERSARIAL_CLEAN` (14/14 mutation matrix; dual-corruption fails both engines; arity scan incl. planted-mismatch detection), followed by a nit remediation (tally hygiene, restored recovery-slice observations, pipeline-routed tooth) re-verified clean.

B.4 Stage 009: Flat-Slab Return Residual (Check A)

Part / anchor. Part II – Gravity (brane$\leftrightarrow$bulk return, Check A)

Claim status. Exact Within Closure

Purpose. Derive, on the postulated flat-slab family, what a finite DC-sink brane$\leftrightarrow$bulk return actually delivers against the Stage-008 cancellation targets: the return transfer $T_\ell(\omega) = \alpha_\ell e^{i\omega\, 2d/c_s}$, the bounded residual orders $p_{\mathrm{res}}(\ell{=}0) = 1$, $p_{\mathrm{res}}(\ell{=}1) = 3$, and the signed local source accounting $Z = -M_0\varepsilon_0/(1+\varepsilon_0)$ – the Check-A component of `RETURN_RESIDUAL_PREDICTION` (Check-B localization = Stage 010).

Status. Exact Within Closure – exact *within the postulated slab family* (brane at $w = 0$, return/absorber at $w = d$; the geometry is postulated, the return response is derived on it). The v3 labeling is carried exactly: $Z < 0$ (drain admissibility) is the **premise** (`Z_is_premise=true`); the formula $Z = -M_0\varepsilon_0/(1+\varepsilon_0)$ is **accounting** (channel sum $Z_{\mathrm{throat}} + Z_{\mathrm{return}}$, sign certificate computed). Open-item #9 is *sharpened, not closed*; the headline is printed only as the Check-A component with the Stage-010 qualifier.

The derived return response. From the bidirectional Helmholtz basis $\Phi_\ell = A_\ell e^{i\omega w/c_s} + B_\ell e^{-i\omega w/c_s}$, the round-trip transit time is *solved* ($\tau = 2d/c_S$, asserted via $e^{i\omega\tau} \equiv$ phase). DC continuity fractions $\alpha_\ell = 1/(1{+}\varepsilon_\ell)$ with independent per-channel transmissions $\varepsilon_0, \varepsilon_1 > 0$; steady circulation balances ($J_{\mathrm{leak}} = J_{\mathrm{return}} + J_{\mathrm{neighbor}}$, both moments) assert-zero. A genuine ω-order scan computes $\nu_\ell = \mathrm{ord}_\omega(1 - T_\ell) = 0$ – a finite DC sink leaves the ω^0 deviation $1 - T_\ell(0) = \varepsilon_\ell/(1+\varepsilon_\ell)$ – so the residual amplitudes $A_{\mathrm{res}} = \mathrm{kernel}_\ell \cdot \mathrm{moment} \cdot (1 - T_\ell)$ sit at **computed** orders $p_{\mathrm{res}} = p_{\mathrm{raw}} + \nu_\ell = 1, 3$: a *bounded monopole/dipole c_s-radiation residual tied to the drain strength* – the falsifiable departure from GR. The Check-A classification booleans are computed from the orders.

Reversibility (per-channel, computed). At $\varepsilon_0 \to 0^+$: $T_0 \to e^{i\omega\tau}$, strict $\nu_0 = 1 \Rightarrow p_{\mathrm{res},0} = 2$, and $Z \to 0$ (a perfect-return channel has no drain). At $\varepsilon_1 \to 0^+$: strict $\nu_1 = 1$ computed *directly from the $\ell = 1$ limit* (the source reused ν_0; the reshape computes it independently) $\Rightarrow p_{\mathrm{res},1} = 4$. The prediction is contingent on the transmissions, not baked in. Consumed from Stage 008 (cited, dual-site integrity-checked): the three radiation kernels; Q_2/kernel$_2$ inert (`T2_applied=false`). Register: rows $\{d, \varepsilon_0, \varepsilon_1\}$ (slab-family parameters; ε's FREE-UNREDUCED on the Gate-6 route) + edge R24.

Verification. Dual-engine, independent routes, zero file I/O – the source pair's JSON-digest bridge, its runtime YAML reads of pathA_28 outputs, and its SHA-256 trace bookkeeping are all retired. SymPy 48 PASS / Mathematica 52 PASS, exit 0, CWD-independent. Eight able-to-fail teeth fire. Full tri-review on fresh agents: `FIDELITY_CLEAN` (all values hand-re-derived; 009/010 boundary clean); the adversarial leg (16-mutant matrix) **caught a genuine rig-class defect** – the consumed-kernel citation-integrity block was set-then-compare-to-self (a single-source corruption passed) – remediated to dual-site integrity (acceptance: all four one-site corruptions now fail in both engines) plus tally-hygiene nits, then re-verified clean. The `Off[Limit::alimv]` suppression was unmasked and shown benign (every silenced limit's result independently asserted).

B.5 Stage 010: Slab Localization $p = 2$ with Reachable NOGO (Check B)

Part / anchor. Part II – Gravity (brane↔bulk return, Check B; completes the pathA_29 fold)

Claim status. Exact Within Closure

Purpose. Show, on the postulated flat-slab family, that Newtonian $1/r^2$ gravity SURVIVES the finite slab: both admissible DC-sink completions carry a normalizable m=0 transverse zero mode whose 3D-radial Green function gives flow $\propto 1/r^2$ ($p = 2$); the anti-localizing warp control gives $p = 3$ and the same classifier returns `RETURN_NOGO` (the able-to-fail companion). This computes Check B and emits the COMPLETED joint verdict `RETURN_RESIDUAL_PREDICTION` = (Check A cited, Stage 009) ∧ (Check B computed here).

Status. Exact Within Closure – exact *within the postulated slab family* (brane at $w = 0$, return/absorber at $w = d$; the geometry is postulated, localization is derived on it). The gravity-range ($1/r^2$) leg passes inside the localizing flat-slab family because both DC-sink completions give $p = 2$; the joint verdict `RETURN_RESIDUAL_PREDICTION` is completed here, with `RETURN_NOGO` genuinely reachable. Open-item #9 is *sharpened, not closed*: localization holding for the FAMILY is not the family being SELECTED by dynamics (R19/W_{slab}); the return-cancellation targets (R23) remain deferred.

The localizing zero mode and $p = 2$. On $0 \le w \le d$ each admissible completion is a transverse eigenproblem $f'' + m^2 f = 0$ with a normalizable constant m=0 zero mode ($f_0 = 1/\sqrt{d}$, $\int_0^d f_0^2\, dw = 1$): the *destructuring/absorbing* (compact cell, Neumann) and *Bloch* (q=0 periodic) cells, with gapped first modes at $m_1 = \pi/d$, $2\pi/d$. The 3D-radial zero-mode equation $g'' + (2/r)g' + ((\omega/c_s)^2 - m^2)g = 0$ is solved by genuine `dsolve` (dynamic route: solved first, outgoing spherical branch selected via C_1/C_2 – a boundary SELECTION, normalization = compact overlap $1/d$ – then $\omega \to 0$); the limit Green $1/(4\pi d r)$ gives flow $1/(4\pi d r^2)$ and *extracted* exponent $p = 2$. The static route ($\omega = 0$ first) gives the same $1/(4\pi d r)$ and $p = 2$; the two routes agree on both the exponent and the Green ($\mathit{dynamic}_{\omega\to 0} = \mathit{static}$). The massive/gapped solve gives a Yukawa $e^{-m_1 r}/r$ – only the m=0 zero mode sets the far field. Both completions agree $p_{\text{abs}} = p_{\text{bloch}} = 2$.

Able-to-fail: the counterfactual and the NOGO warp. Multiplying the solved Green by r^{-4} ($\to 1/r^5$) yields radial-operator residual $5/(\pi d r^7) \neq 0$ – REJECTED, proving the solve has teeth. The mandatory anti-localizing warp $\mu(w) = e^{2k_{\mathrm{warp}}w}$ has a *non-normalizable* zero mode ($\int_0^\infty \to \infty$, computed) and a continuum Green $1/(4\pi r^2)$ giving $p_{\mathrm{delocalizing}} = 3$; the same classifier returns `RETURN_NOGO`. The classifier verdict is a function of the computed exponents (and the cited `quadrupole_survives`), never a stamp; `RETURN_NOGO` is reachable if the return delocalizes OR the quadrupole channel does not survive.

Consumed (cited, dual-site integrity + exact anchor). From Stage 008: $p_{\mathrm{raw}}(\ell{=}2) = 5 \Rightarrow$ `quadrupole_survives` (finite quadrupole order; T_2 not applied, `T2_applied=false`, no $\ell{=}2$ recompute). From Stage 009: the Check-A component `A_residual_pass=true` (joint-composition print only). Register: *zero new counted knobs*; k_{warp} is a control-construction symbol (tracked, not counted); edge R25 (localization $p = 2$ EARNED-within-family, able-to-fail via the warp NOGO – does not discharge R19 or R23).

Verification. Dual-engine, independent routes, zero file I/O – the source pair's JSON-digest bridge and all SHA-256 trace/`structure_id`/`expr_digest` bookkeeping are retired; the `.wl` solves the radial ODEs, the normalizations, the warp divergence, and the classifier natively (own `DSolve` basis, own `Limit`/`Integrate`), with an arity self-check and no silenced messages. SymPy 71 PASS / Mathematica 81 PASS (81 = 71 shared + 10 arity self-checks), exit 0, CWD-independent. Eight able-to-fail teeth fire. Full tri-review on fresh agents: `FIDELITY_CLEAN` (all values hand-re-derived; 009/010 boundary clean – Check-A cited not recomputed) and `ADVERSARIAL_CLEAN` (22-mutant matrix; a hardcoded-wrong-Green mutant confirmed the counterfactual guard is genuine, and the classifier genuinely reaches `RETURN_NOGO`). The adversarial leg's one substantive find – a *coordinated both-sites* corruption of the Stage-008 citation to another finite integer escaped – was folded to Stage-009 parity by an exact-value anchor, and re-verified (`REVERIFY_CLEAN`) as the sole gate closing that escape.

B.6 Stage 011: Frozen-Reduction Certificate (L_s Helmholtz) (Check II-G1a)

Part / anchor. Part II – Gravity (brane↔bulk return; the frozen-throat longitudinal reduction; opens the pathA_30 D/N fold, completed by Stage 012)

Claim status. Reduced / Controlled Reduction

Purpose. Certify that, on the postulated straight finite throat with the wall FROZEN ($R = R_0,\ \eta = 0$) and the matter perturbation left dynamic ($\propto e^{-i\omega t}$), the reduced longitudinal operator collapses to the constant-coefficient Helmholtz resonator operator $L_s\psi = \psi''(s) + (\omega/c_S)^2\psi(s) = 0$, with $c_S^2 = 5K\rho_*^4/m$ on $s \in [0, L_0]$ (cap $R_0(L_0) = 0$) — *produced* by a per-term reduction (projection measure plus every intruding coefficient computed with its vanishing/deferral condition), NOT typed — together with the reduction certificate: the validity window under which the collapse is exact.

Status. Reduced / Controlled Reduction — a controlled reduction with an explicit validity certificate. This stage carries the reduction-certificate component of the joint pathA_30 verdict `DN_UNITTEST_BC_DEPENDENT`; the D/N determinant, the DtN $-(\omega/c_S)\tan(L_0\omega/c_S)$, the half-shifted pole ladder, the Robin counterfactual, and the `BC_DEPENDENT` landing (*bc_derivation_emitted* = false, a banked calibration input) are Stage 012's; the cut is drawn at the general-solution `dsolve` (never called here). The wave speed is *banked* (Part I edge R1, Stage 005, at the straight-reference density ρ_*), so this stage adds NO reduction of its own: its contributions are the two *structural* edges R26 (the validity record) and R27 (the $\xi \neq \ell_c$ firewall), which discharge no knob.

The produced operator (the de-rig). The reduced operator is *assembled* from dimensionally-consistent terms,

$$L_s \;=\; \psi'' \;+\; M\,\psi' \;+\; N\,\psi \;-\; B\,\frac{\hbar^2}{4m^2 c_S^2}\,\psi'''',$$

with $M = \frac{d}{ds}\log\sqrt{\gamma_0}$ ($[M] = L^{-1}$), the ψ-coefficient $N = (\omega/c_s(s))^2$ with $c_s(s)^2 = 5K\rho_0(s)^4/m$ ($[N] = L^{-2}$), and $B \in \{0,1\}$ the Bogoliubov inclusion flag — the k^4 correction being a genuine *fourth-derivative* intrusion (from the identity $\psi'' + (\omega/c_S)^2\psi - (\hbar^2/4m^2c_S^2)\psi'''' = 0$ for the full BdG dispersion), NOT a coefficient shift. The three source checks are replaced by genuine computations that can each fail: `operator_is_helmholtz` is expr_equal($L_s -$ ideal$, 0$) on the *assembled* L_s (not two hand-typed twins); `speed_is_cs` is *extracted* from the ψ-coefficient of L_s ($= (\omega/c_S)^2$, bulk, not wall/healing-renormalized, not literal `True`); `domain_is_L0` is *solved* from the pinch-off $R_0(s) = 0$ on a POSTULATED monotone taper $R_0(s) = R_{\mathrm{mouth}}(1 - s/L_0)$ (root $= L_0$, R_{mouth}-independent; a labeled SELECTION, not `L0==L0`); and `unsuppressed_operator_intrusion` is *computed* from the assembly (not hardwired `False`) and gates the verdict.

The reduction certificate (validity window). Under the frozen background ($\rho_0(s) = \rho_*$, $\sqrt{\gamma_0} = A_{\perp 0}$ constant, $A_{M0} = 0$) every intruding term is computed to vanish or is deferred: the projection-measure coefficient $M = 0$; $\rho_0' = 0 \Rightarrow c_s' = 0 \Rightarrow N = (\omega/c_S)^2$ (the bulk sound speed); $\delta V_{\mathrm{conf}} = 0$; the background quantum potential $Q(\rho_0) = -\frac{\hbar^2}{2m}\partial_s^2\sqrt{\rho_0}/\sqrt{\rho_0}$ gives $\partial_s Q = 0$; and the BdG term is deferred ($B = 0$) only under $k\xi \ll 1$ ($\xi = \hbar/(mc_s)$), with smallness witness $\hbar^2k^2/(4m^2c_S^2) = (k\xi/2)^2$. Thus L_s is const-coefficient Helmholtz *conditional on* $\{\rho_0'/\rho_0 = 0,\ \sqrt{\gamma_0}$ const$,\ \delta V_{\mathrm{conf}} = 0,\ \nabla Q = 0,\ k\xi \ll 1\}$ (edge R26); the BdG k^4 term is DEFERRED, not dropped unconditionally. The healing length $\xi = \hbar/(mc_s)$ and the confinement length ℓ_c (in $V_{\mathrm{wall}}(\Sigma/\ell_c)$) are kept as DISTINCT symbols (edge R27, the firewall analogous to R22's $\mu_R \neq \mu_R^{(4)}$); ℓ_c is INERT here ($\delta V_{\mathrm{conf}} = 0$).

Able-to-fail teeth. Eight teeth, mutations on copies: a nonconstant $\sqrt{\gamma_0}$ makes $M \neq 0$; a retained BdG ($B = 1$) injects the ψ'''' term; a nonuniform $\rho_0(s)$ makes N s-dependent (each $\to$ `FAIL_OPERATOR_INTRUSION`); a wall/healing renormalization $\to$ `FAIL_WRONG_SPEED`; a corrupted taper root $\to$ `FAIL_WRONG_DOMAIN`; an $\ell_c \to \xi$ conflation trips the firewall; a one-site R1 exponent corruption trips the dual-site integrity (a coordinated both-site drift is caught by the explicit frozen-export anchor); and a corrupt-$[K]$ mutation flips $[c_S^2]$ to $(3, 0, -2) \to$ `DN_UNITTEST_FAIL_DIMENSIONAL`. A dedicated witness-path tooth (nonzero δV_{conf} with $M = 0,\ N =$ bulk$,\ B = 0$) drives the verdict to `FAIL_OPERATOR_INTRUSION` through the intrusion flag alone, making the flag independently load-bearing.

Consumed + dimensional leg. The speed law $c_s^2 = 5K\rho^4/m_{\mathrm{GNLS}}$ is CITED (Part I edge R1, Stage 005) evaluated at ρ_*, integrity-checked dual-site (literal `5*K*rho**4/m` vs. the EOS route $\partial_\rho(K\rho^5)/m$) plus an explicit frozen-export anchor; the EOS exponent-5 closure $P = K\rho^5$ stays `IMPOSED`. The c_S^2 dimensional leg uses the $[K] = [P] - 5[\rho]$ chain in four spatial dimensions: $[K] = (18, 1, -2)$, $[c_S^2] = (2, 0, -2)$ (order L, M, T); the corrupt-$[K]$ probe flips it to $(3, 0, -2)$. L_0 (throat depth) is POSTULATED `ACTION`-geometry (like Stage 009's d, not a medium constant); zero new counted knobs.

Verification. Dual-engine, independent routes, zero file I/O — the source pair's scratch-YAML/`_sympy_exprs.wl` export, the Mathematica-YAML re-read, and the `expression_digest/engine_agreement` plumbing are retired; the `.wl` derives the certificate, the L_s assembly, the speed extraction, the domain solve, and the dimensional leg natively (own `D/Solve/Coefficient/Simplify`), with an arity self-check and no silenced messages; the transfer matrix, if present, is only an operator-nature cross-check (no DtN/pole/Robin — Stage 012). SymPy 61 PASS / Mathematica 71 PASS (71 = 61 shared + 8 arity self-checks + 2 native trig-basis confirmations), exit 0, CWD-independent. Full tri-review on fresh agents: `FIDELITY_CLEAN` (all values hand-re-derived; the three booleans confirmed genuinely PRODUCED; the 011/012 boundary clean) and `ADVERSARIAL_CLEAN` (the de-rig proven able-to-fail — operator via $M/N/B$, speed via renormalization, domain via taper — and the BdG confirmed a real fourth-order term). Three teeth/label nits were remediated (the firewall tooth made genuine, a witness-path tooth added so the intrusion flag is independently load-bearing, the arity labels corrected) and re-verified `REVERIFY_CLEAN`.

B.7 Stage 012: DtN Pole Ladder & Robin Falsifier ($Z_{00} = -(\omega/c_S)\tan$) (Check II-G1b)

Part / anchor. Part II – Gravity (brane↔bulk return; the frozen-throat resonator response; COMPLETES the pathA_30 D/N fold opened by Stage 011)

Claim status. Exact Within Closure

Purpose. On Stage 011's cited frozen constant-coefficient Helmholtz operator $L_s\psi = \psi'' + (\omega/c_S)^2\psi = 0$ on $s \in [0, L_0]$, pose and solve the Dirichlet/Neumann boundary-value problem: derive the outward-mouth Dirichlet-to-Neumann map $Z_{00} = -(\omega/c_S)\tan(L_0\omega/c_S)$, its half-shifted resonance ladder $\omega_n = \pi c_S(n + \frac{1}{2})/L_0$, its round-trip $R_{\mathrm{rt}} = 1$, and the Robin counterfactual that proves the D/N determination is not hardcodable — landing the joint verdict at `DN_UNITTEST_BC_DEPENDENT` because the boundary condition itself is imposed (*bc_derivation_emitted* = false, a banked calibration input).

Status. Exact Within Closure — the DtN, ladder, round trip, and Robin counterfactual are exact algebra *within the imposed-BC closure*. This stage COMPLETES the joint pathA_30 verdict `DN_UNITTEST_BC_DEPENDENT`: Stage 011 carried the reduction-certificate component (`REDUCTION_CERTIFIED`), and Stage 012 adds the D/N ladder plus the `BC_DEPENDENT` landing. It is a near-pure *bridge-strip* (contrast Stage 011's de-rig): the physics is already genuinely computed in the source and the `.wl` transfer-matrix route already carries it, so the reshape severs the scratch-YAML bridge and keeps the native content. The operator, domain, and speed are *consumed* from Stage 011 (dual-site citation-integrity); the stage opens at the general-solution `dsolve`. Its register contribution is the single *imposed* edge R28 (the D/N boundary is a banked calibration input), which discharges no knob.

The D/N boundary-value problem and the DtN. The `dsolve` of the cited L_s gives $\psi = C_1\sin(\omega s/c_S) + C_2\cos(\omega s/c_S)$. Imposing the Dirichlet mouth trace $\psi(0) = \psi_M$ and the Neumann cap $\psi'(L_0) = 0$, the 2×2 coefficient matrix (determinant $-\omega\cos(L_0\omega/c_S)/c_S$) is solved by `LUsolve`, and the outward-mouth admittance is

$$Z_{00} \;=\; -\frac{\partial_s\psi|_0}{\psi|_0} \;=\; -\frac{\omega}{c_S}\,\tan\Big(\frac{L_0\omega}{c_S}\Big).$$

`dtn_matches_target` compares this *derived* Z_{00} (from the `LUsolve`) against the typed target $-k\tan(kL_0)$ — a genuine derived-vs-typed comparison, not an $X \equiv X$: perturbing the Neumann RHS recomputes the derived DtN and the comparison fails.

Ladder, static limit, round trip. The DtN denominator's $\cos(L_0\omega/c_S)$ factor gives the pole equation $\cos(L_0\omega/c_S) = 0$, whose roots are the *half-shifted* ladder $\omega_n = \pi c_S(n + \frac{1}{2})/L_0$; `halfshift` is computed as *pole_residual* $= 0$. The small-ω series $-L_0\omega^2/c_S^2 - L_0^3\omega^4/(3c_S^4) + O(\omega^6)$ is the leading static compliance (distinct from the DC limit $\lim_{\omega\to 0^+} Z_{00} = 0$; no separate static solve). With Dirichlet/Neumann reflections $r_D = -1$, $r_N = +1$, the round-trip factor $-\exp(2iL_0\omega/c_S)$ evaluates to $R_{\mathrm{rt}} = 1$ ($\varphi_0 = 0 \bmod 2\pi$) on the ladder.

The Robin falsifier. Without a counterfactual the D/N determination is hardcodable. The Robin cap $\partial_s\psi - \alpha\psi = 0$ interpolates Neumann ($\alpha \to 0$) and Dirichlet ($\alpha \to \infty$); its DtN reads off *robin_denominator_core* $= k\cos(kL_0) - \alpha\sin(kL_0)$. The `counterfactual_guard` has exactly six members, each a *computed residual* (not a stamped truthiness): `robin_determinant_emitted` tied to the computed Robin determinant *robin_det* $\equiv -$*robin_denominator_core*; `recovers_DN_at_alpha0` ($\alpha \to 0$ recovers $-(\omega/c_S)\tan$); `recovers_DD_at_alpha_inf` ($\alpha \to \infty$ recovers the D/D DtN $k\cot(kL_0)$); `halfshift_destroyed_for_DD` (the D/D denominator $\sin(kL_0)$ gives the *integer* ladder $\pi c_S j/L_0$, not the half-shifted one); `numeric_alpha_distinct` ($\alpha = 2/L_0$ distinct from both limits, exact rationals); and `dtn_mismatch`. The artifact *dd_zero_mode_removable* $= 1/L_0$ is held out of the guard. Breaking *robin_denominator_core* flips the recomputed `recovers_DD_at_alpha_inf` and is caught ($\to$`FAIL_COUNTERFACTUAL`).

Imposed BC (honest scope) and the completed joint verdict. The D/N boundary pair is IMPOSED (*bc_provenance* = imposed); the explicit mouth/cap V_{wall} gradient derivation is *not* emitted (*bc_derivation_emitted* = false), so the verdict lands at `DN_UNITTEST_BC_DEPENDENT` rather than `DN_UNITTEST_PASS` (edge R28; earning `PASS` is a deferred upgrade, not fabricated here). The 012-scoped verdict is a function of the computed rungs (`FAIL_DIMENSIONAL` $\to$ `FAIL_POLE_LADDER` $\to$ `FAIL_COUNTERFACTUAL` $\to$ the `BC_DEPENDENT` landing), and the joint verdict is printed as `DN_UNITTEST_BC_DEPENDENT` = (011: `REDUCTION_CERTIFIED`, cited) $\wedge$ (012: DtN ladder + `BC_DEPENDENT` landing, computed here), not typed as 012-earned.

Consumed + dimensional leg. The frozen operator L_s, domain $[0, L_0]$, and speed c_S are CONSUMED from Stage 011 with dual-site citation-integrity: for L_s, site A the typed export form and site B the *null-space reconstruction* from the `dsolve`'d pair $\{\sin, \cos\}$ (posit $y'' + ay' + by$, solve for $(a, b) = (0, (\omega/c_S)^2)$) — a genuinely independent construction (a $\sin/\cos \to \sinh/\cosh$ corruption recovers $b = -(\omega/c_S)^2$ and fires); for $c_S^2 = 5K\rho^4/m$ (R1 at ρ_*), the literal vs the EOS route $\partial_\rho(K\rho^5)/m$. The 012 dimensional legs use the $[K] = [P] - 5[\rho]$ chain: [tan_argument] $= [kL_0] = 1$ (dimensionless), [Z_{00}_prefactor] $= [-k] = L^{-1}$; the corrupt-$[K]$ probe propagates $[c_S^2] \to [c_S] \to [k]$, flipping both legs $\to$ `DN_UNITTEST_FAIL_DIMENSIONAL`. α (Robin cap admittance, $[\alpha] = L^{-1}$) is a control-construction symbol; zero new counted knobs.

Verification. Dual-engine, independent routes, zero file I/O — the source pair's scratch-YAML/`_sympy_exprs.wl` export, the Mathematica-YAML re-read, and the `expression_digest/engine_agreement` plumbing are retired; the `.wl` KEEPS its transfer-matrix route as the independent engine (propagate the mouth state through `transferMatrix`$[L_0]$, impose the Neumann cap, read Z_{00} — a different decomposition from the `.py LUsolve`, computed agreement), preserving its native `robinAlpha0==dtnTransfer/robinAlphaInf==ddTransfer` self-checks, with an arity self-check and no load-bearing message silenced. SymPy 84 PASS / Mathematica 90 PASS (the +6 = the `.wl` arity block net of two SymPy-only DtN-detail checks), exit 0, CWD-independent. Full tri-review on fresh agents: `FIDELITY_CLEAN` (all values hand-re-derived; the six-member Robin guard each a genuine residual; `dtn_matches_target` genuine derived-vs-typed; the L_s null-space dual-site independent; the consume-from-011 boundary clean) and `ADVERSARIAL_CLEAN` (16-mutant matrix — the guard un-stampable, the sin / cos → sinh / cosh corruption fires in both engines, even a hardcoded DtN target is caught, all four verdict rungs reachable, arity clean). Eight able-to-fail teeth, mutations on copies.

B.8 Stage 013: Breathing Harmonic Profiles & M/K Operator Projection (Check II-G2a)

Part / anchor. Part II – Gravity (the frozen-throat $\ell = 0$ breathing mode; the (a, L) collective-coordinate closure; the M/K-projection component of the joint `BREATHING_CALIBRATED`, with truncation consistency = Stage 014 and the legacy-Hessian + Hellmann–Feynman force = Stage 015)

Claim status. Exact Within Closure

Purpose. Reduce the $\ell = 0$ linearized breathing of Gate-1's frozen constant-coefficient wall operator $\mathcal{L}_0 = \mu_\eta^{-1}(-\partial_w(T_w\partial_w\cdot) + K_\eta\cdot)$ on $w \in [0, L_0]$ to a two-mode collective closure $Q = (\delta_a, \delta_L)$: solve the harmonic-lift profiles α_a, α_L (from $\mathcal{L}_0\alpha = 0$ under the collective BCs), project the wall inertia and stiffness onto them by *real* $\int dw$ *operator projection* to build the mass/stiffness matrices M_{AB}, K_{AB}, and assemble the dynamical-EOM left-hand side $M_{AB}\ddot{Q}^B + K_{AB}Q^B = F_A^{(\mathrm{HF})}$. The first Part-II *calibrated* stage: the structure is earned, the wall constants are calibration inputs.

Status. Exact Within Closure — the harmonic-lift profiles, the M_{AB}/K_{AB} operator projection, and the EOM left-hand side are exact algebra *within the calibrated wall closure.* The verdict is the joint `BREATHING_CALIBRATED`; this stage carries its M/K-projection + (a, L)-closure component. It is the **first Part-II calibrated stage**: the gravity-sector algebra was knob-free through Stages 008–012, and the calibration enters here with the breathing-mode wall response. *The structure is EARNED; the values are CALIBRATED.* The frozen throat packet is *consumed* from Gate-1 (Stages 011/012) with dual-site citation-integrity; c_S is *not* consumed (the matter-sector c_s/BdG k^4 is deferred under $k\xi \ll 1$).

The harmonic-lift profiles. The general solution of $\mathcal{L}_0\alpha = 0$ ($\Leftrightarrow \alpha'' - \beta^2\alpha = 0$, with $K_\eta = T_w\beta^2$) is $\alpha = C_1 \sinh(\beta w) + C_2 \cosh(\beta w)$. Under the collective BCs,

$$\alpha_a = \frac{\sinh(L_0\beta - \beta w)}{\sinh(L_0\beta)} \quad (\alpha_a(0) = 1,\ \alpha_a(L_0) = 0), \qquad \alpha_L = \frac{r_{AL}\sinh(\beta w)}{\sinh(L_0\beta)} \quad (\alpha_L(0) = 0,\ \alpha_L(L_0) = r_{AL}).$$

The harmonic-residual assertion $\mathrm{Lop}[\alpha] = (-T_w\alpha'' + K_\eta\alpha)/\mu_\eta \equiv 0$ (a raise on failure) proves the profiles genuinely annihilate the operator. Its able-to-fail tooth uses a *non-kernel* corruption (a wrong wavenumber $\sinh(L_0\beta - 2\beta w)$ or $\alpha + 1$) — a mere rescale or argument sign-flip stays in $\ker(\mathcal{L}_0)$ and leaves the residual 0, so it would be a non-tooth; the adversarial leg's decisive ablation confirmed a kernel-preserving corruption is caught only by the *separate* BC assert.

M/K **by real** $\int dw$ **operator projection.** With $\langle f,g\rangle_\mu = 4\pi\int_0^{L_0}\mu_\eta f g\,dw$,

$$M_{AB} = 4\pi\int_0^{L_0}\mu_\eta\,\alpha_A\alpha_B\,dw, \qquad K_{AB} = 4\pi\int_0^{L_0}\left[T_w\,\alpha'_A\alpha'_B + K_\eta\,\alpha_A\alpha_B\right]dw,$$

computed by `sp.integrate` / native `Integrate`. Closed forms (both engines agreeing): $K_{aa} = 4\pi T_w\beta/\tanh(L_0\beta)$, $K_{aL} = -4\pi T_w\beta r_{AL}/\sinh(L_0\beta)$, $K_{LL} = 4\pi T_w\beta r_{AL}^2/\tanh(L_0\beta)$, $\det K = 16\pi^2 T_w^2\beta^2 r_{AL}^2$; and the M_{AB} forms with $[M_{AB}] = M$, $[K_{AB}] = M\,T^{-2}$. The genuineness spine (the anti-"typed M/K" burden, since one could otherwise type the legacy static Hessian $\{\chi^2\kappa+\sigma_A, -\chi\kappa, \kappa+\sigma_L\}$) is twofold: (i) an *independent re-integration* of the full integrands compared against the hardcoded report closed-forms (not against the same projection call — that would be a vacuous $X \equiv X$); dropping the $T_w\alpha'\alpha'$ gradient makes it mismatch and fires. (ii) the `forbidden_fit_flags` (`K_from_static_hessian`, `M_or_K_typed_from_legacy_values`, `full_matrix_fit`) computed from the *symbol names* of M_{AB}/K_{AB} — robust to SymPy's assumption-fragility (`Symbol('kappa')` $\neq$ `Symbol('kappa',positive=True)`, so an object-set intersection would silently miss): the names must lie in $\{\mu_\eta, T_w, \beta, L_0, r_{AL}, \pi\}$ and contain none of $\{\kappa, \chi, \sigma_A, \sigma_L\}$; typing K_{AB} from the legacy Hessian introduces those names, flips the flag, and is caught in both engines (`K_AB_provenance=operator_projection_not_static_Hessian`).

The dynamical-EOM left-hand side. The collective EOM LHS is $M_{AB}\ddot{Q}^B + K_{AB}Q^B = F_A^{(\mathrm{HF})}$ with $Q = (\delta_a, \delta_L)$, emitted as the two expanded rows. This stage emits the LHS and EOM structure only; the right-hand side $F_A^{(\mathrm{HF})}$ (the Hellmann–Feynman force) is a symbolic placeholder filled by Stage 015 (as is the HF two-route enforcement, the pathA_31 v1 rig locus). A native $M_{\det} \to 0$ degenerate counterfactual ($\alpha_a \to 0$, recomputed via the same projection) is caught by a native `M_det`$\equiv 0$ test, not the Stage-015 structure gate.

Consumed packet and the anti-tautology guard. The frozen wall packet $\{L_0 = 37/20,\ T_w = 1,\ K_\eta = 1,\ \beta = 1\}$ is cited from Gate-1 with two independently-corruptible relations: site A (constitutive) $\beta - \sqrt{K_\eta/T_w} \equiv 0$ and site B (branch) $\beta L_0 - 37/20 \equiv 0$ (the branch-determinable L/a), plus the frozen-export anchor. *The packet values are held as independent cited datums* (`K_eta_cited`, …) — the guard does *not* re-expand K_η through the alias $K_\eta = T_w\beta^2$, which would collapse site A to $\beta - \beta \equiv 0$ (a disguised $X \equiv X$); a lone `K_eta_cited` corruption genuinely breaks site A. L_0 is already registered (Stage 011, ACTION-geometry) — not re-counted.

Calibration and register (the first Part-II counted knobs). The wall constitutive packet contributes **three counted CALIB knobs** $\{\mu_\eta$ (inertia, $M\,L^{-1}$), T_w (tension, $M\,L\,T^{-2}$), β (inverse-length scale, L^{-1})$\}$, with $K_\eta = T_w\beta^2$ a *derived* manifestation (edge R29) — so three counted, not four. β is counted, *not* dressed as geometry: the source is explicit that geometry alone does not derive the wall constants, and $\beta L_0 = 37/20$ is $\beta_{(\mathrm{calibrated})}\cdot L_0 (=$ the L/a geometry), not an independent branch-pin of β (the conservative, non-shrinking read). r_{AL} (dimensionless collective BC ratio) is a control-construction symbol, tracked but not counted. Edge R30 is the named-pending nonlinear-throat reduction that would derive $\{\mu_\eta, T_w, \beta\}$ from the medium (reduction debt).

Verification. Dual-engine, independent routes, zero file I/O — the source pair's scratch-YAML/`_sympy_exprs.wl` export and the hybrid `.wl`'s `Get`/`sympy`*-cross-check/`Export` are retired; the `.wl` KEEPS its native `DSolveValue`+`Integrate` route but its asserts are *re-targeted* to its own native M/K (Mathematica-native `Csch`/`Coth` closed forms, algebraically equal to the SymPy tanh / sinh forms), with a genuine arity self-check. The 3-way cut is clean: no Stage-014 truncation/generalized-eig/β_{L_0}-sweep and no Stage-015 legacy Hessian / Hellmann–Feynman force / static-dynamic limit is computed. SymPy 78 PASS / Mathematica 84 PASS (the +6 = the `.wl` arity block), exit 0, CWD-independent. The directive design-review used the Codex→Grok→Codex bookend; Grok's compute-verification pass caught a kernel-preserving residual-tooth defect (folded) plus hardening nits. Full tri-review on fresh agents: `FIDELITY_CLEAN` (every profile/M/K/det independently re-derived via a different normalization path; `expand_hyperbolic_args` a math-preserving identity) and `ADVERSARIAL_CLEAN` (a ten-ablation matrix, all firing: the free-symbol-name ancestry, the non-kernel residual tooth, the `K_eta_cited` anti-tautology, the native degenerate-M_{det} guard, the corrupt-$[T_w]$ probe). One non-blocking nit (the baseline re-integration was value-wise $X \equiv X$) was remediated to compare against the hardcoded closed-forms, then `REVERIFY_CLEAN`. Six able-to-fail teeth, mutations on copies.

B.9 Stage 014: Truncation Consistency — Combined-Basis Generalized Eigenproblem, β_{L_0} Sweep, and N-Convergence (Check II-G2b)

Part / anchor. Part II – Gravity (the frozen-throat $\ell = 0$ breathing mode; the truncation-consistency component of the joint `BREATHING_CALIBRATED`, with the M/K-projection + (a, L)-closure = Stage 013 and the legacy-Hessian + Hellmann–Feynman force = Stage 015)

Claim status. NUMERICALLY LOCATED

Purpose. Certify that Stage-013's two-mode collective closure $Q = (\delta_a, \delta_L)$ is a *faithful* truncation, not an artifact. Embed the two collective harmonic-lift profiles α_a, α_L in a larger Galerkin basis $B = \{\alpha_a, \alpha_L, g_1 \dots g_N\}$ (N sine lane modes $g_n = \sin((n-\frac{1}{2})\pi w/L_0)$), solve the combined-basis generalized eigenproblem $Kv = \omega^2 Mv$, and measure how much of each of the two lowest generalized modes lives in $\mathrm{span}\{\alpha_a, \alpha_L\}$ — the modal overlaps o_1, o_2 — across a β_{L_0} sweep and as $N \to 16$. The first *numeric / float-bearing* stage of the rebuild.

Status. NUMERICALLY LOCATED — the generalized eigenproblem, the modal overlaps, the β_{L_0} validity window, and the N-convergence are numerically computed (`scipy.linalg.eigh`+`quad`; `Eigensystem`+`NIntegrate`). The verdict is the joint `BREATHING_CALIBRATED`; this stage carries its *truncation-consistency* component (2/3). It **adds no new counted knobs**: the underlying wall constants $\{\mu_\eta, T_w, K_\eta, \beta\}$ are Stage-013's calibration inputs, and the numeric truncation controls are method/tolerance parameters. Stage-013's collective profiles and the frozen packet are *consumed* with dual-site citation-integrity; c_S is *not* consumed (matter-sector deferred, $k\xi \ll 1$).

The combined-basis generalized eigenproblem. Over $B = \{\alpha_a, \alpha_L, g_1 \dots g_N\}$ with normalized $x = w/L_0 \in [0, 1]$, the mass/stiffness Gram is built by numeric quadrature, $M_{ij} = \int_0^1 \phi_i \phi_j$, $K_{ij} = \int_0^1 [\phi_i' \phi_j' + \beta^2 \phi_i \phi_j]$, rank-deficient rows are dropped, and $Kv = \omega^2 Mv$ is solved and sorted ascending. The two collective modes $\alpha_a = \sinh(\beta(1-x))/\sinh\beta$, $\alpha_L = \sinh(\beta x)/\sinh\beta$ are *cited* from Stage 013 (not re-derived); the numeric 2×2 collective block equals Stage-013's M_{AB}/K_{AB} only up to the 4π, L_0-Jacobian, and μ_η-weight prefactors, which cancel in the generalized eigenproblem and the mass-normalized overlaps — so the seam is guarded via the consumed profiles, *not* a naive M_{aa} equality.

The modal overlaps and the truncation floor. Projecting each of the two lowest generalized eigenvectors onto span$\{\alpha_a, \alpha_L\}$ via the mass-Gram gives $o_k = \|P_\alpha v_k\|_M / \|v_k\|_M$. The certificate is

$$\text{pass} \iff o_1 \geq \text{FLOOR} \ \wedge \ o_2 \geq \text{FLOOR} \ \wedge \ \min(\omega_1^2, \omega_2^2) > 0, \qquad \text{FLOOR} = 0.9,$$

a genuine predicate on the *computed* overlaps. At the physical $\beta_{L_0} = 37/20$, $N = 16$: $o_1 = 0.99311$, $o_2 = 0.98776$, $\min(\omega^2) = 3.42252$, gap $= 2.22787$, pass $=$ true. All three conjuncts are separately able-to-fail (the o_2 floor, the $\min(\omega^2) > 0$ positivity leg, and the window edge).

The computed validity window. Sweeping $\beta_{L_0} \in \{0.1, \ldots, 3.0, 5, 10, 18, 30, 50\}$ — *spanning well beyond* the pass region (to $o_1 \approx 0.27$ at $\beta_{L_0} = 50$) — the certificate passes for $\beta_{L_0} \in \{0.1 \ldots 3.0\}$ and **fails** for $\beta_{L_0} \geq 5$ ($o_1 = 0.860 <$ FLOOR already at $\beta_{L_0} = 5$). The window $\{\text{min:}0.1,\ \text{max:}3.0\}$ is *computed* from the passing set; it has a real upper edge. The honest caveat: clean two-mode truncation holds only for order-unity wall stiffness $K_\eta/T_w \lesssim 2.6$ (the emergent edge, since $\beta^2 = K_\eta/T_w$ and $(3/1.85)^2 \approx 2.63$) — an emergent number, not a typed criterion. The physical $\beta_{L_0} = 37/20$ is the *cited* Gate-1 anchor, not the β that maximizes overlap. *The pathA_31 v1 rejection was a gamed truncation threshold; the floor and window here are genuinely computed and able-to-fail.*

N-convergence and the counterfactual overlaps. At $\beta_{L_0} = 37/20$, o_1 is stable across $N \in \{4, 8, 12, 16\}$ ($0.99301 \to 0.99311$, $\max|\Delta| \approx 10^{-4}$), pass $=$ true at every N — so the certificate is not a small-N artifact (the mass condition growth 3.1e3 $\to$ 1.9e5 is a benign conditioning effect). Two counterfactual profiles pin the honest scope: the *degenerate* $\alpha_a \equiv 0$ collapses the collective span (rank deficient, $o_2 = 0.2227 <$ FLOOR) and **fails** the floor; the *constant* $\alpha_a \equiv 1$ — a *wrong* profile — gives $o_1 = 1.0$, $o_2 = 0.9738 \geq$ FLOOR and **passes** the overlap. *The overlap therefore certifies truncation-consistency, not profile-correctness*: Stage 014 records that the constant profile passes and does *not* claim it is rejected — its rejection is Stage-015's Hellmann–Feynman mismatch (and Stage-013's $\mathcal{L}_0[\alpha] = 0$ residual).

Consumed closure and register. The collective profiles and the frozen packet $\{L_0 = 37/20,\ \beta = 1,\ \beta L_0 = 37/20\}$ are cited from Stage 013 with two independently-corruptible relations: site A (branch anchor) $\beta L_0 - 37/20 \equiv 0$, reading an independent cited datum; and site B (profile-consumption) the consumed profiles satisfy *both* the harmonic residual $-\alpha'' + \beta^2\alpha = 0$ *and* the collective BC values — the residual alone misses a kernel-preserving corruption (a rescale leaves the residual 0 but breaks the BC), so residual$\wedge$BC are both able-to-fail. Stage 014 adds **zero new counted knobs**: the numeric controls $\{\text{FLOOR} = 0.9,\ N_{\text{final}} = 16,\ N_{\text{conv}},\ \beta_{L_0}\text{-sweep}\}$ are method/tolerance parameters (tracked, not counted); the wall packet is consumed from Stage 013 (no double-count). One structural edge, R31 (the truncation validity window $K_\eta/T_w \lesssim 2.6$), discharges nothing.

Verification. Dual-engine, independent numeric routes, zero file I/O — the source pair's numeric-WL scratch export and the hybrid `.wl`'s `Get`/`sympy`*-float-diff/`Export` are retired; the `.wl` KEEPS its native `NIntegrate`+`Eigensystem` `galerkinRow` route but its asserts are *re-targeted* to its own computed overlaps, floor, and window (it declares its own sweep grid and the cited profile closed forms natively, not via `DSolveValue`), with a genuine arity self-check. Agreement is transcript-level (both engines independently reach the same pass-pattern, window, and o_k to $\sim 10^{-13}$, tighter than the 0.9 margin). The 3-way cut is clean: no Stage-013 $\int dw$ projection and no Stage-015 Hellmann–Feynman force is computed. SymPy/SciPy 93 PASS / Mathematica 100 PASS (the +7 = the `.wl` arity/leakage block), exit 0, CWD-independent. The directive design-review used the Codex$\to$Grok$\to$Codex bookend (three BLOCKING folds: the site-B residual$\wedge$BC, the $\min(\omega^2) > 0$ tooth, and the N-convergence non-converging ablation; Grok's compute-verification independently re-derived every anchor). Full tri-review on fresh agents: `FIDELITY_CLEAN` (an independent from-scratch Galerkin re-derived every value) and `ADVERSARIAL_ISSUES` — the anti-gaming floor/window proven clean (23/23 genuine ablations firing), but *two* able-to-fail teeth (the constant-passes-overlap honesty leg and the degenerate-bypass leg) were vacuous x:=const; expect_fail(x) stamps in both engines. Remediated to a *shared predicate* read by both the baseline and the tooth's mutated-copy assertion; a fresh-agent re-verify confirmed `REVERIFY_CLEAN` via the deciding coupling meta-test (corrupting the shared predicate makes the tooth stop firing, so the audit correctly fails). Nine able-to-fail teeth, mutations on copies.

B.10 Stage 015: Legacy-Hessian Structure Recovery & the Hellmann–Feynman Force (Check II-G2c)

Part / anchor. Part II – Gravity (the frozen-throat $\ell = 0$ breathing mode; the legacy-structure + Hellmann–Feynman-force component of the joint `BREATHING_CALIBRATED`, which it *completes* — with the M/K-projection + (a, L)-EOM-LHS = Stage 013 and the truncation consistency = Stage 014)

Claim status. Exact Within Closure

Purpose. Complete the calibrated breathing sector. **(Q1)** Show Stage-013's operator-projected closure M_{AB}, K_{AB} has the same structural signature as the legacy static energy Hessian $H_{\text{legacy}} = \partial^2 E_{\text{geom}}/\partial Q^2$ (the (a, L) closure *recovered*, not re-postulated). **(Q2)** Supply the EOM driving force $F_A^{(\text{HF})}$ — the right-hand side Stage 013 deferred — by *two genuinely different* constructions (a distributed source projection and a Hellmann–Feynman parametric derivative) that agree only after simplification. Then take the static-dynamic limit and land the completed joint verdict.

Status. Exact Within Closure — the structure recovery and the Hellmann–Feynman force are exact closed-form algebra (`sp.integrate/FullSimplify`; `expect_zero` residuals, exact rationals $L_0 = 37/20$; the symbolic / float-free inverse of the numeric Stage 014). The verdict is the joint `BREATHING_CALIBRATED`; this stage carries its *legacy-structure + HF-force* component (3/3) and **completes** it, filling the EOM RHS $F_A^{(\text{HF})}$. It adds **one** new counted `CALIB` knob — the breathing driving-force scale V_{p0}/ℓ_c (with V_{p0} and the now-live ℓ_c its manifestations; ρ_* consumed) — making it the second calibration-adding Part-II stage after Stage 013. Stage-013's profiles and M_{AB}/K_{AB} and Stage-014's truncation certificate are *consumed* with dual-site citation-integrity; the legacy constants $\{\kappa, \chi, \sigma_a, \sigma_L\}$ are the legacy-Hessian pattern basis (not counted afresh); c_S is *not* consumed (matter-sector deferred, $k\xi \ll 1$).

The legacy energy and its Hessian. With $E_{\text{geom}} = \frac{1}{2}\kappa(\delta_L - \chi\delta_a)^2 + \frac{1}{2}\sigma_a\delta_a^2 + \frac{1}{2}\sigma_L\delta_L^2$,

$$H_{\text{legacy}} = \frac{\partial^2 E_{\text{geom}}}{\partial Q^2} = \begin{pmatrix} \chi^2\kappa + \sigma_a & -\chi\kappa \\ -\chi\kappa & \kappa + \sigma_L \end{pmatrix},$$

symmetric, negative off-diagonal ($-\chi\kappa < 0$ for $\chi, \kappa > 0$), det $= \kappa\sigma_a + \kappa\chi^2\sigma_L + \sigma_a\sigma_L > 0$. The legacy constants enter *here*; a firewall echo confirms the cited M_{AB}/K_{AB} free symbols exclude them.

The structure recovery. The gate reads the cited M_{AB}/K_{AB} and matches the structural signature of H_{legacy}: M positive-definite by Sylvester — $M_{aa} > 0$ and $\det M > 0$ for $B = \beta L_0 > 0$ — discharged by *exact-identity certificates* (not `is_positive`, which returns `None` on the transcendentals, and *not* the $M_{aa} - m^{+}_{aa} = 0$ form-equality, which is $X \equiv X$ once M is cited): $f_1 = \sinh B \cosh B - B$ has $f_1(0) = 0$, $f_1' = \cosh 2B - 1 = 2\sinh^2 B \geq 0$; $f_2 = \sinh^2 B - B^2 = (\sinh B - B)(\sinh B + B) > 0$ via $g = \sinh B - B$, $g' = \cosh B - 1 = 2\sinh^2(B/2)$. Then K symmetric, $K_{aL} < 0$, $\operatorname{rank} K = \operatorname{rank} H_{\text{legacy}} = 2$, and matching zero-pattern give $K_{\text{structure ok}}$; full matrix fit = false (the recovery is structural, $M_{aa} \neq H_{aa}$). The probes $M_{aa} \to -M_{aa}$ and $K_{aL} \to -K_{aL}$ flip the gate. *The operator-projected closure has the same structural form as the legacy energy Hessian.*

The Hellmann–Feynman force (both routes). With the confinement force density src $= \rho_* V_{p0}/\ell_c$ and the action measure $4\pi \int dw$:

$$\underbrace{F_A^{\text{dist}} = 4\pi \int_0^{L_0} \text{src}\, \alpha_A \, dw}_{\text{distributed projection}} \;=\; \underbrace{F_A^{\text{leg}} = -4\pi \int_0^{L_0} \rho_* \frac{\partial V_{\text{conf}}}{\partial q_A}\Big|_0 dw}_{\text{Hellmann–Feynman parametric}},$$

$V_{\text{conf}} = (V_{p0}/\ell_c)(r - R_0(w) - q_a\alpha_a - q_L\alpha_L)$, Route 2 a genuine $\partial/\partial q$ (the double-negative). They agree (hf force reduces = true, $F_a^{\text{dist}} = 4\pi V_{p0}\rho_*(\cosh L_0\beta - 1)/(\beta\ell_c \sinh L_0\beta)$), yet the *raw* unsimplified trees are computed-distinct: unsimplified routes identical = false ($+4\pi\int(+\cdots)$ vs $-4\pi\int(-\cdots)$). *This is the pathA_31 v1 rejection's other locus — the HF $x{-}x$ tautology — emitted here as two genuinely different constructions; the verdict rung places the anti-$x{-}x$ in the OR: typing Route 2 as a copy of Route 1 trips `BREATHING_FAIL_HF_FORCE`.* This fills the deferred RHS $F_A^{(\text{HF})}$.

Static-dynamic limit and the profile guard. $\ddot{Q} \to 0$ in $M_{AB}\ddot{Q} + K_{AB}Q = F_A$ gives $K_{AB}Q = F_A$, the static $\partial E_{\text{geom}}/\partial Q = 0$ limit (the exact $\ddot{Q} \to 0$ EOM residual identity, not a separate solve). The HF guard rejects both wrong profiles ($\alpha_a \equiv 0$: $F_a^{\text{dist}} = 0$; $\alpha_a \equiv 1$: $F_a^{\text{dist}} = 4\pi L_0 V_{p0}\rho_*/\ell_c$; both $\neq F_a^{\text{leg}}$, so hf mismatch = true). *The 014↔015 boundary:* the constant profile *passes* Stage-014's modal overlap but *fails* Stage-015's HF — the Hellmann–Feynman mismatch is the profile guard the overlap could not supply (with Stage-013's $\mathcal{L}_0[\alpha] = 0$ residual).

Consumed closure and register. Stage-013's profiles, M_{AB}/K_{AB}, and the frozen packet $\{L_0 = 37/20,\ \beta = 1,\ \beta L_0 = 37/20\}$ are cited with dual-site integrity: site A the branch anchor $\beta L_0 - 37/20 \equiv 0$; site B the profiles satisfy *both* the harmonic residual $-\alpha'' + \beta^2\alpha = 0$ *and* the BC values (residual alone misses a kernel-preserving corruption); and the M/K citation covers every entry — two det-identities for the diagonals, and, because det carries M_{aL}^2 (blind to an off-diagonal sign flip), explicit sign checks $M_{aL} > 0$ (via $h = B - \tanh B$) and $K_{aL} < 0$. Register: **one** new counted `CALIB` knob, the driving-force scale V_{p0}/ℓ_c; the legacy $\{\kappa, \chi, \sigma_a, \sigma_L\}$ are the pattern basis (not counted afresh). Two structural edges: R32 (the legacy-Hessian structure recovery, discharges nothing) and R33 (the confinement-drive reduction debt, a sibling of R30's wall-response debt, `PENDING`).

Verification. Dual-engine, independent symbolic routes, zero file I/O — the source pair's `sympy`* export and the hybrid `.wl`'s `Get`/`checks`/`Export`/digest are retired; the `.wl` KEEPS its native HF `Integrate`, Hessian `D`, and structure `Det`/`MatrixRank`, re-targeted to its own values, citing the profiles and M/K as native literals (not `DSolveValue`, not re-`Integrate`) and building its own H_{legacy}, HF routes, and raw-tree anti-$x{-}x$ comparison, with an arity self-check. Agreement is transcript-level (the `.wl` $\tanh(\beta L_0/2)$ form equals the `.py` $(\cosh L_0\beta - 1)/\sinh L_0\beta$ via the half-angle identity). The 3-way cut is clean: no Stage-013 $\int dw$ projection, no Stage-014 generalized eig. SymPy 95 PASS / Mathematica

102 PASS, exit 0, CWD-independent. The directive design-review used the Codex→Grok→Codex bookend (two BLOCKING folds: the det-blind off-diagonal sign hole $M_{aL} \to -M_{aL}$, added explicit sign checks; and the float-free positivity-certificate requirement — $M_{aa} > 0$ is not dischargeable by `is_positive`, so exact-identity certificates replace the vacuous form-equality). Full tri-review on fresh agents: `FIDELITY_CLEAN` (an independent read hand-verified all seven certificate identities and every value) and `ADVERSARIAL_ISSUES` — the three central burdens proven genuine by live copy-mutations, but *one* able-to-fail tooth (the HF profile-guard leg) was a vacuous $x-x$ self-compare in both engines. Remediated to read the real wrong-profile forces (with three lower-severity literal/scaffolding flags made-genuine or honestly de-counted); a fresh-agent re-verify confirmed `REVERIFY_CLEAN` via the coupling meta-test (each remediated construct fires under a mutation of its named object). Eight able-to-fail teeth, mutations on copies.

B.11 Stage 016: The $\ell = 2$ SO(3) Covariance Theorem (Check II-G3a)

Part / anchor. Part II – Gravity (the frozen-throat $\ell = 2$ sector; the EARNED-FIRST leg of a two-way split of pathA_32 — the SO(3) covariance-theorem component of the joint `ISOTROPY_CALIBRATED`, with the grouped-P2 lane isotropy + the calibration partition = Stage 017)

Claim status. Exact Within Closure

Purpose. Earn the $\ell = 2$ angular structure the grouped-lane isotropy rides. Show the five real $\ell = 2$ harmonics form one orthonormal SO(3)-irrep (Gram = I_5) with a single *computed* $-\Delta_{S^2}$ eigenvalue $\lambda_m = \ell(\ell+1) = 6$ for every m (Laplace–Beltrami + Rayleigh quotient + eigenfunction residual), and that the K_2 *angular* stiffness $K_2 = \tilde{K} + \lambda_m \tilde{T}_\Omega$ uses that computed eigenvalue. Land the joint verdict as a partial component.

Status. Exact Within Closure — the covariance theorem is exact symbolic algebra (genuine S^2-integral Gram, the real $-\Delta_{S^2}$ operator, Rayleigh quotients, eigenfunction residuals, exact $\{L, M, T\}$ dimension vectors; `expect_zero`/`expect_bool` residuals, no `scipy`/`numpy`/ floats/tolerances). The verdict is the joint `ISOTROPY_CALIBRATED`; this stage carries its *SO(3) covariance-theorem* component (1/2, the EARNED slice) and lands the joint as **partial** (016 earned, 017 pending) — not complete. It adds **zero** new counted knobs (a structural covariance edge, like Stages 011/012/014); the new symbols first appearing here — $T_\Omega/\tilde{T}_\Omega$ (the angular-stiffness density) and $\beta_2(w)$ (the frozen $\ell = 2$ radial profile) — have their counting *deferred* to Stage-017's calibration partition. c_S is *not* consumed (the angular structure is speed-free).

The real basis and orthonormality. The five real $\ell = 2$ harmonics $Y_{20} = \sqrt{5/16\pi}\,(3\cos^2\theta - 1)$, $Y_{21c} = -\sqrt{15/4\pi}\,\sin\theta\cos\theta\cos\phi$, Y_{21s} ($\cos\phi \to \sin\phi$), $Y_{22c} = \sqrt{15/16\pi}\,\sin^2\theta\cos 2\phi$, Y_{22s} ($\cos 2\phi \to \sin 2\phi$) are orthonormal on the sphere:

$$\mathrm{Gram}_{ij} = \int_{S^2} Y_i Y_j\, d\Omega = \int_0^\pi \int_0^{2\pi} Y_i Y_j\, \sin\theta\, d\phi\, d\theta = \delta_{ij}, \qquad \mathrm{Gram} = I_5.$$

This is a computed integral — a wrongly-normalized harmonic breaks gram is identity and drives `covariant` to `FAIL_NOT_COVARIANT`.

The computed eigenvalue (the crux). With the real Laplace–Beltrami operator $-\Delta_{S^2} f = -[(1/\sin\theta)\partial_\theta(\sin\theta\,\partial_\theta f) + (1/\sin^2\theta)\partial_\phi^2 f]$, two independent computations certify λ_m is *computed*, not typed: the Rayleigh quotient $\lambda_m = \int Y_m(-\Delta_{S^2})Y_m\,d\Omega / \int Y_m^2\,d\Omega = 6$ for every m, *and* the eigenfunction residual $(-\Delta_{S^2})Y_m - \lambda_m Y_m = 0$ (proving each Y_m is a genuine eigenfunction). So $\lambda_m = \ell(\ell+1) = 6$ is the same across all five m: the $\ell = 2$ sector is one 5-dimensional SO(3)-irrep with a single eigenvalue — the angular degeneracy that *is* the covariance.

The K_2 angular stiffness. The angular stiffness $K_2 = \tilde{K} + \lambda_m \tilde{T}_\Omega$ (with $M_2 = \tilde{M}$) is assembled with the *live* computed eigenvalue (`build_K2(lambdas)`, not a typed 6). Its computed-ness is able-to-fail through a residual on the assembled coefficient ($(-\Delta_{S^2})Y - (\text{coeff in } K_2)\,Y = 0$, which a wrong typed coefficient breaks) *and* the *bare* forced eigenvalue probe (coefficient $2 \neq 6 \Rightarrow$ `FAIL_NOT_COVARIANT`). *The source's $k_{\text{coeff equal}}$ self-compare $\lambda - \lambda \equiv 0$ is vacuous and is de-counted*; its role is carried by these two genuine checks.

Angular dimensional consistency. On pathA_32's own convention — volume densities on the wall measure $dV = a^2\,dw\,d\Omega$ (dimension L^3), β_2 dimensionless, $\beta_2' = L^{-1}$, $[\mu_\eta] = ML^{-3}$, $[T_w] = ML^{-1}T^{-2}$, $[K_\eta] = [T_\Omega] = ML^{-3}T^{-2}$ — one finds $[M_2] = M$, each K_2 term MT^{-2}, $[K_2/M_2] = T^{-2}$. The dimensions are *sourced* from the density integrals (not back-solved). Corrupting the sourced $[T_\Omega]$ (and $\tilde{T}_\Omega$) by one power of L makes the K_2 sum inhomogeneous $\Rightarrow$ `FAIL_DIMENSIONAL`; the same holds for a one-power corruption of $[\mu_\eta]$, $[T_w]$, or $[K_\eta]$.

The able-to-fail battery. The scoped verdict runs 016's gates only (dimensional, covariant, tautology, able-to-fail; 017's stability/denominator/lane/dynamic gates are not computed here). The aggregate battery is 016's three probes — wrong eigenvalue (coeff $2 \neq 6 \Rightarrow$ `FAIL_NOT_COVARIANT`), tautology hash collision (identical harmonic inputs $\Rightarrow$ non-distinct SHA-256 hashes $\Rightarrow$ `FAIL_TAUTOLOGICAL`), and dimensional corrupt T_Ω ($\Rightarrow$`FAIL_DIMENSIONAL`) — each a computed conjunction of (its mutation fires its verdict) $\wedge$ (its self-ablation suppresses); *neutering any one flips the aggregate false.*

Honest scope. Angular earned / radial calibrated: 016 derives the angular structure, but the radial profile $\beta_2(w)$ and the radial/support scalars remain frozen calibration inputs (not derived from the Gate-1 R_0 support equation) — which is *why* the joint is `ISOTROPY_CALIBRATED` rather than `ISOTROPY_PASS`; the calibration partition is Stage-017's. The 54/5 quadrupole normalization, outgoing odd-N coefficients, and the solved nonlinear branch ($G =$`GENUINE_BLOCKED`) are deferred Gate 4/5/6 work.

Consumed closure and register. *Provenance, not a cross-stage relation:* pathA_32's volume-density / dimensionless-β_2 convention differs from Stage-013's line-density / $\beta = L^{-1}$ convention (related by an $\int a^2 d\Omega \approx L^2$ bridge, not equal), so the Stage-013 relation $K_\eta = T_w\beta^2$ does *not* transfer. Hence μ_η, T_w are cited as the same physical wall constants counted at Stage 013 (provenance), the frozen radial reference and the Gate-1 D/N boundary (Stage 012 R28) are cited, and the genuine able-to-fail integrity is pathA_32's *own* $[M_2] = M/[K_2] = MT^{-2}$ dimensional consistency. Register: **zero** new counted knobs; the new T_Ω/β_2 counting is deferred to Stage 017; structural edge **R34** (the $\ell = 2$ SO(3) covariance provenance — five real harmonics = one SO(3)-irrep with $\lambda_m = 6$; discharges nothing).

Verification. Dual-engine, independent symbolic routes, zero file I/O — the source pair's scratch-YAML engine-agreement handoff is severed; the `.wl` KEEPS its native `Integrate`/ `D`/`Hash` route (its own `dimOf`/`scopedVerdict`), not a mirror, agreeing at the transcript level (Gram $= I_5$, $\lambda_m = 6$, the three probe verdicts, the 016 aggregate), with an arity self-check. SymPy 82 PASS / Mathematica 91 PASS, exit 0, CWD-independent. The directive design-review used the Codex→Grok→Codex bookend; the Grok compute-verify pass caught the volume-vs-line dimensional-convention mismatch (the cross-stage dual-site was non-transferable → reframed to provenance + self-contained integrity), and strengthened the $k_{\text{coeff equal}}$ de-count to a residual on the assembled K_2 coefficient. Full tri-review on fresh agents: `FIDELITY_CLEAN` (an independent read hand-re-derived the harmonics, the Laplace–Beltrami operator, all Rayleigh $\lambda = 6$ with zero residuals, and the full $L/M/T$ dim walk) and `ADVERSARIAL_CLEAN` (per-tooth ablation: all 82 SymPy checks + 4 native `.wl` ablations fired at their own assert) — with one low-severity stamped literal (participates in verdict) remediated to a computed verdict-propagation check; a fresh-agent `REVERIFY_CLEAN` confirmed it fires under its own ablation. Symbolic per-tooth ablation, mutations on copies.

B.12 Stage 017: Grouped-P2 Lane Isotropy (Check II-G3b)

Part / anchor. Part II – Gravity (the frozen-throat $\ell = 2$ sector; the COMPLETING leg of the two-way split of pathA_32 — the grouped-lane isotropy + calibration partition that consumes Stage 016's covariance theorem, lands the joint `ISOTROPY_CALIBRATED`, and exports the $\ell = 2$ port kernel)

Claim status. Exact Within Closure

Purpose. Show that the grouped $\{20, 21, 22\}$ $\ell = 2$ lanes, assembled from Stage 016's consumed covariance ($\lambda_m = 6$, $K_2 = \tilde{K} + \lambda_m \tilde{T}_\Omega$), respond isotropically — the P_2-projection raw-D defects vanish (primary) and the normalized-u response defects vanish (cross-check) — and that the response is genuinely able-to-fail (the six-probe battery). Land the joint verdict as complete and export the port kernel.

Status. Exact Within Closure — mixed exact-symbolic + numeric-sample: the lane algebra, D-lanes, raw-D/normalized-u defects, and the c/s degeneracy are exact symbolic (`compact(expr)==0`, `expect_zero/expect_bool`); the response-probe "moves under ε" tests and the stability/denominator windows are numeric over a calibration sample/window (tolerance-guarded, as the source). The verdict is the joint `ISOTROPY_CALIBRATED`; this stage carries its *grouped-lane isotropy* component (2/2) and lands the joint as **complete** (016 earned-cited ∧ 017 earned-here). It is the *third* calibration-adding Part-II stage — it adds two counted `CALIB` knobs $\{T_\Omega, \beta_2\}$.

The grouped-lane assembly. Each channel is assembled from the consumed eigenvalue $\lambda_m = 6$ and the K_2 form:

$$M_2 = c_{\text{self}} \tilde{M}, \quad K_2 = c_{\text{self}} \, (\tilde{K} + \lambda_m \tilde{T}_\Omega), \quad c_{\text{self}} = \int Y_m^2 \, d\Omega = 1 \text{ (the Gram diagonal, cited from 016)},$$

with D-lanes $D_0 = K_2 - (\tilde{B}_0 + \tilde{Z}_0)$, $D_2 = -(\tilde{M} + \tilde{B}_2 + \tilde{Z}_2)$, $D_4 = -(\tilde{B}_4 + \tilde{Z}_4)$. The three grouped lanes are 20, 21 = avg($21c$, $21s$), 22 = avg($22c$, $22s$); this grouped $\{M_2, K_2, D_{0,2,4}, \tilde{B}/\tilde{Z}\}$ *is* the exported $\ell = 2$ port kernel. $\lambda_m = 6$ and the K_2 form are cited from Stage 016, not re-derived.

Raw-D defects = 0 — **the primary isotropy test.** With the P_2-projection defect_pair$(t) = ((2t_{20} - t_{21} - t_{22})/10, \, (t_{21} - t_{22})/2)$, the grouped D-lane defects vanish at every order, $\{a_{D0}, b_{D0}, a_{D2}, b_{D2}, a_{D4}, b_{D4}\} = 0$ (exact), with the within-group degeneracy $D_{21c} = D_{21s}$, $D_{22c} = D_{22s}$. This is the isotropy statement: the grouped $\ell = 2$ response is angularly degenerate because the lanes ride 016's single eigenvalue $\lambda_m = 6$.

Normalized-u defects $= 0$ — the cross-check. The DtN-observable normalized responses $u_2 = -D_2/D_0$, $u_4 = (D_2^2 - D_0D_4)/D_0^2$ also have vanishing P_2-projection defects. This is a *cross-check*, not the primary test: a pure-prefactor anisotropy *moves* the raw-D defects but leaves the normalized-u defects zero (the pure prefactor probe), so normalized-u alone would miss a real anisotropy — raw-D is decisive.

The consumed covariance — a genuine cross-stage dual-site. Because Stages 016 and 017 share the same pathA_32 convention (volume densities, dimensionless β_2) — unlike 016's provenance-only cite of Stage 013 — the cited $\lambda_m = 6$ and K_2 form are a *checkable* relation. Site A: the λ_m fed into the lane assembly equals 016's exported eigenvalue. Site B: the *assembled* lane K_2's $\tilde{T}_\Omega$ coefficient equals the cited λ_m (reads the assembled lane, guarding the assembly formula). The λ-dimensionless trap: a corruption $\lambda : 6 \to 4$ is dimensionally silent, keeps the windows positive, and *keeps raw-D isotropy incidentally unbroken* (all-lanes $\lambda = 4$ gives raw-D$= 0$) — so an incidental gate cannot catch it. A one-site corruption fires an explicit integrity guard; the coordinated-both-sites corruption is caught by a single-harmonic $(-\Delta_{S^2})Y_{20} = \lambda Y_{20}$ echo. Stage 017 does not re-derive the covariance theorem.

The able-to-fail battery. 017's aggregate is its six response probes, each a computed conjunction; *neutering any one flips* able to fail ok *false*: pure prefactor/sector selective/ m dependent ($\to$`FAIL_ANISOTROPIC_BRANCH`, load-bearing on the computed move-flags — their forced verdict via `case_verdict(lane_collapse_ok=False)` is vacuous by design), degenerate β_2 ($\to$`FAIL_STABILITY`), singular denominator ($\to$`FAIL_SINGULAR_RESPONSE`), static drop inertia ($\to$`FAIL_STATIC_RESPONSE`). The three 016-owned probes are cited via 016's three-probe aggregate; the joint able to fail is 017's six $\wedge$ 016's cited three.

Honest scope and register. Calibrated not `PASS`: 017 earns the grouped-lane isotropy, but the wall profile $\beta_2(w)$, $R_0(w)$, the radial scalars $\tilde{M}/\tilde{K}/\tilde{T}_\Omega$, the support scalars $\tilde{B}/\tilde{Z}$, and the calibration window are frozen calibration inputs — *not* derived from the Gate-1 R_0 support equation (edge **R36**, the `ISOTROPY_PASS` target). Register: two new counted `CALIB` knobs $\{T_\Omega, \beta_2\}$ (the $\ell = 2$ angular-stiffness density — a genuinely new physical DOF absent from the $\ell = 0$ breathing packet — and the frozen $\ell = 2$ radial profile). The radial scalars $\tilde{M}/\tilde{K}/\tilde{T}_\Omega$ are *derived* manifestations $= \int \rho\, \beta_2^2\, dV$ (edge **R35**, the $\ell = 2$ analogue of R29). The port-kernel support/Maxwell scalars $\{\tilde{B}, \tilde{Z}\}$ are frozen inputs first load-bearing here but tracked/downstream-pinned (the isotropy verdict is value-independent of them; the $\tilde{Z}$ Maxwell couplings pin at Stages 018–024). μ_η, T_w are cited provenance (Stage 013; $K_\eta = T_w\beta^2$ R29 non-transferable across the volume-vs-line convention); c_S is not consumed. Edge **R34** (016's covariance provenance) is backfilled. Part-II counted `CALIB` set $= \{\mu_\eta, T_w, \beta\}(013) + \{V_{p0}/\ell_c\}(015) + \{T_\Omega, \beta_2\}(017) = 6$. Deferred (Gate 4/5/6): the 54/5 normalization, outgoing odd-N coefficients, and the solved nonlinear branch ($G =$`GENUINE_BLOCKED`).

Verification. Dual-engine, independent symbolic + native routes, zero file I/O — the source pair's scratch-YAML handoff is severed; the `.wl` keeps its native `Integrate/FullSimplify/ D/N` route (its own `verdictFromGates/caseVerdict/dimOf`), not a mirror, agreeing at the transcript level (the grouped lanes, raw-D$= 0$, normalized-u$= 0$, the six probe verdicts, the joint `ISOTROPY_CALIBRATED`), with an arity + unevaluated-leakage self-check. SymPy 118 PASS / Mathematica 127 PASS, exit 0, CWD-independent. The directive design-review used the Codex$\to$Grok$\to$Codex bookend; the Grok compute-verify pass caught a blocking gap (the harmonics/`intS2` infrastructure 017 needs for its anisotropy coefficients had been mislabeled for exclusion) plus two nits (the export coefficient pinned to $\tilde{K}+6\tilde{T}_\Omega$; the three anisotropy probes' forced verdict flagged vacuous so the load rides the move-flags), and independently confirmed that all-lanes $\lambda = 4$ keeps raw-D$= 0$ — proving the explicit dual-site necessary. Full tri-review on fresh agents:

FIDELITY_CLEAN (independent re-derivation of the anisotropy coefficients $\{2/7, 1/7, -2/7\}$, the Y_{20} echo, the pure-prefactor discriminator, the six probe verdicts, and the joint landing) and ADVERSARIAL_ISSUES (42 per-tooth ablations, 41 firing at their own assert) with one genuine finding — the .wl Site-B (K_2-form) residual reconstructed K_2 from lambdaByChannel instead of reading the assembled lane, so an assembly-formula corruption fired only downstream. Remediated: the .wl Site-B now reads the assembled lane (mirroring SymPy), with matching assembly-formula ablation teeth in both engines. A fresh-agent REVERIFY_CLEAN confirmed the coupling meta-test (the fixed tooth fires at its own named assert on an assembly-formula corruption in both engines; neutering the fix makes it vacuous), with no regression. Symbolic/numeric per-tooth ablation, mutations on copies.

B.13 Stage 018: The Outgoing $\ell = 2$ DtN Hankel Fingerprint and χ_Q Sign (Check II-G4a)

Part / anchor. Part II – Gravity (the frozen-throat $\ell = 2$ radiative sector; the EARNED-FIRST leg of a four-way split of pathA_33 — the outgoing DtN Hankel-fingerprint + χ_Q-sign component of the joint QUAD_CALIBRATED, with the prefactor algebra = Stage 019, the $54/5 = 2 \cdot 27/5$ provenance partition + the CALIBRATED label = Stage 020, and the $\hat{\mu}_0$-free dimensional closure = Stage 021)

Claim status. Exact Within Closure

Purpose. Earn the exterior-wave radiative signature the quadrupole normalization rides. Show the outgoing $\ell = 2$ DtN response $\hat{Y}_2^{\text{out}}(z) = -3/\Lambda_2^{\text{out}}(z)$, $\Lambda_2^{\text{out}} = z\, h_2^{(1)\prime}/h_2^{(1)}$, $z = a\omega/c_s$, yields the DERIVED rational fingerprint $u_2 = a^2/9c_s^2$, $u_4 = 4a^4/81c_s^4$, $v_5 = a^5/27c_s^5$ (the 27 computed from $v_5^z = 1/27$), and the sign $\chi_Q = +1$ outgoing / -1 incoming computed from $j_2 \pm i\, y_2$. Land the joint verdict as a partial component.

Status. Exact Within Closure — the fingerprint is exact symbolic algebra (the outgoing $\ell = 2$ spherical-Hankel logarithmic derivative and its small-frequency series are exact expansions; expect_zero/expect_bool/expect_fail residuals, no scipy/numpy/floats/tolerances). The verdict is the joint QUAD_CALIBRATED; this stage carries its *outgoing-fingerprint + χ_Q-sign* component (1/4, the EARNED-FIRST slice) and lands the joint as **partial** (018 earned; 019/020/021 pending) — not complete. It adds **zero** new counted knobs (an EARNED/structural fingerprint slice, like Stages 016/011/012/014); c_s is a units carrier only (the earned rationals and χ_Q are c_s-free), cited from Stage-005 R1 as provenance, not a consumed value.

The outgoing DtN log-derivative. The exterior $\ell = 2$ outgoing solution is the spherical Hankel function $h_2^{(1)}(z) = j_2(z) + i\, y_2(z)$, $z = a\omega/c_s$, with $j_2 = (3/z^3 - 1/z)\sin z - (3/z^2)\cos z$, $y_2 = (1/z - 3/z^3)\cos z - (3/z^2)\sin z$ built from explicit rational$\cdot\sin/\cos$ (no stored artifact). The DtN eigenvalue and normalized response are

$$\Lambda_2^{\text{out}}(z) = z\, \frac{h_2^{(1)\prime}(z)}{h_2^{(1)}(z)}, \qquad \hat{Y}_2^{\text{out}}(z) = -\frac{3}{\Lambda_2^{\text{out}}(z)},$$

where $-3 = -(\ell + 1)$ for $\ell = 2$ is the outgoing static DtN slot (self-derived). This is self-contained exterior-wave algebra: it does *not* consume the interior port kernel (that enters Stage-019's prefactor).

The fingerprint (the EARNED headline). Expanding $\hat{Y}_2^{\text{out}}$ about $z = 0$,

$$\hat{Y}_2^{\text{out}}(z) = 1 + \tfrac{1}{9}z^2 + \tfrac{4}{81}z^4 + i\,\tfrac{1}{27}z^5 + O(z^6),$$

so $u_2^z = 1/9$, $u_4^z = 4/81$, $v_5^z = 1/27$ (the odd z^1, z^3 vanish; the z^5 term is purely imaginary — the radiating part, coeff$(z, 5)/i$). Restoring units through $z = a\omega/c_s$: $u_2 = a^2/9c_s^2$, $u_4 = 4a^4/81c_s^4$, $v_5 = a^5/27c_s^5$. The 27 is EARNED as $1/v_5^z$ — the radiative-slot denominator, series-derived, *not* a typed literal (this is the `derived_in_gate` 27 that Stage-020's $54/5 = 2 \cdot 27/5$ partition rides). The coefficients are read off the *actual* symbolic series of $-3/(z\,h_2^{(1)\prime}/h_2^{(1)})$ with $h_2^{(1)}/\Lambda_2^{\text{out}}/\hat{Y}_2^{\text{out}}$ emitted — not hardcoded and compared. Perturbing the typed target (u_2^z expected $1/9 \to 1/8$) fires the assert, and mutating the *derivation* ($h \cdot (1 + z^4) \to u_4^z = 112/81$, leaving u_2^z, v_5^z) fires exactly that coefficient's `expect_fail` — a genuine per-coefficient firewall.

The χ_Q sign. Against the canonical radiative slot $a^5/(27c_s^5)$, $\chi_Q = v_5/(a^5/27c_s^5) = +1$ outgoing ($h_2^{(1)}$) and -1 incoming ($h_2^{(2)} = j_2 - i\,y_2$); the sign *emerges* from $j_2 \pm i\,y_2$ (the z^5 coefficient flips), never a typed $+1$. The standing branch j_2 gives $\Lambda_{\text{stand}}(0) = +2$ ($= +\ell$, not the outgoing -3), $\hat{Y}_{\text{stand}}(0) = -3/2 \neq 1$, and no radiating term — proving the $+1/27$ rational is outgoing-BC-selected, not universal.

Passivity and the units-restored dim leg. The radiating $v_5\omega^5$ arises from the genuine outgoing BC: the imposed-dissipation probe replaces it with a phenomenological sink and is rejected (`FAIL_NOT_OUTGOING`) by source-label provenance, with a genuine *dynamic* self-ablation re-run (with $\neq$ without, not a constant — the pathA_33 v1 trip-up avoided); the static$\leftrightarrow$dynamic consistency is the fingerprint's own static slot $\hat{Y}_2^{\text{out}}(\omega \to 0) = 1$ (*not* the prefactor $P_0 = N_0/D_0$, which is 019's). The units-restored dim leg runs on the real physical coefficients ($[u_2] = T^2$, $[u_4] = T^4$, $[v_5] = T^5$, with $[a] = L$, $[c_s] = LT^{-1}$); corrupting an a/c_s power fires `FAIL_DIMENSIONAL`. This is *not* the $\hat{\mu}_0$-free $[P_0^{\text{phys}}] = 1$ gate (that is 021's) — the 018 slice contains no $\hat{\mu}_0/N_0/D_0/P_0/$`build_dimensions` construct, keeping the back-solved-$\hat{\mu}_0$ v1 rig out.

Honest scope. Earned exterior-wave signature / calibrated magnitude: 018 derives the fingerprint shape $(1/9, 4/81, 1/27)$ and the $\chi_Q = +1$ sign, but the 54/5 normalization magnitude and G are calibrated ($G =$`GENUINE_BLOCKED`), landed at Stage 020; the prefactor algebra is Stage 019; the $\hat{\mu}_0$-free dimensional closure is Stage 021. The actual-branch a-scaling and the numerical port scalars N_n/D_n are deferred Gate 4/5/6 work. The χ_Q reconciliation (pathA_33 +1 vs pathA_22b ≈ 0.712 — same name, different computations) is a tracked Part-VII item, not merged here.

Consumed closure and register. *Provenance only, no dual-site:* 018 is self-contained exterior-wave algebra, so — unlike Stage-017's genuine cross-stage dual-site — there is no checkable relation to guard. 017's $\ell = 2$ port kernel (the interior wall mode; literal $N_n/D_n = 019$'s), 009/010's bulk Helmholtz mode, and c_s (Stage-005 R1) are cited as narrative provenance with no guard on any object the fingerprint never uses; c_s is an inert units carrier, and c_S (the frozen-wall speed) is not a live symbol here. Register: **zero** new counted knobs; structural edge **R37** (the outgoing-DtN $\ell = 2$ Hankel-fingerprint provenance — $u_2 = a^2/9c_s^2$, $u_4 = 4a^4/81c_s^4$, $v_5 = a^5/27c_s^5$ with the 27 computed from v_5^z, and $\chi_Q = +1$ outgoing / -1 incoming; discharges nothing). Part-II counted CALIB set unchanged at $\{\mu_\eta, T_w, \beta\}(013) + \{V_{p0}/\ell_c\}(015) + \{T_\Omega, \beta_2\}(017) = 6$.

Verification. Dual-engine, independent symbolic routes, zero file I/O — the source pair's scratch-YAML handoff is severed; the `.wl` KEEPS its native $j_2/y_2/$`Series/ Coefficient` route (its own `dimOf`), not a mirror, agreeing at the transcript level ($\Lambda_2^{\rm out}/\hat{Y}_2^{\rm out}$, the derived $u_2^z = 1/9/u_4^z = 4/81/v_5^z = 1/27$, $\chi_Q = +1$/incoming -1, the standing static slot, the PARTIAL landing), with an arity + unevaluated-leakage self-check. SymPy 59 PASS / Mathematica 65 PASS, exit 0, CWD-independent. The directive design-review used the Codex→Grok→Codex bookend (both engines compute-verified the series $1 + z^2/9 + 4z^4/81 + i\,z^5/27$, the incoming $\chi_Q = -1$, and the standing $\Lambda_{\rm stand}(0) = +2$; three non-blocking clarity nits folded — the P_0 name collision, the passivity scope, the 3a/3b self-ablation scoping). Full tri-review on fresh agents: `FIDELITY_CLEAN` (an independent read hand-re-derived j_2/y_2, $\Lambda_2^{\rm out}$, $\hat{Y}_2^{\rm out}$, the full series, $\chi_Q = \pm 1$, the standing slot) and `ADVERSARIAL_CLEAN` (per-tooth ablation: 30/32 targeted mutations fired at their own assert, the 2 non-fires harness artifacts re-covered by corrected runs; no vacuous tooth) — no remediation required. Symbolic per-tooth ablation, mutations on copies.

B.14 Stage 019: The Squared-Denominator Prefactor Algebra $P(\omega) = D_0N/D^{\rm cons\,2}$ (Check II-G4b)

Part / anchor. Part II – Gravity (the frozen-throat $\ell = 2$ radiative sector; the SECOND leg of a four-way split of pathA_33 — the squared-denominator prefactor-algebra component of the joint `QUAD_CALIBRATED`, with the DtN Hankel fingerprint = Stage 018 (done), the $54/5 = 2 \cdot 27/5$ provenance partition + the CALIBRATED label = Stage 020, and the $\hat{\mu}_0$-free dimensional closure = Stage 021)

Claim status. Exact Within Closure

Purpose. Earn the prefactor structure the quadrupole normalization rides. Show the squared-denominator prefactor $P(\omega) = D_0\,N(\omega)/D^{\rm cons}(\omega)^2$, $N = N_0 + N_2\omega^2 + N_4\omega^4$, $D^{\rm cons} = D_0 + D_2\omega^2 + D_4\omega^4$, Taylor-expands about $\omega = 0$ to $P_0 = N_0/D_0$, $P_2 = (D_0N_2 - 2D_2N_0)/D_0^2$, $P_4 = (D_0^2N_4 - 2D_0(D_2N_2 + D_4N_0) + 3D_2^2N_0)/D_0^3$ — with the $-2D_2N_0$ factor-of-2 the signature of the squared denominator (plain N/D gives only $-D_2N_0$, provably wrong). Land the joint verdict as a partial component.

Status. Exact Within Closure — the prefactor algebra is exact symbolic: the coefficients are exact rational functions of the abstract port scalars, read off a symbolic `series(...)` of $D_0N/D^{\rm cons\,2}$ (`expect_zero`/ `expect_bool`/`expect_fail` residuals, no `scipy`/`numpy`/ floats; the light integer sample substitution is a transcript-level cross-check only). The verdict is the joint `QUAD_CALIBRATED`; this stage carries its *squared-denominator prefactor-algebra* component (2/4) and lands the joint as **partial** (018 done, 019 earned here; 020/021 pending) — not complete. It adds **zero** new counted knobs (an EARNED/structural prefactor-algebra slice, like Stages 018/016/011/012/014); the abstract D_0, D_2, D_4 are Stage-017's exported D-lanes and the N_0, N_2, N_4 are the deferred port N-moments, cited as provenance, not consumed values.

The squared-denominator prefactor object. The port numerator and the conservative ("consumption") denominator are

$$N(\omega) = N_0 + N_2\omega^2 + N_4\omega^4, \qquad D^{\rm cons}(\omega) = D_0 + D_2\omega^2 + D_4\omega^4,$$

and the prefactor is the *squared*-denominator object

$$P(\omega) = \frac{D_0\,N(\omega)}{D^{\rm cons}(\omega)^2} \qquad \text{(not the plain } N/D\text{)}.$$

The D_n are Stage-017's exported $\ell = 2$ port-kernel D-lanes and the N_n are the concrete port N-moments of `build_port_moments`; here they are carried as abstract, port-agnostic symbols — the algebra below holds for *any* nonzero $D_0 \dots N_4$, so no port value is consumed (the numerical (D_n, N_n) are deferred Gate-6 branch data).

The prefactor algebra (the EARNED headline). Expanding $P(\omega)$ about $\omega = 0$ and reading the coefficients off the actual series,

$$P_0 = \frac{N_0}{D_0}, \quad P_2 = \frac{D_0 N_2 - 2D_2 N_0}{D_0^2}, \quad P_4 = \frac{D_0^2 N_4 - 2D_0(D_2 N_2 + D_4 N_0) + 3D_2^2 N_0}{D_0^3}.$$

The $-2D_2N_0$ term is the *signature* of the squared denominator: it arises only from the $-2x$ term of $(1+x)^{-2}$ with $x = (D_2/D_0)\omega^2 + (D_4/D_0)\omega^4$. The coefficients are coeff(ω, n) off the *actual* symbolic series of $D_0 N/D^{\mathrm{cons}\,2}$ — with the object and its series emitted — and are checked against an *independent* typed reference (a genuine derive-then-check that could disagree), not hardcoded and compared. Mutating the object (e.g. to plain N/D, or perturbing a D_n/N_n power) changes the extracted coefficient and fires exactly that coefficient's `expect_fail`; a swapped-in *correct* object flips the discriminator to true — a genuine positive control.

The N/D self-check. The plain (non-squared) object exposes the factor of two directly: N/D^{cons} series-expands to $P_2^{\mathrm{plain}} = (D_0N_2 - D_2N_0)/D_0^2$, so the computed gap

$$P_2^{\mathrm{plain}} - P_2 = \frac{D_2 N_0}{D_0^2} \neq 0,$$

i.e. $-D_2N_0$ versus the correct $-2D_2N_0$. The gap is *computed* (not asserted); the counterfactual probe `3g_wrong_prefactor_object` replaces $D_0N/D^{\mathrm{cons}\,2}$ with plain N/D, so `plain_equals_correct_P2` = `False` and the probe fires `FAIL_PREFACTOR_ALGEBRA`, while the correct object does *not* fire. Its self-ablation is a genuine *dynamic* re-run of an 019-local verdict gate (the prefactor match $\wedge$ the N/D self-check; with $\neq$ without, never a constant, and never the joint `base_verdict`) — the pathA_33 v1 trip-up avoided.

Provenance-only consumption and units-free scope. This is where Stage-018's deferred literal N_n/D_n port-kernel consumption lands — at the *provenance* level, not a value-consumption dual-site. Because the algebra is port-agnostic and `build_port_moments` is emitted-but-never-checked (deferred Gate-6 branch data), there is no checkable cross-stage relation to guard: Stage-017's D-lanes, the concrete N-moments, and Stage-018's fingerprint context are cited as narrative provenance with no guard/dual-site on any of them (the same provenance-only landing as Stage 018; a guard on the unused/deferred moments would be a vacuous tooth). The stage is *units-free*: it introduces no $c_s/a/G/\hat{\mu}_0$ and has no dimensional leg (the $\hat{\mu}_0$-free $[P_0^{\mathrm{phys}}] = 1$ closure is Stage-021's). A runtime free-symbol guard asserts the earned expressions' symbols are exactly $\{\omega, D_0, D_2, D_4, N_0, N_2, N_4\}$.

Honest scope. Earned prefactor structure / calibrated magnitude: 019 derives the port-agnostic algebra (P_0, P_2, P_4 and the N/D factor-of-two self-check), but the $54/5 = 2{\cdot}27/5$ normalization magnitude and G are calibrated ($G =$`GENUINE_BLOCKED`), landed at Stage 020, and the numerical port scalars (D_n, N_n) from the solved nonlinear branch remain deferred Gate-6 work. The DtN fingerprint is Stage 018; the $\hat{\mu}_0$-free dimensional closure is Stage 021.

Consumed closure and register. *Provenance only, no dual-site:* the abstract D_0, D_2, D_4 are Stage-017's exported D-lanes and the N_0, N_2, N_4 are `build_port_moments`' concrete port N-moments (deferred Gate-6, asserted against nothing); Stage-018's fingerprint context is a narrative cite. Register: **zero** new counted knobs; structural edge **R38** (the squared-denominator prefactor-algebra provenance — $P_0 = N_0/D_0$, $P_2 = (D_0N_2 - 2D_2N_0)/D_0^2$, $P_4 = (D_0^2N_4 - 2D_0(D_2N_2 + D_4N_0) + 3D_2^2N_0)/D_0^3$ with the $-2D_2N_0$ factor-of-2 the squared-denominator signature; discharges nothing). Part-II counted CALIB set unchanged at $\{\mu_\eta, T_w, \beta\}$(013) + $\{V_{p0}/\ell_c\}$(015) + $\{T_\Omega, \beta_2\}$(017) = 6.

Verification. Dual-engine, independent symbolic routes, zero file I/O — the source pair's scratch-YAML handoff is severed; the `.wl` KEEPS its native `serW/Coefficient/ FullSimplify` route on its own $P(\omega)$/plain objects (typing its own expected $P_0/P_2/P_4$), not a mirror, agreeing at the transcript level (the series-extracted coefficients, the plain $P_2 = (D_0N_2 - D_2N_0)/D_0^2$, and `plainEqualsCorrectP2=False`), with an arity + unevaluated-leakage self-check. SymPy 18 PASS / Mathematica 24 PASS, exit 0, CWD-independent. The directive design-review used the Codex→Grok→Codex bookend (both engines compute-verified the series $N_0/D_0 + (D_0N_2 - 2D_2N_0)\omega^2/D_0^2 + \cdots$, the plain N/D gap D_2N_0/D_0^2, and the emitted-but-never-checked reading of `build_port_moments`; four nits folded — a line-ref fix, the D-lane/N-moment split, the sibling-symbol wording, and the D0..N4-only sample substitutions). Full tri-review on fresh agents: `FIDELITY_CLEAN` (an independent read hand-re-derived $P_0/P_2/P_4$ and the N/D gap D_2N_0/D_0^2) and `ADVERSARIAL_ISSUES` (per-tooth ablation: the central firewall, N/D self-check, and dynamic 019-local ablation all genuine; two subsumed/mirror teeth flagged — one made-genuine as a swapped-in correct-object positive control, one honestly de-counted — plus a de-obfuscation of narrative tokens and a legibility fix, all re-verified `REVERIFY_CLEAN` by the coupling meta-test). Symbolic per-tooth ablation, mutations on copies.

B.15 Stage 020: The $54/5 = 2 \cdot 27/5$ Provenance Partition and the CALIBRATED Verdict Label (Check II-G4c)

Part / anchor. Part II – Gravity (the frozen-throat $\ell = 2$ radiative sector; the THIRD leg of a four-way split of pathA_33 — the $54/5 = 2 \cdot 27/5$ provenance-partition + the CALIBRATED-verdict-label component of the joint `QUAD_CALIBRATED`, with the DtN Hankel fingerprint = Stage 018 (done), the squared-denominator prefactor algebra = Stage 019 (done), and the $\hat{\mu}_0$-free dimensional closure = Stage 021 (the completing leg))

Claim status. Exact Within Closure

Purpose. Land the CALIBRATED headline. Show the assembled $\ell = 2$ quadrupole magnitude $54Gc_s^5/(5a^5c^5)$ decomposes as $54/5 = 2 \cdot 27/5$, with the 27 earned (`derived_in_gate`, Stage-018's fingerprint $1/v_5^z$) and the $2/5 + G$ calibrated (`external_bridge_input`, the GR Burke–Thorne $2G/5c^5$, G =`GENUINE_BLOCKED`); a four-way provenance partition classifies the assembled 54/5 as `external_bridge_input` (external dominates), so the source-faithful verdict lands `QUAD_CALIBRATED` (not `QUAD_PASS`) — driven by provenance, not by $G \to \lambda G$ invariance.

Status. Exact Within Closure — units-bearing but exact symbolic/rational for the earned content: the a^{-5} scaling (exact a-power via Stage-018's frozen slot), the Γ_5/χ_Q equivalence (`compact`$(\ldots) = 0$ residuals), the $54/5 = 2 \cdot 27/5$ SymPy rational identity (bound to `target_rhs/v5_slot`), and the four-way provenance partition (exact class comparison via tag dominance); `expect_zero/expect_bool/`

`expect_fail` residuals, no floats/tolerances in the earned content. The verdict is the joint `QUAD_CALIBRATED`; this stage carries its *provenance-partition + CALIBRATED-label* component (3/4) and lands the joint as **partial** (018/019 done, 020 earned here; 021 the completing leg) — and it is the leg where the CALIBRATED *classification* is determined. It adds **zero** new counted knobs (a classification slice, like Stages 018/019).

The assembled magnitude and the Γ_5/χ_Q equivalence. The $\ell = 2$ quadrupole normalization the gravity sector must match is the Burke–Thorne form

$$\hat{m}_0^2 P_0 = \frac{54\,G\,c_s^5}{5\,a^5\,c^5} \quad\Longleftrightarrow\quad \gamma_{\text{quad}}^{\text{eff}} = \frac{2G}{5c^5}.$$

The equivalence is a self-contained algebraic bridge. With $P_0^{\text{phys}} = (c_s/a)^2(N_0/D_0)$ (Stage-019's prefactor) and Stage-018's outgoing $\chi_Q = +1$, the forward residual

$$\text{forward} = \frac{54Gc_s^5}{5a^5c^5}\cdot\chi_Q\cdot\frac{a^5}{27c_s^5} - \frac{2G}{5c^5} = \frac{2G(\chi_Q - 1)}{5c^5}$$

vanishes *iff* $\chi_Q = 1$ and $54/27 = 2$ (the 27 is Stage-018's radiative slot $1/v_5^z$). So the outgoing sign Stage 018 derived is exactly the sign a positive quadrupole demands; a wrong $2/5 \to 3/5$ breaks the bridge (`3e`→`FAIL_EQUIVALENCE`).

The $54/5 = 2\cdot 27/5$ decomposition (the headline — a bound SymPy identity). The assembled coefficient factors as

$$\frac{54}{5} = 2\cdot\frac{27}{5},$$

with the 27 =`derived_in_gate` (Stage-018's fingerprint $1/v_5^z$) and the $2/5$+G = `external_bridge_input` (the GR Burke–Thorne $2G/5c^5$). This is a SymPy-verified rational identity *bound to the stage's own assembled expressions* — not the source's typed string `"54/5 = 2*27/5"`, and not a bare-literal $\text{Rational}(54, 5) - 2\cdot\text{Integer}(27)/5$ arithmetic tautology. The magnitudes are extracted: $\text{mag} = \texttt{compact}(\texttt{target\_rhs}/(Gc_s^5/(a^5c^5))) = 54/5$ and $27_{\text{slot}} = \texttt{compact}(a^5/(c_s^5\cdot\texttt{v5\_slot})) = 27$ (from Stage-018's frozen slot $\texttt{v5\_slot} = a^5/27c_s^5$), then $\texttt{compact}(\text{mag} - 2\cdot 27_{\text{slot}}/5) = 0$. Mutating `target_rhs` ($54 \to 55$) or the identity ($2\cdot 26/5$) fires the assert.

The a^{-5} target scaling. The assembled magnitude carries a^{-5} as a *derived* consequence of a-cancellation, not a typed target power: the Burke–Thorne coefficient $2G/5c^5$ is a-free, and Stage-018's radiative slot $\texttt{v5\_slot} = a^5/27c_s^5$ supplies the a^5, so derived_power $= a\text{-power}(\gamma_{\text{target}}) - a\text{-power}(\texttt{v5\_slot}) = 0 - 5 = -5$, and the reconstructed $\gamma_{\text{target}}/(\chi\cdot\texttt{v5\_slot}) = \texttt{target\_rhs}$ (a-power -5). The a^5 is typed *once*; the wrong-scaling probe ($a^5 \to a^4$) fires at the assembled residual (`3c`→`FAIL_SCALING`). The actual branch a-scaling (from N_0/D_0) is deferred Gate-6.

The four-way provenance partition and the PROVENANCE-driven verdict. Every quantity is classified from its provenance *tags* by a tag-dominance rule `deferred` > `external` > `derived` > `convention`: `fingerprint_27` → `derived_in_gate`, `PN_2_over_5` → `external_bridge_input`, G → `external_bridge_input`, and the assembled 54/5 (mixing the earned 27 tag with the external 2/5 tag) → `external_bridge_input` because *external dominates.* The 020-local verdict then reads the classes with the source-faithful rule

$$\texttt{if}\ g\_\text{class} = \texttt{derived}\ \wedge\ \text{mag}\_\text{class} = \texttt{derived}:\ \texttt{QUAD\_PASS};\quad \texttt{else}:\ \texttt{QUAD\_CALIBRATED},$$

so with G and the assembled 54/5 both external, the verdict lands `QUAD_CALIBRATED`: the PDE delivers the form/branch (the 27), not Newton's G ($G =$`GENUINE_BLOCKED`). The classifier's dominance rule is proven by a truth-table + key-class + tag-mutation tooth (the source `partition_ok` alone is trivially true), and the verdict is proven to *read* provenance by a `QUAD_PASS` positive control (a coherent all-derived partition) *and* a required mixed control (one external, one derived → `QUAD_CALIBRATED`, the case an inverted rule would mislabel).

The verdict is provenance-driven, not $G \to \lambda G$-invariance-driven. A separate, non-verdict-driving diagnostic makes the point sharp: 54/5 is a pure number, hence G-invariant ($G \to \lambda_G G$ leaves it unchanged), yet it is `external_bridge_input` by provenance. An invariance-only test would *mislabel* the calibrated 54/5 as earned (`invariance_only_trap_catches_54_over_5 = True`); the classification is set by the provenance tags, and `3f_partition_mislabel` fires `FAIL_PROVENANCE_PARTITION` in both directions (G external→derived and 27 derived→external).

Units-bearing scope (no dimensional-homogeneity gate). Unlike Stage 019 (units-free), Stage 020 is units-bearing — $\{c_s, a, c, G\}$ live in the scaling/equivalence algebra — but it does algebra + provenance, *not* the dimensional homogeneity gate $[P_0^{\mathrm{phys}}] = 1$ (that is Stage-021's). The cut is by operation, enforced by a runtime free-symbol allowlist (no $\hat{\mu}_0$ symbol; the source's legacy `gamma_quad_eff` $\hat{\mu}_0$ string is dropped), and a structural no-dimensional-dependency cut (no `dim_of` construct; a structural assertion that the 020-local verdict's signature/bytecode carries no dim parameter/dependency — the `.wl` checks its definition arity/text).

Honest scope. Earned classification / calibrated magnitude: 020 earns the a^{-5} scaling, the Γ_5/χ_Q equivalence, the $54/5 = 2{\cdot}27/5$ decomposition, and the four-way provenance partition; the 54/5 magnitude and G are calibrated ($G =$`GENUINE_BLOCKED`). The c^5 is the GR/λ_γ units bridge ($P_0 \propto c_s^5/c^5 = 1/\lambda_\gamma^5$) — c is the light cone c_γ in its GR-bridge role, a benchmark carrier, not EM propagation and not a fresh knob. The actual branch a-scaling and the numerical (D_n, N_n) port scalars remain deferred Gate-6. The dimensional closure $[P_0^{\mathrm{phys}}] = 1$ is Stage-021's.

Consumed closure and register. *Provenance, no dual-site:* Stage-018's $\chi_Q = +1$ and 27 ($= 1/v_5^z$) genuinely enter 020's self-contained equivalence bridge (a corrupt cited value breaks `forward` — the equivalence tooth itself, not a manufactured cross-stage guard); Stage-019's $P_0 = N_0/D_0$ appears only in the Γ_5 definition, absent from the equivalence `ok` residual (a $\{N_0, D_0\} \subseteq \Gamma_5 \wedge \notin$residual firewall guards this); Stage-017's D-lanes are the provenance of the abstract port scalars. Register: **zero** new counted knobs; structural edge **R39** (the $54/5 = 2 \cdot 27/5$ provenance-partition + the CALIBRATED-verdict provenance; discharges nothing — a classification landing, not a reduction). $G =$`GENUINE_BLOCKED`/external (already registered); $c =$ the c_γ GR-units-bridge (cited benchmark); the $2/5 =$ GR bridge; the $27 =$ Stage-018's derived fingerprint. Part-II counted CALIB set unchanged at $\{\mu_\eta, T_w, \beta\}(013) + \{V_{p0}/\ell_c\}(015) + \{T_\Omega, \beta_2\}(017) = 6$.

Verification. Dual-engine, independent symbolic routes, zero file I/O — the source pair's scratch-YAML handoff is severed. Unlike Stages 018/019 (which kept the `.wl`'s already-native blocks), the source `.wl` had *zero* 020 content, so the 020 `.wl` is a genuinely fresh independent authoring: native `Exponent/Together/Cancel/FullSimplify` for the scaling + the Γ_5 bridge, native `Rational/Simplify` for the identity bound to its own `targetRHS/ v5Slot`, a native `Association`-based provenance classifier using a rank-based `MaximalBy` (a genuinely different mechanism from the `.py` if-chain, same dominance), and `expr/.` $G \to \lambda_G G$ for the g-invariance trap — no `Series`, no `.py` mirroring — agreeing at the transcript level (the a^{-5}, `forward` $= 0$, the bound $54/5 = 2 \cdot 27/5$, the provenance classes, the CALIBRATED verdict with the `QUAD_PASS` and mixed controls, both `3f` directions), with an arity + unevaluated-leakage self-check. SymPy 74 PASS / Mathematica 82 PASS, exit 0, CWD-independent. The directive design-review used the Codex→Grok→Codex bookend: Codex folded four blocking genuineness gaps; Grok's compute-verify then caught the CALIBRATED verdict-rule *inversion* (the source default is `else`→`QUAD_CALIBRATED`; the directive had inverted it to `PASS` unless both external — a shippable rig, fixed by a required mixed control) plus a bare-literal-identity nit; a Codex confirm then swept two consistency gaps, all folded to `DIRECTIVE_CLEAN`. Full tri-review on fresh agents: `FIDELITY_CLEAN` (an independent read hand-re-derived the Γ_5 bridge, the bound identity, the classify-provenance dominance, and the g-invariance trap) and `ADVERSARIAL_ISSUES` (per-tooth ablation: the four key genuineness risks — the bound identity, the mixed control catching the inverted rule, the proven classifier, the genuinely-authored `.wl` — all clean; three low-severity vacuous/subsumed teeth flagged, all remediated make-genuine with no de-counts — the structural dimensional cut, the $\{N_0, D_0\} \subseteq \Gamma_5$ before/after firewall, the independently-classified baseline-external side — re-verified `REVERIFY_CLEAN` by the coupling meta-test). Symbolic per-tooth ablation, mutations on copies.

B.16 Stage 021: The $\hat{\mu}_0$-free $[P_0^{\mathrm{phys}}] = 1$ Dimensional Closure (Check II-G4d)

Part / anchor. Part II – Gravity (the frozen-throat $\ell = 2$ radiative sector; the FOURTH, completing leg of a four-way split of pathA_33 — the $\hat{\mu}_0$-free $[P_0^{\mathrm{phys}}] = 1$ dimensional-closure component of the joint `QUAD_CALIBRATED`, with the DtN Hankel fingerprint = Stage 018 (done), the squared-denominator prefactor algebra = Stage 019 (done), and the $54/5 = 2 \cdot 27/5$ provenance partition + the CALIBRATED verdict label = Stage 020 (done))

Claim status. Exact Within Closure

Purpose. Complete the joint `QUAD_CALIBRATED`. Certify the assembled $\ell = 2$ quadrupole prefactor $P_0^{\mathrm{phys}} = (c_s/a)^2(N_0/D_0)$ is dimensionless from the sourced port dimensions $[N_0] = L^{-1}M$, $[D_0] = L^{-1}T^{-2}M$ — a $\hat{\mu}_0$-FREE gate that catches the natural-units trap (the handoff's $P_0 = N_0/D_0$ silently drops $(c_s/a)^2$) and reads the sourced port dimensions (corrupt-$[N_0]$ fires `FAIL_DIMENSIONAL`) rather than a back-solved free carrier $\hat{\mu}_0$ (corrupt-$[G]$ does not fire).

Status. Exact Within Closure — exact symbolic (L, M, T)-triple dimensional-vector algebra, float-free: the $\hat{\mu}_0$-free gate (`dim_of`(P_0^{phys}) = `ZERO_DIM`), the natural-units-trap catch (dropping the frequency normalization leaves $T^2 \neq 1$), the corrupt-dim scope truth-table, and the anti-v1 back-solve-tautology demotion; `expect_zero`/ `expect_bool`/`expect_fail` residuals, no floats/tolerances. The verdict is the joint `QUAD_CALIBRATED`; this stage carries its *dimensional-closure* component (4/4) and lands the joint **complete** (018/019/020 done, 021 earned here). It adds **zero** new counted knobs (a dimensional-closure slice, like Stages 018/019/020). $\hat{\mu}_0$ is a free-carrier non-verdict diagnostic, not a counted knob.

The $\hat{\mu}_0$-free $[P_0^{\mathrm{phys}}] = 1$ gate. With the sourced port dimensions $[N_0] = L^{-1}M$ (the density/continuity-port numerator) and $[D_0] = L^{-1}T^{-2}M$ (the carried reduced static conservative denominator $D_0 = K - B_0 - Z_0$), and the frequency normalization $(c_s/a)^2$,

$$[P_0^{\mathrm{raw}}] = [N_0/D_0] = T^2, \qquad [(c_s/a)^2] = T^{-2}, \qquad [P_0^{\mathrm{phys}}] = [(c_s/a)^2 N_0/D_0] = 1,$$

so `dimensional_ok = True`. The verdict-bearing check is `dimensional_ok` $=$ (`dim_of`(P_0^{phys}) $=$ `ZERO_DIM`) — a computed (L, M, T)-triple equality reading the sourced dims, not a typed `"1"`. Crucially $\hat{\mu}_0$ does *not* enter the verdict: it is absent from `raw_symbol_dims` (the gate's dimension map), and the verdict's computed read-set excludes $\hat{\mu}_0/\mu_{\text{dim}}$/`homogeneity_pass`.

The natural-units-trap catch (the dimensional milestone). The handoff's $P_0 = N_0/D_0$ silently drops the $(c_s/a)^2$ frequency-normalization factor (a natural-units artifact where c_s/a was set to 1). The gate catches it: $[P_0^{\text{raw}}] = T^2 \neq 1$, so the `3d` drop-normalization probe fires `FAIL_DIMENSIONAL`. The frequency normalization is required for $[P_0^{\text{phys}}] = 1$.

The corrupt-dim scope truth-table. The $\hat{\mu}_0$-free gate fires precisely on the dims that enter P_0^{phys}:

corrupt $[N_0] \to 0$	$[P_0^{\text{raw}}] = (1, -1, 2)$,	$[P_0^{\text{phys}}] = (1, -1, 0) = L\,M^{-1} \neq 1$	$\to$ `FAIL_DIMENSIONAL` (`3d`$'$)
corrupt $[D_0] \to 0$		$[P_0^{\text{phys}}] = (-1, 1, -2) \neq 1$	$\to$ `FAIL_DIMENSIONAL`
corrupt $[c_s] \to 0$		$[P_0^{\text{phys}}] = (-2, 0, 2) \neq 1$	$\to$ `FAIL_DIMENSIONAL`
corrupt $[G] \to 0$		$[P_0^{\text{phys}}] = 1$ still ($G \notin$ free)	$\to$ `NO_FAIL` (scope)
correct		$[P_0^{\text{phys}}] = 1$	$\to$ `NO_FAIL`

The corrupt-$[N_0]$ result is $[P_0^{\text{phys}}] = (1, -1, 0)$ — the $(c_s/a)^2$ factor *remains* (it is not $-[D_0] = (1, -1, 2)$, which is $[P_0^{\text{raw}}]$). The gate is G-independent; the corrupt-$[G]$ row is a dependency-scope diagnostic, asserted via the runtime test $G \notin$ `free_symbols`(P_0^{phys}).

The $\hat{\mu}_0$ homogeneity diagnostic is NON-verdict — the anti-v1 discriminator. The full closure $\hat{m}_0^2 P_0 = 54Gc_s^5/(5a^5c^5)$ can be made dimensionally homogeneous by back-solving the free carrier $[\hat{\mu}_0] = L^{-1}T^{-1}M^{-1/2}$ from (`rhs_dim` $-$ `p0_dim`)/2, giving [lhs] = [rhs] = $L^{-2}T^{-2}M^{-1}$, `homogeneity_pass` $=$ `True`. This is a *tautology*: re-solving $[\hat{\mu}_0]$ after *any* corruption keeps `homogeneity_pass` $=$ `True` (the back-solve absorbs the change), so it can never fail — exactly the v1-rejected rig, correctly demoted here to a labelled diagnostic (`participates_in_verdict` $=$ `False`, computed from the verdict read-set). The decisive anti-v1 tooth is two-part: (a) a computed read-set assertion that the verdict excludes $\hat{\mu}_0/\mu_{\text{dim}}$/`homogeneity_pass` (and $\hat{\mu}_0 \notin$ `raw_symbol_dims`), and (b) a deliberately-wired mutant verdict $=$ the back-solved `homogeneity_pass`, re-run over all four corruptions and shown to stay `NO_FAIL` on all of them — proving it is vacuous, so the real verdict must be the $\hat{\mu}_0$-free $[P_0^{\text{phys}}] =$ `ZERO` gate (which fires on three of the four). The gate's $[N_0]/[D_0]/[c_s]$ FAIL rows, not the $[G]$ row, are what reject the all-`NO_FAIL` back-solved gate.

The $\hat{Y}$ dimensionless check. The outgoing physical expansion $\hat{Y} = 1 + u_2\omega^2 + u_4\omega^4 + i\,v_5\omega^5$ with Stage-018's frozen fingerprint $u_2 = a^2/9c_s^2$, $u_4 = 4a^4/81c_s^4$, $v_5 = a^5/27c_s^5$ (a provenance-frozen local fixture, not the Stage-018 machinery) is dimensionless (each term $T^2 \cdot T^{-2} = 1$, etc.), so $[\hat{Y}] = 1$. The check is wrapped in a structured `try_dim_of/Catch` so a corrupted ω-power ($u_2\omega^3$) fires the named $\hat{Y}$-dimensionless assert (a caught mismatch), not an uncaught `DimError` that would bypass the tooth.

The completing landing. Stage 021 lands the joint `QUAD_CALIBRATED` **complete** (018 fingerprint $\wedge$ 019 prefactor $\wedge$ 020 provenance-partition + CALIBRATED $\wedge$ 021 dim closure). **Complete $\neq$ pass**: the joint token stays `QUAD_CALIBRATED` (calibrated, not `QUAD_PASS`) — Stage 020 already determined the CALIBRATED classification and $G =$`GENUINE_BLOCKED`; 021 supplies the dimensional closure that makes the assembled magnitude dimensionally sound, but does not flip the verdict. The PDE delivers the form/branch (the 27 fingerprint + the dimensionally-sound closure), not Newton's G.

Honest scope. Earned closure / calibrated magnitude: 021 earns the $\hat{\mu}_0$-free $[P_0^{\text{phys}}] = 1$ closure, the natural-units-trap catch, and the corrupt-$[N_0]/[G]$ discriminator (the gate reads the sourced port dims, not a back-solved $\hat{\mu}_0$); the 54/5 magnitude and G are calibrated ($G =$`GENUINE_BLOCKED`). 021 is the other half of the 020/021 operation-level cut — Stage 020 did algebra + provenance, Stage 021 does the $[\cdot]$ dimensional-homogeneity gate, the only pathA_33 leg with a dimensional gate. The sourced port dims $[N_0] = L^{-1}M$ (pathA_43 density-port numerator) and $[D_0] = L^{-1}T^{-2}M$ (the carried reduced static conservative denominator) are sourced dimensional inputs (Cluster C stages 024/027 + pathA_34 not yet built in build order); they genuinely enter the gate (so the corrupt-$[N_0]$ tooth is genuine), but 021 does not re-derive them. The actual branch a-scaling and the numerical (D_n, N_n) port scalars remain deferred Gate-6.

Consumed closure and register. *Provenance, no dual-site:* the sourced port dims (pathA_43 numerator + the carried conservative denominator), Stage-018's $u_2/u_4/v_5$ (a local frozen fixture for the $\hat{Y}$ check, not `build_fingerprint`), Stage-019's $P_0 = N_0/D_0$ (entering P_0^{raw}), and Stage-020's assembled `target_rhs` (the $\hat{\mu}_0$ diagnostic's rhs only). The sourced port dims genuinely enter the $\hat{\mu}_0$-free gate (the corrupt-$[N_0]$ probe is the genuine able-to-fail tooth). Register: **zero** new counted knobs; structural edge **R40** (the $\hat{\mu}_0$-free $[P_0^{\text{phys}}] = 1$ dimensional closure; discharges nothing — a dimensional-consistency closure, not a reduction; records that 021 completes the joint `QUAD_CALIBRATED`, which stays CALIBRATED not PASS). $\hat{\mu}_0$ is a free-carrier non-verdict diagnostic; the sourced port dims are dimensional provenance; c/G are already registered. Part-II counted CALIB set unchanged at $\{\mu_\eta, T_w, \beta\}$(013) + $\{V_{p0}/\ell_c\}$(015) + $\{T_\Omega, \beta_2\}$(017) = 6.

Verification. Dual-engine, independent symbolic routes, zero file I/O — the source pair's scratch-YAML handoff is severed. Like Stages 018/019 (and unlike Stage 020's fresh authoring), the source `.wl` has the 021 dimensional block, so the 021 `.wl` keeps its native, already-independent Wolfram dimensional-vector engine (a `Which`-based `dimOf` over `Times`/`Power`/`Plus`, native $\langle|\ldots|\rangle$ `rawDims`, `Series`-free dimension algebra) and severs only the YAML machinery; the Stage-018 fingerprint (j_2/y_2) and Stage-019 prefactor blocks are removed and the $\hat{Y}$ coefficients become local `u2Sourced/u4Sourced/v5Sourced` fixtures. It adds (the source `.wl` lacked both) the corrupt-$[G]$-does-not-fire scope control and the dynamic 021-local self-ablation; the `.wl`'s local pass-guard uses a split-out `yhatOk` and excludes `homogeneityPass` (a print-only diagnostic), with an arity + unevaluated-leakage self-check. Agreement is at the transcript level ($[P_0^{\text{raw}}] = T^2$, $[(c_s/a)^2] = T^{-2}$, $[P_0^{\text{phys}}] = 1$, `3d/3d′` `FAIL_DIMENSIONAL`, corrupt-$[G]$ `NO_FAIL`, `homogeneity_pass = True` diagnostic). SymPy 42 PASS / Mathematica 50 PASS, exit 0, CWD-independent. The directive design-review used the Codex→Grok→Codex bookend: Codex folded six blocking findings — most notably that the originally-proposed corrupt-$[G]$ anti-v1 discriminator was *backwards* (a back-solved $\hat{\mu}_0$ gate re-solves μ_{dim} after every mutation, so it fires on nothing, and it is the $\hat{\mu}_0$-free gate's $[N_0]/[D_0]/[c_s]$ FAIL rows that reject it) — then a Grok compute-verify pass independently confirmed the $\hat{\mu}_0$-free gate dims, the corrupt-dim truth-table, the back-solve-is-a-tautology crux, and the $\hat{Y}$/structured-catch, and a final Codex confirm closed it to `DIRECTIVE_CLEAN`. Full tri-review on fresh agents: `FIDELITY_CLEAN` (an independent read hand-re-derived the $\hat{\mu}_0$-free gate dims, the corrupt-dim truth-table, and the back-solve-tautology table, confirming no 018/019/020 leakage and a genuine independent `.wl`) and `ADVERSARIAL_ISSUES` (per-tooth ablation: the two key teeth — the anti-v1 read-set/wired-mutant and the corrupt-$[G]$ scope control — genuine; five low-severity stamped/subsumed teeth flagged, remediated two make-genuine (`rerun_gate_logic` derived from the actual re-run, `participates_in_verdict` derived from the computed read-set) and three de-count (the QUAD-landing tautology and two subsumed summaries retained as labeled prints) — re-verified `REVERIFY_CLEAN` by the coupling meta-test, tallies 45/53→42/50 from the honest de-counts). Symbolic per-tooth ablation, mutations on copies.

B.17 Stage 022: The Cross-ℓ $-(\ell+1)/\Lambda_\ell$ Fingerprints + the Gate-4 Non-Regression (Check II-G5a)

Part / anchor. Part II – Gravity (the frozen-throat cross-ℓ radiative sector; the EARNED-FIRST leg of a two-way split of pathA_34 — the cross-ℓ outgoing DtN $-(\ell{+}1)/\Lambda_\ell$ fingerprints ($\ell = 0, 1, 2$) + the Gate-4 quadrupole non-regression component (1/2) of the joint `FAIL_UNDERDETERMINED_NOT_PREDICTIVE`). The native nullspace underdetermination departure — dim-8 / return-nullity-2, the residuals vs pathA_29, the $Z_0^{\mathrm{ret}}/Z_1^{\mathrm{ret}}$ Gate-6 selector need — that *delivers* the FAIL is Stage 023 (II-G5b).

Claim status. Exact Within Closure

Purpose. Generalize Stage 018's $\ell = 2$-only DtN Hankel fingerprint across ℓ: certify that the outgoing DtN responses $\hat{Y}_\ell^{\mathrm{out}} = -(\ell+1)/\Lambda_\ell$ for $\ell = 0, 1, 2$ — computed from the spherical Hankel h_ℓ — series-expand to the radiative fingerprint $\{1,\ 1/2,\ 1/27\}$ at leading orders $\{\omega^1, \omega^3, \omega^5\}$, with the static-DtN slot $\Lambda_\ell(z \to 0) = -(\ell+1)$ derived from the Hankel logarithmic derivative and the radiative order verified by scanning; and that the $\ell = 2$ leg reproduces the completed pathA_33 quadrupole fingerprint $\{u_2, u_4, v_5\} = \{1/9, 4/81, 1/27\}$ (the Gate-4 non-regression — the cross-ℓ generalization does not regress the earned quadrupole).

Status. Exact Within Closure — exact symbolic, float-free, dimensionless z-space ($z = a\omega/c_s$): the cross-ℓ spherical-Hankel logarithmic derivatives and their small-frequency series are exact `sympy.series` / native `Series` expansions; `bool_zero`/`expect_bool`/`expect_fail` residuals, no floats/tolerances, no units-restoration leg. The verdict is the joint `FAIL_UNDERDETERMINED_NOT_PREDICTIVE`; this stage carries its *fingerprint + non-regression* component (1/2) and lands it as an **EARNED-FIRST PARTIAL** (022 earned; 023 pending and it is 023 that delivers the FAIL). It adds **zero** new counted knobs (an earned exterior-wave signature slice, like Stages 016/018). c_s is an R1-derived units carrier, not a counted knob; the $\ell = 0/1$ stiffnesses and the return admittances $Z_0^{\mathrm{ret}}/Z_1^{\mathrm{ret}}$ are Stage 023's, not counted here.

Two distinct verdicts, both printed. The script emits an *022-local* exit-0 gate and, separately, the joint physics label:

`LOCAL_AUDIT_VERDICT: CROSS_L_FINGERPRINT_OK` (fingerprints + non-regression; the exit-0 gate)
`JOINT_LANDING_LABEL (PARTIAL; FAIL not evaluated by 022): FAIL_UNDERDETERMINED_NOT_PREDICTIVE`

022's script passes (exit 0) — the earned content is correct and every tooth fires; the FAIL token is the joint gate's physics label carried as a 1/2 partial. 022 is the earned half of a gate that ultimately fails (cf. Stage 003 = earned transverse photons + the characterized `FAIL_CAUCHY` departure). The 022-local verdict's computed read-set is exactly {`cross_l_fingerprints`, `ell2_non_regression`} and provably excludes any nullspace / return / `base_verdict` object (none is built here) — so the default-verdict-is-PASS trip-up is avoided: 022 does not compute or print a FAIL from any rigged mechanism.

The cross-ℓ outgoing DtN fingerprints (the earned headline). For each ℓ the exterior outgoing solution is the spherical Hankel function of the first kind $h_\ell^{(1)} = j_\ell + i\,y_\ell$ ($z = a\omega/c_s$); the normalized radiative response is $\hat{Y}_\ell^{\rm out}(z) = -(\ell+1)/\Lambda_\ell(z)$ with $\Lambda_\ell(z) = z\, h_\ell^{(1)\prime}/h_\ell^{(1)}$. Series-expanding about $z=0$,

$$\begin{aligned}
\ell=0:&\quad \hat{Y}_0 = 1 + i\,z + \cdots && \Rightarrow\ \texttt{radiative_coeff} = 1 \text{ at order } \omega^1,\\
\ell=1:&\quad \hat{Y}_1 = 1 + \tfrac{1}{2}z^2 + i\,\tfrac{1}{2}z^3 + \cdots && \Rightarrow\ \texttt{radiative_coeff} = 1/2 \text{ at order } \omega^3,\\
\ell=2:&\quad \hat{Y}_2 = 1 + \tfrac{1}{9}z^2 + \tfrac{4}{81}z^4 + i\,\tfrac{1}{27}z^5 + \cdots && \Rightarrow\ \texttt{radiative_coeff} = 1/27 \text{ at order } \omega^5.
\end{aligned}$$

The leading radiative coefficient is $\{1, 1/2, 1/27\}$ at order $2\ell{+}1 = \{\omega^1, \omega^3, \omega^5\}$, extracted as the imaginary coefficient $\texttt{coeff}(z, 2\ell+1)/i$ and checked derive-vs-typed — the derived side is the actual series of $-(\ell+1)/(z\,h'/h)$, emitted, not hardcoded. The static-DtN slot $\Lambda_\ell(z\to 0) = -(\ell+1)$ is *derived* from the Hankel logarithmic derivative ($h_\ell \sim z^{-(\ell+1)}$ as $z \to 0$), read off $\texttt{lam\_series.coeff}(z,0)$ — not the hand-set numerator (the source's set-then-compare is de-rigged). The radiative order is verified by *scanning* the imaginary series for its first nonzero power and asserting all lower imaginary coefficients vanish (not the source's dodgeable "preselected $2\ell+1$ coefficient is nonzero": a $+i\,z$ corruption of the $\ell=2$ series leaves z^5 nonzero yet the true leading imaginary order becomes 1; the scan catches it). The static value $\hat{Y}_\ell(0) = 1$ is a de-counted printed diagnostic ($= -(\ell+1)/\Lambda_{\rm static}$, subsumed by the derived $\Lambda_{\rm static}$).

The incoming-sign-flip (the genuine sign tooth). The incoming branch $h_\ell^{(2)} = j_\ell - i\,y_\ell$ flips *only* the radiative (imaginary) sign — $\texttt{radiative\_coeff} \to \{-1, -1/2, -1/27\}$ with every lower real coefficient unchanged ($\texttt{bool\_zero}$(incoming+outgoing) $= 0$). This computed sign-flip is the genuine sign content. The $\chi_Q = \texttt{radiative\_coeff}/$(canonical slot) $= +1$ equality is a de-counted printed diagnostic — algebraically subsumed by the $\ell=2$ non-regression ($27v_5 - 1 \equiv 27(v_5 - 1/27)$), so once the $\ell=2$ slot passes $\chi_Q = 1$ is automatic; not an independent tooth.

The Gate-4 quadrupole non-regression. The $\ell=2$ leg ($-(\ell+1)/\Lambda_\ell|_{\ell=2} = -3/\Lambda_2$) reproduces Stage 018's earned quadrupole fingerprint $u_2 = 1/9,\ u_4 = 4/81,\ v_5 = 1/27$. This is a derive-vs-typed non-regression: 022 re-derives the $\ell=2$ tuple from its own Hankel and checks it against Stage 018's independently-earned typed literals (not a same-machinery reconstruction of the typed side — no subsumed $X \equiv X$). The cross-ℓ $-(\ell+1)$ normalization gives the right -3 at $\ell=2$, so the general machinery does not regress the quadrupole. Each of u_2, u_4, v_5 has its own coefficient-isolating mutant (the `3e` incoming mutation flips only v_5, so u_2/u_4 need their own derivation-copy mutants — the Stage 018 pattern); breaking the $\ell=2$ branch to incoming fires `FAIL_QUAD_REGRESSION`, whose self-ablation is a dynamic two-verdict re-run (with $\neq$ without) re-scoped to the 022-local verdict.

Honest scope. Earned cross-ℓ signature + non-regression / the FAIL is 023's: 022 derives the outgoing cross-ℓ fingerprints and the Gate-4 quadrupole non-regression — the report's earned fingerprint / raw-order / Gate-4 subset. The rest of the report's earned content (the residual form/sign/order, conditional on a positive bounded branch) plus the FAIL-delivering native nullspace underdetermination (dim-8 / return-nullity-2, $\epsilon_{\rm eff}$ left free at the linear Gate-5 level) and the deferred scalar/dipole magnitude/nonzero prediction belong to Stage 023. The joint FAIL is deferred to 023, not evaluated here (the `JOINT_LANDING_LABEL` is a printed string; 022 builds no nullspace and does not back-solve $\epsilon_{\rm eff}/Z$ — the `FAIL_TAUTOLOGICAL` firewall, 023's, is not reintroduced). The fingerprints are dimensionless z-space rationals; the $z = a\omega/c_s$ units realization and the residual dimensional gate are 023's. The $Z_0^{\rm ret}/Z_1^{\rm ret}$ selector (the first concrete Gate-6 input) is the deferred nonlinear solve — 023's characterized departure.

Consumed closure and register. The one *checkable* consumption is Stage-018's fingerprint $\{u_2, u_4, v_5\} = \{1/9, 4/81, 1/27\}$, which 022's re-derived $\ell=2$ fingerprint reproduces (a per-slot / `3e`-tooth-backed derive-vs-typed non-regression; the second site is Stage-018's independently-earned literal, not a same-machinery reconstruction). *Provenance / context (cited, not re-derived):* Stage 019/020/021 + the completed pathA_33 `QUAD_CALIBRATED` joint (the prefactor $P(\omega) = D_0 N/D^2_{\text{cons}}$, the 54/5 magnitude, the $\hat{\mu}_0$-free dimensional closure); Stage 008's raw $\ell = 0, 1, 2$ outgoing amplitudes; Stage 009/010's bulk Helmholtz mode; and c_s (the density sound speed, Stage 005 R1 $c_s^2 = 5K\rho^4/m$) as the units symbol in $z = a\omega/c_s$ (distinct from the frozen-wall Helmholtz speed c_S, 011–017). Register: **zero** new counted knobs; structural edge **R41** (the cross-ℓ outgoing-DtN $-(\ell+1)/\Lambda_\ell$ fingerprint provenance — the $\ell = 0, 1, 2$ radiative signature $\{1, 1/2, 1/27\}$ at $\{\omega^1, \omega^3, \omega^5\}$ with $\Lambda_{\text{static}} = -(\ell + 1)$ derived + the scanned order, plus the Gate-4 quadrupole non-regression reproducing the Stage-018 fingerprint component of the now-completed pathA_33 fold; discharges nothing — earned exterior-wave structure + a consistency non-regression, not a reduction). Part-II counted CALIB set unchanged at $\{\mu_\eta, T_w, \beta\}(013)$ + $\{V_{p0}/\ell_c\}(015)$ + $\{T_\Omega, \beta_2\}(017) = 6$.

Verification. Dual-engine, independent symbolic routes, zero file I/O — the source pair's scratch-YAML handoff is severed. The `.wl` is a genuinely independent route, *re-authored* (not a kept transliteration): it uses Wolfram's built-in `SphericalHankelH1`/`SphericalHankelH2` + `SeriesCoefficient` — a materially different construction from the `.py`'s hand-built sin/cos j_ℓ/y_ℓ + `Series`/`Coefficient` (the source `.wl`'s `branchData` was a near-transliteration of the `.py` `dtn_branch`, so it was discarded — the mirror-policy transliteration screen). Agreement is at the transcript level (the derived $\{1, 1/2, 1/27\}$ at $\{\omega^1, \omega^3, \omega^5\}$, $\Lambda_{\text{static}} = \{-1, -2, -3\}$ with $h_{\text{mut}} = z\,h$ firing each and $H1 \leftrightarrow H2$ shown inert, the scanned order, the $\ell = 2$ non-regression, the per-slot mutants, the incoming sign-flip, and `3e`→`FAIL_QUAD_REGRESSION`); SymPy 55 PASS / Mathematica 64 PASS, exit 0, CWD-independent. The directive design-review used the Codex→Grok→Codex bookend: Codex folded five blocking findings (the `.wl` must be re-authored independent, not the transliterated `branchData`; the normalization and raw-order teeth must be genuinely derived — a Λ_{static} from the log-derivative + a coefficient scan replacing the source's $X \equiv X$ set-then-compare and preselected-order check; the $\ell = 2$ tuple was double-counted and χ_Q was subsumed, so the duplicate was removed, χ_Q de-counted, and per-slot u_2/u_4 mutants added; the earned/deferred partition is a subset relation with an explicit local/joint output; the checkable consumption is Stage-018's fingerprint), then a Codex confirm-pass caught a further directive-level reasoning error (the proposed Λ_{static} ablation "outgoing→incoming" is *inert* — both branches give $-(\ell+1)$ — so the mutant must be a pole-order corruption $h_{\text{mut}} = z\,h$; and Λ_{static}'s subsumed $\hat{Y}_\ell(0) = 1$ is de-counted). A Grok-4.5 compute-verify of the folded directive returned `DIRECTIVE_CLEAN`, independently confirming the cross-ℓ series, the pole-order-vs-inert mutant, the scan-vs-preselect counterexample, the χ_Q subsumption identity, and the u_2/u_4 isolation; a final Codex confirm closed the bookend. Full tri-review on fresh agents (arbiter re-run reproduced the build SymPy 56 / Mathematica 65, exit 0, CWD-independent): `FIDELITY_CLEAN` (an independent read hand-re-derived the cross-ℓ fingerprints, the pole-order Λ_{static} mutant vs the inert branch swap, the scan-vs-preselect order, the χ_Q subsumption, and the per-slot isolation — all computed not stamped, no 019/023 leakage, a genuine built-in-`SphericalHankelH1` `.wl`) and `ADVERSARIAL_ISSUES` (both confirm-pass reasoning catches held live-proven, all core physics teeth genuine; two low-severity redundancy teeth — a vacuous `3e rerun_gate_logic` constant-length check and a subsumed read-set-exclusion tooth — were remediated, the first make-genuine [compare the two traced verdicts] and the second an honest de-count [the adjacent exact-equality tooth retains the guard], plus two `.wl` nits [a symbol typo and a mutation-aware parity fix]), re-verified `REVERIFY_CLEAN` by the coupling meta-test (each remediated construct fires under a mutation of its object and goes vacuous when neutered; no regression, both engines). Tallies 56/65 → 55/64 (net −1 per engine from the honest de-count). Symbolic per-tooth ablation, mutations on copies.

B.18 Stage 023: The Native Nullspace Underdetermination Departure (Check II-G5b)

Part / anchor. Part II – Gravity (the frozen-throat cross-ℓ return sector; the COMPLETING, FAIL-delivering leg of a two-way split of pathA_34 — Stage 022 earned the cross-ℓ $-(\ell+1)/\Lambda_\ell$ fingerprints + the Gate-4 non-regression (the EARNED-FIRST 1/2), and this stage builds the native nullspace underdetermination that *delivers* the joint `FAIL_UNDERDETERMINED_NOT_PREDICTIVE` (2/2)). This COMPLETES the pathA_34 fold (022∧023) and closes the Gate-1–5 gravity ladder; the only remaining gravity item, Gate 6 (the nonlinear return-selector), is sim-deferred (a Part-VII open-item, R42).

Claim status. Open

Purpose. Certify that the linear Gate-5 return law is genuinely underdetermined: over 11 genuine generator dofs $[\Omega_U, \Omega_W, R_{\text{mix}}, g_U, g_W, D_0, K_{0c}, K_\eta, T_\Omega, Z_0^{\text{ret}}, Z_1^{\text{ret}}]$ the three collected named constraints $\{P_0^{\text{raw}}, K_{0c}, K_\eta + 2T_\Omega\}$ have Jacobian rank 3 $\Rightarrow$ native nullspace dimension 8, and augmenting with the return-transfer gradients $\{\partial T_0, \partial T_1\}$ gives return-augmented rank 5 $\Rightarrow$ *return-moving nullity 2*: the return admittances $\{Z_0^{\text{ret}}, Z_1^{\text{ret}}\}$ survive every collected constraint (Z is a pathA_29 premise) $\Rightarrow$ `FAIL_UNDERDETERMINED_NOT_PREDICTIVE`; and that a derived counterfactual selector collapses the return-moving nullity to 0 $\Rightarrow$ `CROSS_L_RESIDUAL_PREDICTION` (the able-to-fail witness).

Status. Open. FAIL-headline stage (cf. Stage 003 = earned transverse photons + the characterized `FAIL_CAUCHY` departure): the top-line verdict is a `FAIL_*`, but the earned content — a genuinely-computed native nullspace and the able-to-fail selector control — is the headline, and the FAIL is the first-class characterized landing (no softening). The computation is exact symbolic, float-free: the constraint Jacobian, its rank/nullspace, the return-transfer gradients, the A_0/A_1 residuals, and the (L, M, T)-triple dimensional algebra are exact `sympy` (`sp.Matrix(rows).rank()`) / native Wolfram (`NullSpace`, `MatrixRank`) computations; `expect_zero`/`expect_bool`/`expect_fail` residuals, no floats/tolerances. It adds **zero** new counted CALIB knobs (the set stays 6) and $Z_0^{\text{ret}}/Z_1^{\text{ret}}$ add zero new free dofs (aliases of the existing $\varepsilon_0/\varepsilon_1$ debt); the effective $\ell = 0/1$ stiffnesses K_{0c}/K_1 add *counted* `FREE-UNREDUCED` `PENDING` reduction-debt.

Two distinct labels, both printed. The script emits a script/teeth status separately from the earned physics verdict:

`AUDIT_STATUS=PASS` (the script ran, all teeth fired, exit 0)
`PHYSICS_VERDICT=FAIL_UNDERDETERMINED_NOT_PREDICTIVE (2/2, COMPLETING)` (the earned characterized-FA

The script PASSES (exit 0) by correctly *computing* the earned FAIL — the FAIL is the earned physics, not an execution failure.

The native nullspace (a genuine rank, not a rigged construction). The three collected named constraints are the $\ell = 2$ port kernel $P_0^{\text{raw}} = (\Omega_U^2 g_W + R_{\text{mix}} g_U)^2/(\Omega_U^2\Omega_W^2 - R_{\text{mix}}^2)^2/D_0$ (handoff §10.2–10.3), the $\ell = 0$ collective stiffness K_{0c} (§8.2 / Gate-2), and the $\ell = 1$ harmonic stiffness $K_1 = K_\eta + 2T_\Omega$ (§9.4). Their Jacobian J over the 11 dofs has

rank$(J) = 3$ (a genuine `sp.Matrix(rows).rank()` on symbolic ∂-rows — no zero-padding, no hardcoded 8/2), Augmenting J with the return-transfer gradients $\partial T_0, \partial T_1$ — where $T_\ell = K_\ell/(K_\ell + Z_\ell^{\mathrm{ret}})$ is the DC return transmission — gives

$$\mathrm{rank}(J \cup \{\partial T_0, \partial T_1\}) = 5, \qquad \texttt{return\_moving\_nullity} = 5 - 3 = 2 \;\Rightarrow\; \texttt{FAIL\_UNDERDETERMINED\_NOT\_PREDICTIVE}.$$

The two return admittances $\{Z_0^{\mathrm{ret}}, Z_1^{\mathrm{ret}}\}$ are untouched by every collected constraint ($\partial c/\partial Z_\ell^{\mathrm{ret}} = 0$) yet move T_0/T_1; this is witnessed constructively — the unit vectors $e_{Z_0^{\mathrm{ret}}}, e_{Z_1^{\mathrm{ret}}}$ each lie in the constraint nullspace yet give $\Delta T_0 = -K_{0c}/(K_{0c} + Z_0^{\mathrm{ret}})^2 \neq 0$ and $\Delta T_1 = -(K_\eta + 2T_\Omega)/(\cdots)^2 \neq 0$. Why the return survives: pathA_29 records Z as a *premise* (`Z_is_premise=True`, `boundary_dof=none`) — the linear theory supplies no equation to fix it.

The counterfactual selector control (the able-to-fail). Adding the two selector rows $\{Z_0^{\mathrm{ret}} = K_{0c},\ Z_1^{\mathrm{ret}} = K_\eta + 2T_\Omega\}$ raises the constraint rank $3 \to 5$ (native nullity $8 \to 6$) and makes $T_0 = T_1 = 1/2$ (constants, zero gradient), so `return_moving_nullity`: $2 \to 0$ and the verdict flips to `CROSS_L_RESIDUAL_PREDICTION`. Because `base_verdict` defaults to the predictive token and returns the FAIL only when the computed return-moving nullity exceeds 0, **the FAIL is earned from the computed nullity, not baked**. $\{Z_0^{\mathrm{ret}} = K_{0c}, Z_1^{\mathrm{ret}} = K_\eta + 2T_\Omega\}$ is a *counterfactual rank-collapse witness* (`derived_from_named_pde=False`, `control_only=True`), *not* a proven Gate-6 selector; the earned export is the *need* — Gate 6 must supply two independent equations fixing the two return directions (or an equivalent return law).

The A_0/A_1 residuals (forward-built; consuming Stage-022's $\{1, 1/2\}$). The $\ell = 0/1$ return residual amplitudes are the cross-ℓ/port continuation of Stage 009's `RETURN_RESIDUAL_PREDICTION` ($A_{\mathrm{res}} \propto \varepsilon_\ell/(1+\varepsilon_\ell)$), built *forward* from Stage 022's earned coefficients $v_0 = 1,\ v_1 = 1/2$ and the transmission $(1 - T_\ell)$:

$$A_0 = i\, v_0\, (a\omega/c_s)\, M_0\, (1 - T_0), \qquad A_1 = i\, v_1\, (a\omega/c_s)^3\, D_1\, (1 - T_1), \qquad \varepsilon_\ell = Z_\ell^{\mathrm{ret}}/K_\ell,$$

with $1 - T_\ell = \varepsilon_\ell/(1+\varepsilon_\ell)$ *forward* from the transfer definition — never back-solved from a residual. They are checked against the *independent* pathA_29 form expected $A_0 = i\, a\omega M_0 \varepsilon_0/(c_s(1+\varepsilon_0))$, expected $A_1 = i\, a^3\omega^3 D_1 \varepsilon_1/(2c_s^3(1+\varepsilon_1))$, and $A_0 - \text{expected}\, A_0 = 0$, $A_1 - \text{expected}\, A_1 = 0$ at orders ω^1, ω^3. The consumption is checkable: $A_1 - \text{expected}\, A_1 = i(a^3\omega^3/c_s^3) D_1 (\varepsilon_1/(1+\varepsilon_1))(v_1 - 1/2)$, zero iff $v_1 = 1/2$ — the $2c_s^3$ encodes the consumed $1/2$, so corrupting v_1 ($1/2 \to 1/3$) fires the A_1 form residual. The residual class (form/sign/order) is earned; the magnitude is deferred because the nullspace leaves $\varepsilon_{\mathrm{eff}}$ free.

The dimensional gate and the anti-back-solve firewall. A recursive (L, M, T)-triple `dim_of` (raising on a sum-mismatch) certifies, from the sourced dims, $[A_0] = (0, 1, -1)$, $[A_1] = (1, 1, -1)$, $[T_\ell] = [\varepsilon_\ell] = [P_0^{\mathrm{phys}}] = (0, 0, 0)$ with $[P_0^{\mathrm{phys}}] = (c_s/a)^2 [P_0^{\mathrm{raw}}]$; corrupting a sourced dim ($[M_0]$) fires `FAIL_DIMENSIONAL` while the unconstrained free carrier q_{free} gives `NO_FAIL` (the free-carrier-independence control). The `FAIL_TAUTOLOGICAL` firewall classes the $\varepsilon_{\mathrm{eff}}$ magnitude `deferred_branch_data` while underdetermined (so emitting it as derived is forbidden); a dedicated `emit_epsilon_magnitude_as_derived` mutation (and `assert_not_derive` on the genuinely-023-derived forward T_0/T_1 map) fire `FAIL_TAUTOLOGICAL` — the guard against the v1 rejection's $\varepsilon_{\mathrm{eff}}/Z$ back-solve.

Honest scope. The FAIL is the earned physics: the genuinely-computed native nullspace (dim 8 / return-moving nullity 2) shows the linear theory cannot pin the $\ell = 0/1$ return; Gate 6's nonlinear closure (sim-deferred) must supply two independent return equations (or an equivalent return law). The raw nullity 8 is partly bookkeeping — two of the three constraints (K_{0c}, K_1) self-pin their own stiffness, so

P_0^{raw} alone gives rank 1 / nullity 10, but the return-moving nullity is still 2; the verdict-bearing quantity is `return_moving_nullity`, the raw nullity a reported diagnostic. The residual *class* (form/sign/order) is earned; the *magnitude* is deferred (ε_{eff} left free), never back-solved.

Consumed closure and register. Two checkable relations: Stage-022's $\{1, 1/2\}$ (corrupting the cited v_1 fires A_1) and the pathA_29 residual form (Stage 009/010, $A_{\text{res}} \propto \varepsilon_\ell/(1+\varepsilon_\ell)$, from which expected A_0/A_1 is built independently). *Provenance (cited, not re-derived):* pathA_29's `Z_is_premise`; Stage 008's $R_0 = -M_0$ / $R_1 = -D_1$ targets and the M_0/D_1 moments; Stage 017's port kernel P_0^{raw}; Stage 013/017 as context for the effective stiffnesses; and c_s (Stage 005 R1 $c_s^2 = 5K\rho^4/m$) as the units symbol in $z = a\omega/c_s$ (distinct from the frozen-wall c_S, 011–017). Register: **zero** new counted CALIB knobs (set stays 6); $Z_0^{\text{ret}}/Z_1^{\text{ret}}$ add **zero** new free dofs; but K_{0c}/K_1 add *counted* `FREE-UNREDUCED PENDING` reduction-debt. The return admittances $\{Z_0^{\text{ret}}, Z_1^{\text{ret}}\}$ are coordinate *aliases* of the existing $\varepsilon_0/\varepsilon_1$ `FREE-UNREDUCED` debt ($\varepsilon_\ell = Z_\ell^{\text{ret}}/K_\ell$ invertible — no double-count, no new dof); the effective stiffnesses K_{0c} and the $\ell = 1$ sector $\{K_\eta, T_\Omega\}$ (via $K_1 = K_\eta + 2T_\Omega$, dims MT^{-2}) are `FREE-UNREDUCED PENDING` scalar-reduction *counted as reduction debt* (pending debt stays counted until derived), *not* `DERIVED`, not `CALIB` — their dims do not match registered Stage 013 $K_\eta = T_w\beta^2$ ($ML^{-1}T^{-2}$) or Stage 017 T_Ω ($ML^{-3}T^{-2}$), which Stage 017 records as non-transferable across the volume-vs-line convention (the Stage-016 lesson), so they are not identified with the raw densities (an explicit scalar-reduction to the wall packet would discharge the debt). New structural/obligation edge **R42** (the cross-ℓ nullspace underdetermination departure + the sharpened Gate-6 return-selector obligation; it sharpens the existing $\varepsilon_0/\varepsilon_1$ R24-family debt and adds no free dofs). Part-II counted CALIB set unchanged at $\{\mu_\eta, T_w, \beta\}(013)$ + $\{V_{p0}/\ell_c\}(015)$ + $\{T_\Omega, \beta_2\}(017) = 6$. This stage completes the pathA_34 fold (022$\wedge$023) and closes the Gate-1–5 gravity ladder.

Verification. Dual-engine, independent symbolic routes, zero file I/O — the source pair's scratch-YAML handoff is severed. The `.wl` is a genuinely independent route, *re-authored* (the source `rankAuditFor` block was structurally parallel to the `.py`'s `augRank−rank0`, so it was discarded per the mirror-policy screen): it proves the nullspace constructively — `Length[NullSpace[`J_{base}`]]` $= 8$, `MatrixRank[`basis·Greturn$^\top$`]` $= 2$, the explicit Z_ℓ^{ret} unit directions lie in `NullSpace[`J_{base}`]`, and (with the selector) `Length[NullSpace[`J_{sel}`]]` $= 6$ with the original return gradients giving Greturn·`NullSpace[`J_{sel}`]`$^\top$ $= 0$ — a materially different algorithm from the `.py` rank-subtraction. Agreement is transcript-level (`native_nullspace_dimension=8`, `return_moving_nullity=2`, `baseline_verdict=FAIL_UNDERDETERMINED_NOT_` `selector_control_verdict=CROSS_L_RESIDUAL_PREDICTION`, the A_0/A_1 residual-zero, the dim gate, and the probe tokens); SymPy 111 PASS / Mathematica 117 PASS, exit 0, CWD-independent. The directive design-review used the Codex→Grok→Codex bookend on the pathA_34 v1 rejection locus (the sharpest in Part II — v1 was rejected for a rigged zero-padded nullity, flag-driven probes, and a headline-only `.wl`): Codex folded seven blocking findings (the `.wl` decisive re-author; the "selector collapses the return-moving nullity, not the nullity" math fix — native nullity 8 → 6, rank 3 → 5 — with corrected isolated ablation teeth; the counterfactual-witness-not-proven-selector relabel; the provenance-cut fix — $\{1, 1/2\}$ as `cited_earned_input`, `assert_not_derive` rewired to the 023-derived T_0/T_1 map, the `gate4_prefactor` tag dropped; the strengthened firewall — the `emit_epsilon` tooth, the de-counted `rerun_gate_logic`, the fixed `able_to_fail_bad`; the K_{0c}/K_1 `PENDING`-not-`DERIVED` register correction (dims $(0, 1, -2) \neq$ 013/017 — the Stage-016 convention trap); and the $Z_0^{\text{ret}}/Z_1^{\text{ret}}$ aliases-not-new-dofs register correction), with confirm-passes folding the consistency-sweep gaps. A Grok-4.5 compute-verify returned `DIRECTIVE_CLEAN`, independently confirming (with its own SymPy) the rank/nullity/return-moving, the selector flip, $A_1 -$ expected $A_1 = 0$ iff $v_1 = 1/2$, the dims, the register conventions, and the `.wl` constructive identity, plus one honest-scope note (the raw nullity 8 includes K_{0c}/K_1 self-constraint bookkeeping; the verdict rides return-moving 2); a closing Codex confirm closed the bookend. Full tri-review on fresh agents (arbiter re-run reproduced the build SymPy 116 / Mathematica 123, exit 0, CWD-independent): `FIDELITY_CLEAN` (an independent read hand-re-derived the rank audit, the selector flip, the A_0/A_1 forward-build + the $v_1 = 1/2$ consumption, and the dims — all faithful, genuine rank not zero-padded, no ε back-solve, a materially-different constructive `.wl`) and `ADVERSARIAL_CLEAN` (per-tooth ablation, 15 mutations across both engines: hardcoding the rank, zero-padding the Jacobian, and faking the `.wl NullSpace` basis all make the audit FAIL; the four KEY anti-rig properties confirmed; four non-blocking de-count nits). Codex remediated two make-genuine (the witness-preservation assert recomputes each Jacobian-row dot product from the stored witness vector; the neutralized-mutation meta-test uses a cache-distinct inert context with an independence check) and two honest de-counts (the provenance/premise documentation asserts $\to$ labeled prints; the T/ε identity $\to$ a self-consistency check), re-verified `REVERIFY_CLEAN` by the coupling meta-test (each made-genuine tooth fires under a mutation of its object and goes vacuous when neutered; no regression, both engines). Tallies 116/123 $\to$ 111/117 (net $-5/-6$ per engine from the honest de-counts). Symbolic per-tooth ablation, mutations on copies. With this stage the pathA_34 fold is complete (022$\wedge$023) and the Gate-1–5 gravity ladder is closed.

B.19 Stage 024: The Density-Native $\ell = 2$ Quadrupole Two-Port Derivation (Check II-P1)

Part / anchor. Part II – Gravity (the frozen-throat $\ell = 2$ radiative-port sector, Cluster C; the EARNED-FIRST derivation leg of a four-way split of pathA_43 — the density-native $\ell = 2$ quadrupole two-port derivation component of the joint `DENSITY_PORT_HOSTED`, with the COMPUTED vector-freedom taint = Stage 025, the continuity-moment lineage token-check = Stage 026, and the able-to-fail port checks + closure overlay (the COMPLETING leg) = Stage 027)

Claim status. Exact Within Closure

Purpose. Earn the density-native coupling STRUCTURE the quadrupole port rides. Show the reduced $\ell = 2$ two-port over (q_2 wall, Φ_2 bulk-density), with static operator $M = \left(\begin{smallmatrix} \varpi_{q_2} & -\lambda_c \\ -\lambda_c & \varpi_{\Phi_2} \end{smallmatrix}\right)$ and source $g_{\text{base}}(\eta_q, \eta_\phi)$, $g_{\text{base}} = \sqrt{\rho_{\text{eff}}}\, c_s^2 I_{25} \Xi_Q / a^{7/2}$, yields the load-bearing Phi$_2$-response $(M^{-1}\text{source})[\Phi_2] = P_{\text{den}}/\Delta$ and the port numerator $N_{0,\text{den}} = P_{\text{den}}^2/\Delta^2 = I_{25}^2 \Xi_Q^2 c_s^4 \rho_{\text{eff}} (\eta_\phi \varpi_{q_2} + \eta_q \lambda_c)^2 / (a^7(\lambda_c^2 - \varpi_{\Phi_2}\varpi_{q_2})^2)$, density-hosted and vector-free. Export $N_{0,\text{den}}$; land the joint verdict as a partial component.

Status. Exact Within Closure — the derivation is exact symbolic algebra (the 2×2 static-operator inverse and the resulting rational port numerator are exact `Matrix.inv()/Solve` algebra; `expect_zero/expect_bool` residuals, no `scipy/numpy`/floats/tolerances). The verdict is the joint `DENSITY_PORT_HOSTED`; this stage carries its *density-native two-port derivation* component (1/4, the EARNED-FIRST slice) and lands the joint as **partial** (024 earned; 025/026/027 pending) — not complete. **`DENSITY_TWO_PORT_DERIVED`.** The LOCAL audit verdict (exit-0 gate): $N_{0,\text{den}}$ is genuinely built from the inverse/elimination, the factorization cross-check holds, the $\Delta \neq 0$ guard + the coupling-vanishes + the density-only host-set teeth fire, and the two routes agree. It adds **zero** new counted knobs (a DERIVATION/structure slice, like Stages 018/016/022); the operator entries $\varpi_{q_2}/\varpi_{\Phi_2}/\lambda_c$, the moment I_{25}, Ξ_Q, η_q/η_ϕ, and ρ_{eff} are SIM_DEFERRED/GAP tracked-not-counted.

The density two-port operator. The reduced, mass-normalized static Lagrangian of the $\ell = 2$ sector is a symmetric 2×2 quadratic form over (q_2, Φ_2) (the wall angular amplitude and the bulk-density amplitude), driven by a continuity/interface source:

$$M = \begin{pmatrix} \varpi_{q_2} & -\lambda_c \\ -\lambda_c & \varpi_{\Phi_2} \end{pmatrix}, \qquad (g_q, g_\phi) = g_{\text{base}}(\eta_q, \eta_\phi), \quad g_{\text{base}} = \frac{\sqrt{\rho_{\text{eff}}}\, c_s^2\, I_{25}\, \Xi_Q}{a^{7/2}},$$

with a negative off-diagonal $-\lambda_c$ (the wall↔bulk mixing). The operator entries are abstract symbols whose literal throat values are SIM_DEFERRED — the two-port structure holds for any admissible $\varpi_{q_2}, \varpi_{\Phi_2}, \lambda_c$. This is the $\ell = 2$ continuation of the consumed bulk mode (009/010) and wall mode (016/017).

The port numerator (the EARNED headline). The radiative amplitude is the Φ_2 component of the inverted response. With determinant Δ and adjugate·source numerator P_{den},

$$\Delta = \varpi_{q_2}\varpi_{\Phi_2} - \lambda_c^2, \qquad P_{\text{den}} = \varpi_{q_2} g_\phi + \lambda_c g_q, \qquad (M^{-1}\text{source})[\Phi_2] = \frac{P_{\text{den}}}{\Delta},$$

$$N_{0,\text{den}} = \left(\tfrac{P_{\text{den}}}{\Delta}\right)^2 = \frac{I_{25}^2\, \Xi_Q^2\, c_s^4\, \rho_{\text{eff}}\, (\eta_\phi \varpi_{q_2} + \eta_q \lambda_c)^2}{a^7\, (\lambda_c^2 - \varpi_{\Phi_2}\varpi_{q_2})^2}.$$

The host-set is $\subset \{q_2, \Phi_2, c_s, a, \rho_{\text{eff}}, I_{25}, \Xi_Q, \eta_q, \eta_\phi, \varpi_{q_2}, \varpi_{\Phi_2}, \lambda_c\}$ and contains *none* of the retired vector symbols $\{A_w, F_{\mu w}, J^w, U, W, \Omega_U, \Omega_W, R_{\text{mix}}, g_U, g_W\}$. $N_{0,\text{den}}$ is computed via `Matrix.inv()` (SymPy) / `Solve`+eliminate (WL), not typed.

The load-bearing inverse and the two routes. The two engines derive the same $N_{0,\text{den}}$ by materially different algorithms: SymPy takes the full `static_operator.inv()` and reads response = $(M^{-1}\text{source})[\Phi_2]$; Mathematica keeps a native Green-DtN elimination — solve the wall row for q_2 ($q_2 \to (\lambda_c\Phi_2 + g_q)/\varpi_{q_2}$), substitute into the bulk row, `Solve` the reduced scalar interface equation, then square. In the current pathA_43 source the SymPy inverse was *decorative* ($N_{0,\text{den}}$ assembled from a typed $P_{\text{den}}^2/\Delta^2$); the reshape makes it LOAD-BEARING — $N_{0,\text{den}} =$ `make_N0`(response) with `make_N0`$(r) := r^2$, so the dataflow is response $\to N_{0,\text{den}}$, with P_{den}/Δ retained only as the trace form guarded by the factorization cross-check compact(response $- P_{\text{den}}/\Delta) = 0$. Because a typed $P_{\text{den}}^2/\Delta^2$ passes the cross-check and prints an identical transcript, a code-read cannot catch a decorative inverse; the enforcement is the RUNTIME dataflow probe `make_N0`(response $+ \delta_{\text{probe}}) \neq$ `make_N0`(response), which fires on a response-independent builder (the analog of Stage-023's genuine `Matrix.rank()`). The two-route agreement is the ARBITER's oracle — simplify$(N_{0,\text{den}}^{\text{py}} - N_{0,\text{den}}^{\text{wl}}) = 0$, canonical not raw-string — neither engine reads the other.

Nonsingular guard, coupling boundary, host-set. The inversion is defined only where $\Delta = \varpi_{q_2}\varpi_{\Phi_2} - \lambda_c^2 \neq 0$ (positivity of the entries does not imply it: $\lambda_c^2 = \varpi_{q_2}\varpi_{\Phi_2}$ is admissible), so a named $\Delta \neq 0$ guard runs before the inversion/`Solve` in both engines, with a singular control ($\lambda_c^2 \to \varpi_{q_2}\varpi_{\Phi_2}$) that fails at that guard; the stronger $\Delta > 0$ stability reading is SIM_DEFERRED, not gated. The derivation boundary is the coupling: $g_{\text{base}} \to 0 \Rightarrow N_{0,\text{den}} = 0$ COMPUTED (the zero-control recomputes with $g_{\text{base}} = 0$ and yields exactly 0), while baseline $N_{0,\text{den}} \neq 0$. The density-only host-set is a manifest-construction check ($N_{0,\text{den}}$`.free_symbols` $\subset$ the density set, $\cap$ the vector set $= \emptyset$), not the computed taint graph (that is 025's). Per-tooth ablation: teeth A (inverse factorization + dataflow probe), B ($\Delta \neq 0$ guard), C (coupling-vanishes, two named asserts), D (density-only host-set), E (`.wl` arity + unevaluated-leakage) each fire at their own assert.

Honest scope. Earned structure / SIM_DEFERRED magnitude: 024 derives the reduced coupling STRUCTURE (the two-port form, its host-set, the inverse/elimination), but the literal magnitudes (I_{25}, Ξ_Q, η_q, η_ϕ, the λ_c throat value) are SIM_DEFERRED, and G, 2/5, 54/5 are calibrated ($G =$`GENUINE_BLOCKED`) — the 54/5 partition + closure overlay + PN match-back are Stages 027/028, not 024. c_s is the density/sound (ripple) speed — the analog "speed of light" of the gravitational sector (Stage-005 R1) — and a is the `CONV` pin. ρ_{eff} is the effective reduced-3D *mass* density ($[ML^{-3}]$) that mass-normalizes (q_2, Φ_2); it is *not* Stage-005's ρ_0 (a 4D number density $[L^{-4}]$), and the reduction $\rho_{\text{eff}} \leftarrow \{\rho_0, m, \text{geometry}\}$ is SIM_DEFERRED/GAP.

Consumed closure and register. *Provenance only, no cross-stage dual-site:* 024's genuineness check is intra-stage (the two routes + the load-bearing factorization cross-check). The `physical_relations` trace exhibits $\varpi_{q_2} = K_2/M_2 = (c_s/a)^2\kappa_q$ (pathA_32 wall; 016/017), $\varpi_{\Phi_2} = (c_s/a)^2(6 + (ma)^2)$ (pathA_29 bulk, $6 = \ell(\ell+1)$; 009/010/016), and $\lambda_c = (c_s/a)^2\hat{\lambda}_Q$ (projected continuity/interface); the COMPUTED taint-set proof that these tags are genuine is 025's. I_{25} is a typed $\ell = 2$ continuity-moment input (lineage validated in Stage 026, a forward ref). Register: **zero** new counted knobs; $\varpi_{q_2}/\varpi_{\Phi_2}/\lambda_c$, $I_{25}/\Xi_Q/\eta_q/\eta_\phi$, and ρ_{eff} are SIM_DEFERRED/GAP tracked-not-counted; structural edge **R43** (the density-native $\ell = 2$ two-port algebraic derivation provenance, scoped to 024's algebra — *not* "the vector scaffold retired," which is the joint result proven by 025/027; discharges nothing). Symbol dims recorded as provenance ($[I_{25}] = L^{5/2}$, $[c_s] = LT^{-1}$, $[\rho_{\text{eff}}] = ML^{-3}$, $[a] = L$, $[\varpi_{q_2}] = [\varpi_{\Phi_2}] = [\lambda_c] = T^{-2}$, $[\eta_q] = [\eta_\phi] = [\Xi_Q] = 1 \Rightarrow [N_{0,\text{den}}] = L^{-1}M$); the units-restored dim CHECK is 027's. Part-II counted CALIB set unchanged at $\{\mu_\eta, T_w, \beta\}(013) + \{V_{p0}/\ell_c\}(015) + \{T_\Omega, \beta_2\}(017) = 6$. $N_{0,\text{den}}$ is exported to 025 (taint) / 026 ($I_{25} \in$`free_symbols`) / 027 (dimension, a-scaling, sign, closure).

Verification. Dual-engine, independent symbolic routes, zero file I/O — pathA_43 was already contract-clean (no `Export`/`Put`/`Import`; the `results.yaml` was written by an external orchestrator), so the reshape "sever the bridge" step is a no-op (the lightest reshape in Part II); the work was decomposition + the genuineness upgrade + the LOCAL/PARTIAL framing. The `.wl` KEEPS its native Green-DtN `Solve`+eliminate-q_2 route (its own `clean/fmt`, not a mirror), agreeing at the transcript level ($N_{0,\text{den}}$ canonical factored, the Φ_2-response $= P_{\text{den}}/\Delta$, the $\Delta \neq 0$ guard status, the `physical_relations` provenance, the density-only host-set, the LOCAL verdict, the JOINT PARTIAL), with an arity + Module-result-shape + unevaluated-leakage self-check. SymPy 7 PASS / Mathematica 10 PASS, both exit 0, CWD-independent (repo root + `/tmp`). The directive design-review used the Codex→Grok→Codex bookend (a Codex xhigh pass folded 4 BLOCKING — the $\Delta \neq 0$ guard + singular control, the $\rho \to \rho_{\text{eff}}$ mis-provenance correction, the per-tooth independent-oracle restructure, R43 scoped to 024's algebra; a confirm-pass folded 2 further BLOCKING — tooth A's runtime dataflow probe and tooth C's two named asserts; both engines compute-verified $N_{0,\text{den}}$, response $\equiv P_{\text{den}}/\Delta$, $[N_{0,\text{den}}] = L^{-1}M$, and the a-powers $-3 \to -5$). Full tri-review on fresh agents: `FIDELITY_CLEAN` (an independent read re-derived $N_{0,\text{den}}$ by a third `linsolve` route, diff = 0) and `ADVERSARIAL_CLEAN` (per-tooth ablation: every tooth A–E fired at its own named assert, the decorative-inverse rig caught at runtime by the δ_{probe} in both engines; no vacuous tooth) — zero remediation required, the contract-clean lightest-reshape stage. Symbolic per-tooth ablation, mutations on copies.

B.20 Stage 025: The Density-Port Vector-Freedom Taint Proof (Check II-P2)

Part / anchor. Part II – Gravity (the frozen-throat $\ell = 2$ radiative-port sector, Cluster C; the ANTI-RIG PROOF leg of a four-way split of pathA_43 — the vector-freedom taint proof component of the joint

`DENSITY_PORT_HOSTED`, with the density two-port DERIVATION = Stage 024, the continuity-moment lineage token-check = Stage 026, and the able-to-fail port checks + closure overlay (the COMPLETING leg) = Stage 027)

Claim status. Exact Within Closure

Purpose. Prove Stage-024's exported $\ell = 2$ density-port numerator $N_{0,\mathrm{den}}$ is computationally VECTOR-FREE — that it cannot hide a disguised copy of the retired EM vector port $P_A = \Omega_U^2 g_W + R_{\mathrm{mix}} g_U$. Compute the provenance taint over $N_{0,\mathrm{den}}$.`free_symbols` via a fixed tag map and show it equals exactly the density ancestry {`continuity_interface`, `pathA_29_bulk`, `pathA_32_wall`} with `vector_port` $\notin$ it, source-map complete, and no vector-host symbol. The decisive check is the COMPUTED taint over provenance TAGS — never a name-check — which catches density-LOOKING relabels a name-check misses. Export the vector-freedom verdict; land the joint as a partial component.

Status. Exact Within Closure — exact symbolic set-algebra (the taint is a union of provenance tags over the actual $N_{0,\mathrm{den}}$.`free_symbols`; `expect_bool` residuals, no `scipy`/`numpy`/floats/tolerances). The verdict is the joint `DENSITY_PORT_HOSTED`; this stage carries its *vector-freedom taint proof* component (2/4) and lands the joint as **partial** (024/025 earned; 026/027 pending) — not complete. **`DENSITY_PORT_VECTOR_FREE`.** The LOCAL audit verdict (exit-0 gate, CONDITIONAL on the cited `moment_valid` = True that Stage 026 earns): the computed taint is exactly the three density tags (P1), and source-map completeness $\wedge$ no vector-host $\wedge$ `vector_port` $\notin$ taint (P2). It adds **zero** new counted knobs (a proof/taint slice, like Stages 018/016/022); the vector/relabel/free-carrier symbols are control fixtures, tracked-not-counted.

The subject and the consumption oracle. Stage 025 CITES Stage-024's factored export (it does not re-derive the inverse — that would blur the 024/025 cut):

$$N_{0,\mathrm{den}} = \frac{I_{25}^2\, \Xi_Q^2\, c_s^4\, \rho_{\mathrm{eff}}\, (\eta_\phi \varpi_{q_2} + \eta_q \lambda_c)^2}{a^7\, (\lambda_c^2 - \varpi_{\Phi_2} \varpi_{q_2})^2}.$$

The consumption-integrity oracle computes $N_{0,\mathrm{den}}$.`free_symbols` off the actual cited expression and asserts it equals the exported 10-symbol host contract $\{a, c_s, \rho_{\mathrm{eff}}, I_{25}, \Xi_Q, \eta_q, \eta_\phi, \varpi_{q_2}, \varpi_{\Phi_2}, \lambda_c\}$ (the coordinate hosts q_2, Φ_2 are projected out of the numerator — they extend the *allowable* density-host universe to 12 but are not in $N_{0,\mathrm{den}}$.`free_symbols`). A dropped/renamed symbol breaks the contract (tooth I).

The computed taint and the two separated predicates. Each symbol carries a fixed provenance tag set (faithful to the pathA_43 source with the 024 renames): $c_s, \rho_{\mathrm{eff}}, \varpi_{\Phi_2} \to$ {`pathA_29_bulk`}; $\varpi_{q_2} \to$ {`pathA_32_wall`}; $\Xi_Q, \eta_q, \eta_\phi \to$ {`continuity_interface`}; $a \to$ {`pathA_29_bulk`, `pathA_32_wall`}; $\lambda_c \to$ all three; $I_{25} \to$ {`continuity_interface`, `pathA_32_wall`} (installed from `moment_valid`); the vector, relabel, and σ_{hidden} symbols $\to$ {`vector_port`}; `free_carrier` $\to \emptyset$. The taint is COMPUTED — the union of `tag_map`$[s]$ over $s \in N_{0,\mathrm{den}}$.`free_symbols` — not typed. Two predicates are separated: **P1** `baseline_ancestry_ok` (the genuine numerator's taint equals *exactly* the three density tags, baseline only) and **P2** `vector_free` (`source_map_complete` $\wedge$ `vector_host_symbols` $= \emptyset \wedge$ `vector_port` $\notin$ taint). $N_{0,\mathrm{den}}$ satisfies both; `LOCAL` = P1 $\wedge$ P2 $\wedge$ `moment_valid`.

Why the taint is over TAGS (the two caught rigs), and the de-counted witness. The pathA_43 history holds two rigs a name-check would pass by fiat: (Rig 1) a density-LOOKING two-port built from $\{\omega_{\text{wall}}, \omega_\rho, r_{\text{mix}}, g_\rho, g_{q,\text{old}}\}$ — symbols not in `VECTOR_SYMBOLS` but tagged `vector_port`; and the hidden-vector rig $N_{0,\text{den}} \cdot \sigma_{\text{hidden}}$ (σ_{hidden} tagged `vector_port`, not in the name set). A `free_symbols` $\cap$ `VECTOR_SYMBOLS` name-check passes both; the COMPUTED taint catches them (`vector_port` $\in$ taint). The expression-level ablation (substitute vector-tainted symbols $\to 0$, compare) is RETAINED as a printed witness but is *de-counted*: once P2 holds the ablation set is empty so it cannot independently fail (a subsumed guard; also `nan` on the singular P^2/Δ^2 vector-port fixtures). The decisive gate is the taint-set (P1/P2) + `source_map_complete`. (The self-asserted-`continuity_interface`-tag rig, "Rig 2," is Stage-026's lineage token-check; 025 cites `moment_valid`.)

The able-to-fail battery (per-tooth ablation, coupling meta-test). Each rig fires at its OWN named assert and goes vacuous when neutered: A relabel and B hidden-vector via the *computed-taint* assert (a name-check misses them — the anti-fiat core); C vector-injection and G raw-vector-port via the *vector-host* assert; D provenance-less and E missing-symbol via the two halves (empty / missing) of *source-map completeness*; F tagged-carrier PASSES P2 with a legitimate four-tag taint (not the exact-three-tag P1) and FLIPS to fail when its tag is neutered (the reversibility control); I subject-integrity via the exact host-contract assert; and the two `.wl` runtime scanners (a call-arity mismatch, a leaked unevaluated authored-helper head). The adversarial re-review proved the decisive property load-bearing: downgrading the machinery to a name-check fires the relabel/hidden asserts in both engines.

Honest scope, consumed, register. 025 proves the vector-freedom STRUCTURE, CONDITIONAL on Stage 026 earning `moment_valid`; the EM $A_w/U, W$ scaffold RETIREMENT is the joint 025$\wedge$026$\wedge$027 result, recorded at Stage 027 (if 027 fails dimension/scaling/sign/closure, the port has not displaced the scaffold). Magnitudes ($\Xi_Q, \lambda_c, \rho_{\text{eff}}$ throat values; $G, 2/5, 54/5$) are SIM_DEFERRED/CALIBRATED downstream. *Consumed:* Stage-024's $N_{0,\text{den}}$ (the subject; the checkable host contract) + Stage-026's `moment_valid` (typed forward ref) + the density symbols' provenance tags (024's `physical_relations` + Stage-005 c_s + a `CONV`). Register: **zero** new counted knobs; new structural edge **R44** — the density-port vector-freedom PROOF (the computed taint = the density ancestry, source-map complete, no vector host), recording only the vector-freedom *conjunct* (the retirement is the joint 025/027 result, 027 records it); discharges nothing, like R43/R41/R39. No new dims (the $[N_{0,\text{den}}] = L^{-1}M$ check is 027's). Part-II counted CALIB set unchanged at $\{\mu_\eta, T_w, \beta\}(013) + \{V_{p0}/\ell_c\}(015) + \{T_\Omega, \beta_2\}(017) = 6$.

Verification. Dual-engine, independent symbolic routes, zero file I/O — pathA_43 was already contract-clean, so the reshape "sever the bridge" step is a no-op; the work was decomposition + the `.wl` RE-AUTHOR + the LOCAL/PARTIAL framing. *Unlike Stage 024 (which kept its `.wl` native for the derivation lane), Stage 025 RE-AUTHORS its `.wl`*: the source `.wl` taint machinery was a transliteration of the `.py`, so the mirror policy required a genuinely independent computation — the decisive taint + source-map completeness use a native directed `Graph` of `DirectedEdge`["sym:" $\to$ "tag:"] + `VertexOutComponent` reachability (not `Variables`+`Lookup`+`Union`), and the ablation witness uses $D[\text{expr}, v]$ partial-derivative independence. Transcript-level agreement, neither engine reading the other. SymPy 18 PASS / Mathematica 18 PASS, both exit 0, CWD-independent (repo root + `/tmp`). The directive review used the Codex$\to$Grok$\to$Codex bookend (a Codex xhigh pass folded 5 BLOCKING — most importantly that the expression ablation is logically subsumed by the taint-set gate, reframed to a de-counted witness with the computed taint-set identity + source-map completeness the decisive gate; the two-predicate P1/P2 split; R44 recording only the vector-freedom conjunct; the cite-the-factored consumption; the tooth-H split into a review-acceptance and two runtime scanners — then Grok-4.5 compute-verified the subsumption, the relabel/hidden taint vs the name-check, and the reversibility control `DIRECTIVE_CLEAN`, a Codex confirm-pass folded 4 consistency-sweep gaps, and a Codex final re-confirm returned `DIRECTIVE_CLEAN`). Full tri-review on fresh agents: `FIDELITY_CLEAN` (an independent read re-derived $N_{0,\mathrm{den}}$ from Stage-024's 2×2 inverse, diff $= 0$; confirmed the tag map, the P1/P2 separation, the de-counted witness, the genuine independent `.wl`) and `ADVERSARIAL_CLEAN` (per-tooth ablation on both engines; the name-check-dodge test proved the tag-taint load-bearing) — two **[minor]** non-blocking checks that could not independently fail (a subsumed universe-membership assert; a masked `moment_valid` assert) were de-counted to labeled diagnostic prints (tally $20 \to 18$ per engine, no other check touched), then fresh-agent `REVERIFY_CLEAN`. Symbolic per-tooth ablation, mutations on copies.

B.21 Stage 026: The Continuity-Lineage Token-Check (Check II-P3)

Part / anchor. Part II – Gravity (the frozen-throat $\ell = 2$ radiative-port sector, Cluster C; the CONTINUITY-LINEAGE leg of a four-way split of pathA_43 — the continuity-moment lineage validation component of the joint `DENSITY_PORT_HOSTED`, with the density two-port DERIVATION = Stage 024, the vector-freedom taint proof = Stage 025, and the able-to-fail port checks + closure overlay (the COMPLETING leg) = Stage 027)

Claim status. Exact Within Closure

Purpose. Prove the $\ell = 2$ moment I_{25} that sources Stage-024's density-port numerator $N_{0,\mathrm{den}}$ is a genuine $\int Y_2^* S_{\mathrm{leak}}$ continuity moment DESCENDED from pathA_29's projected-continuity operator (the same operator that produced pathA_28's $\ell = 0$ M_0 and $\ell = 1$ D_1), not a self-asserted / relabeled tag. Validate the $\ell_0 \to \ell_1 \to \ell_2$ ancestry by a COMPUTED exact lexical-token check over the moment strings (never a self-asserted flag, never raw substring), and earn the genuine moment symbol via the I_{25}-vs-I_{wrong2} gate. This DISCHARGES two forward references: it earns `moment_valid` = True (consumed by Stage 025) and validates I_{25}'s lineage (cited by Stage 024). Land the joint as a partial component.

Status. Exact Within Closure — exact symbolic lexical-token set-algebra (the lineage gate is an exact-token subset over the $\ell_0 \to \ell_1 \to \ell_2$ moment strings; the earning gate is a symbol-identity distinction; `expect_bool` residuals, no `scipy`/`numpy`/floats/tolerances). The verdict is the joint `DENSITY_PORT_HOSTED`; this stage carries its *continuity-lineage validation* component (3/4) and lands the joint as **partial** (024/025/026 earned; 027 pending) — not complete. **`CONTINUITY_LINEAGE_EARNED`.** The LOCAL audit verdict (exit-0 gate) $= G_1 \wedge G_2$: the lineage exact-token check holds and the earning gate returns (I_{25}, `moment_valid` = True). It adds **zero** new counted knobs (a lineage/proof slice, like Stages 018/016/022/024/025); I_{25} is already tracked (SIM_DEFERRED magnitude) and I_{wrong2} is a control-scar, tracked-not-counted.

The subject and the $\ell_0 \to \ell_1 \to \ell_2$ lineage. Stage 026 CITES Stage-024's factored export (it does not re-derive the inverse — that would blur the 024/026 cut); the consumption-integrity oracle (tooth I) computes $N_{0,\mathrm{den}}$`.free_symbols` off the actual cited expression and asserts it equals the exported 10-symbol host contract $\{a, c_s, \rho_{\mathrm{eff}}, I_{25}, \Xi_Q, \eta_q, \eta_\phi, \varpi_{q_2}, \varpi_{\Phi_2}, \lambda_c\}$ — with I_{25} the external factor (dropping it removes I_{25}; dropping $\eta_q \lambda_c$ removes only η_q, λ_c surviving in the denominator). The lineage is the structured record that I_{25} descends from pathA_29's operator: operator id `pathA_29_projected_continuity`; ℓ_0 $M_0 = \int S_{\mathrm{leak}}\, d^3x$ (pathA_28 mass rate, target $R_0 = -M_0$); ℓ_1 $D_{1,i} = \int x_i S_{\mathrm{leak}}\, d^3x + \int j_i\, d^3x$ (pathA_28 dipole rate, target $R_1 = -D_1$); ℓ_2 $Q_{2,m} = \int Y_2^* S_{\mathrm{leak}}\, d^3x$ (the $\ell = 2$ continuation, reduced value I_{25}); kernel $Y_2^* S_{\mathrm{leak}}$. The ℓ_0/ℓ_1 tokens are pathA_28's exported moments (Stage 008) and the operator is pathA_29's (Stage 009/010) — provenance; 026 verifies the lineage *exhibits* them, it does not rebuild them.

The two decisive gates: exact tokens (G_1) and the earning gate (G_2). G_1 `continuity_lineage_valid` computes `operator_id` = `pathA_29_projected_continuity` and, per ancestry level, that the required token *set* is a subset of the string's actual token set ($\ell_0 \supseteq \{M_0, \texttt{Integral}, S_{\text{leak}}, d3x\}$, etc.). This UPGRADES the source's raw substring `contains_all`, which a token-STUFFED forgery (`NOT_M0` $\supset M_0$, `FakeIntegral` $\supset$ `Integral`, `S_leakage` $\supset S_{\text{leak}}$, `d3xyz` $\supset d3x$) defeats. The token alphabet is LOAD-BEARING — identifier tokens `[A-Za-z0-9_]+` (Python `\w`; Mathematica `StringCases`$[s, ($`WordCharacter`$\,|\,$"_"$)..]$, *not* bare `WordCharacter..`, which excludes _ and would fragment the genuine tokens $S_{\text{leak}}/Y_2^*/D_{1,i}$ and false-fail the lineage). G_2 `continuity_moment_symbol` is source-faithful: `moment_valid` $\equiv$ `lineage_valid`; `earned_moment` $= I_{25}$ iff (`lineage_valid` $\wedge$ `a_power` $= -7/2$) else I_{wrong2}. An invalid lineage substitutes the *different* symbol I_{wrong2} (dimension $(2, 0, 0) \neq I_{25}$'s $L^{5/2}$) into the port — so a mis-lineaged port is a different object, which is what makes the token-check matter. $G_1 \wedge G_2$ earn the two forward-referenced outputs; 026 uses the $-7/2$ baseline only (the wrong-power scaling verdict is 027's).

Why the check is over tokens, never a flag (Rig 2), and the de-counted witness. The pathA_43 history holds **Rig 2** — a self-asserted `continuity_interface` tag: a devious executor writes a new equation $\equiv$ the retired vector port $P_A = \Omega_U^2 g_W + R_{\text{mix}} g_U$, tags it `continuity_interface`, and wires it as N_0's ancestor. The validator must reject it by exhibited structural TOKENS, not a shared `valid` flag. Since the source lineage dicts have no `valid` field, the rig battery self-enforces this: every negative lineage carries a DECOY `valid` = True + a self-asserted `continuity_interface` claim and is passed through the SAME validator; a coupling meta-test (substitute a flag-only validator) makes the decoy-flagged negatives wrongly pass, so the audit fails — the exact-token check is proven load-bearing. The first-arm dependency witness `earned_moment` $\in N_{0,\text{den}}$`.free_symbols` is a de-counted printed witness (subsumed by G_1 + the contract oracle — no non-subsumed mutation exists; the drop-I_{25} mutation routes to tooth I); the second OR-arm ($N_{0,\text{den}} = 0 \wedge$ `coupling_zero`) is the mandatory tooth G.

The able-to-fail battery (per-tooth ablation, coupling meta-test). Each rig fires at its OWN named assert and goes vacuous when neutered: A `fake_continuity` and B `attack2` (correct id + garbage ℓ_2) via the G_1 lineage-gate assert; C the per-level exact-token ablations (operator id / ℓ_0 / ℓ_1 / ℓ_2-kernel, each independently load-bearing) plus the token-stuffing forgery (passes a substring check, fails the exact-token check — reverting to substring makes it wrongly pass, the genuineness-upgrade meta-test); D the earning gate (neuter the substitution so a corrupted lineage wrongly earns I_{25}); F the baseline-valid positive (passes, flips to fail on any token corruption); G(i–iv) the four isolated vanished-coupling probes on synthetic (`expr`, `coupling_zero`) inputs (isolated from tooth I's fixed $N_{0,\text{den}}$); I subject-integrity via the exact host-contract assert (drop I_{25}^2 / drop $\eta_q\lambda_c$ / replace ρ_{eff}); and the two `.wl` runtime scanners (a call-arity mismatch, a leaked unevaluated authored-helper head). The adversarial re-review proved the decisive properties load-bearing: reverting to substring makes the stuffing forgery pass, a flag-only validator makes the decoy negatives pass, and bare-`WordCharacter` false-fails the genuine lineage.

Honest scope, consumed, register. 026 validates the LINEAGE (that I_{25} is a genuine $\int Y_2^* S_{\text{leak}}\, \ell = 2$ continuity moment); the literal moment value ($I_{25}, \Xi_Q, \hat{\lambda}_Q, \rho_{\text{eff}}$ throat values) remains SIM_DEFERRED, and $G, 2/5, 54/5$ are CALIBRATED (027). The joint `DENSITY_PORT_HOSTED` and the EM $A_w/U, W$ scaffold retirement land at Stage 027, which also ASSEMBLES the full `origin_ok` from the 025 + 026 certificates — 026 asserts only the lineage predicates + the baseline dependency witness, not `tags` / `source_map_complete` (025's) or a full `origin_ok` (027's). *Consumed:* Stage-024's $N_{0,\mathrm{den}}$ (the subject; the checkable host contract; $\rho \equiv \rho_{\mathrm{eff}}$) + Stage-008's M_0/D_1 (the ℓ_0/ℓ_1 ancestor moments) + Stage-009/010's projected-continuity operator + Stage-005 c_s + a `CONV` — all provenance. Register: **zero** new counted knobs; new structural edge **R45** — the density-port continuity-lineage validation (I_{25} a genuine $\int Y_2^* S_{\mathrm{leak}}$ moment descended from pathA_29's operator, computed over the exact lexical tokens + the earning gate), scoped as validation of the cited lineage certificate (not a re-derivation of the upstream operator); discharges nothing at the knob level, like R43/R44. It discharges two forward references: the I_{25} row's "lineage → 026" note (now validated; magnitude stays SIM_DEFERRED) and Stage-025's conditional `moment_valid` = True. No new dims (the $[N_{0,\mathrm{den}}] = L^{-1}M$ check is 027's). Part-II counted CALIB set unchanged at $\{\mu_\eta, T_w, \beta\}(013)$ + $\{V_{p0}/\ell_c\}(015)$ + $\{T_\Omega, \beta_2\}(017) = 6$.

Verification. Dual-engine, independent symbolic routes, zero file I/O — pathA_43 was already contract-clean, so the reshape "sever the bridge" step is a no-op; the work was decomposition + the exact-token genuineness UPGRADE + the `.wl` RE-AUTHOR + the LOCAL/PARTIAL framing. *Like Stage 025 (and unlike Stage 024's keep-native derivation lane), Stage 026 RE-AUTHORS its `.wl`:* the source `.wl` lineage machinery was a transliteration of the `.py`, so the mirror policy required a genuinely independent computation — the decisive gates use a native `Graph` + `FindPath` + `FoldList` $\ell_0 \to \ell_1 \to \ell_2$ ancestry walk whose terminal ℓ_2 node produces the earned moment I_{25} only after the full walk validates (with `StringCases`[s, (`WordCharacter` | "_")..] + `SubsetQ` exact-token subsets), materially different from the `.py`'s flat token-set conjunction. Transcript-level agreement, neither engine reading the other. SymPy 27 PASS / Mathematica 27 PASS, both exit 0, CWD-independent (repo root + `/tmp`). The directive review used the Codex→Grok→Codex bookend, which caught a directive-level error at every leg: a Codex xhigh pass folded 5 BLOCKING (the substring→exact-token upgrade; the decoy-flag self-enforcement; the subsumed dependency binding → de-counted witness; the source-faithful `moment_valid` ≡ `lineage_valid` semantics; the `continuity_interface`∈tags removal), a Grok-4.5 compute-verify pass caught 1 BLOCKING (the load-bearing [`A-Za-z0-9_`]+ token alphabet), a Codex confirm-pass caught 1 BLOCKING (mandatory tooth G) + 2 nits, a Codex final-confirm caught 1 BLOCKING (scoping the de-count to the first-arm witness), and a Codex final re-confirm returned `DIRECTIVE_CLEAN`. Full tri-review on fresh agents: `FIDELITY_CLEAN` (an independent read re-derived $N_{0,\mathrm{den}}$ from Stage-024's 2×2 inverse, diff = 0; verified the exact-token tokenizer, the source-faithful G_2, no leakage, and the genuine independent `.wl`) and `ADVERSARIAL_CLEAN` (per-tooth ablation on both engines — 22 mutations, each fired at its own assert; the three decisive proofs held) — one [**minor**] non-blocking subsumed tally (`baseline continuity_dependency_ok is True`) was de-counted to a labeled diagnostic print (tally 28 → 27 per engine, no other check touched), then fresh-agent `REVERIFY_CLEAN`. Symbolic per-tooth ablation, mutations on copies.

B.22 Stage 027: The Density-Port Checks + Closure (Check II-P4)

Part / anchor. Part II – Gravity (the frozen-throat $\ell = 2$ radiative-port sector, Cluster C; the COMPLETING leg of a four-way split of pathA_43 — the port checks + closure component that OWNS the joint `DENSITY_PORT_HOSTED`, with the density two-port DERIVATION = Stage 024, the vector-freedom taint proof = Stage 025, and the continuity-lineage token-check = Stage 026)

Claim status. Exact Within Closure

Purpose. Run the six able-to-fail port checks + the $\bar{K}$ closure over the density-port numerator $N_{0,\mathrm{den}}$ (derived at Stage 024, proven vector-free at Stage 025, continuity-lineaged at Stage 026): the dimension

$[N_{0,\mathrm{den}}] = L^{-1}M$, the a^{-5} scaling $P_0^{\mathrm{phys}} = (c_s/a)^2 N_{0,\mathrm{den}}/D_0$ with a-power -5, the outgoing DtN sign $+i\,z^5/27$ ($\chi_Q = +1$), a nonzero port, the `deferred_scalar` SIM-inconclusive branch, and the $\bar{K}$ closure ($\bar{K}_4 - 4\bar{K}_2^2/\bar{K}_0 = 0$, $\bar{\Gamma}_5 - 2G/(5c^5) = 0$). Assemble the joint `origin_ok/DENSITY_PORT_HOSTED` from the 025 (vector-freedom) + 026 (continuity-lineage) certificates + these checks + a 027-local subject binding, and record the EM $A_w/U, W$ vector-scaffold retirement with the completed joint.

Status. Exact Within Closure — exact symbolic dimension-vector (L, M, T) algebra, spherical-Hankel series, and exact rationals (`expect_bool`/`expect_zero` residuals, no `scipy`/`numpy`/floats/tolerances). The verdict is the joint `DENSITY_PORT_HOSTED`, which **this stage OWNS** (the completing leg; the Stage-023 pattern, unlike the partial legs 024/025/026). `DENSITY_PORT_HOSTED`. **★★ COMPLETE ≠ PASS: the joint stays CALIBRATED not PASS** — G =`GENUINE_BLOCKED` (020's provenance; 54/5 and $\bar{\Gamma}_5$ =external_bridge_input, the 27 =018's derived_in_gate); the magnitude is SIM_DEFERRED, only the STRUCTURE is hosted. It adds **zero** new counted knobs (a checks+closure/proof slice) and completes the pathA_43 density-port fold (024∧025∧026∧027).

The subject and the six checks. Stage 027 CITES Stage-024's factored export (it does not re-run the inverse); the consumption-integrity oracle computes $N_{0,\mathrm{den}}$`.free_symbols` off the actual cited expression and asserts it equals the 10-symbol host contract $\{a, c_s, \rho_{\mathrm{eff}}, I_{25}, \Xi_Q, \eta_q, \eta_\phi, \varpi_{q2}, \varpi_{\Phi_2}, \lambda_c\}$ ($\{q_2, \Phi_2\}$ are coordinate metadata, not in the numerator). The six checks: (1) **DIM** $[N_{0,\mathrm{den}}] = (-1, 1, 0) = L^{-1}M$ in the (L, M, T) tuple convention (via the `dim_of/BASE_DIMS` engine; ★ the (L, M, T) order differs from the register header's $[L, T, M]$, a cross-convention transfer trap; μ-free discipline: corrupt-$[N_0]$ fires, corrupt-$[G]$ is NO_FAIL since $G \notin$ `free_symbols`). (2) **SCALE** $P_0^{\mathrm{phys}} = (c_s/a)^2 N_{0,\mathrm{den}}/D_0$ has a-power -5; ★ the `scaling` control (a-power -3, moment $I_{\mathrm{wrong}2}$ dim $(2, 0, 0)$) compensates in the L-slot ($2-3 = 5/2-7/2 = -1$) so `dim_ok` stays True and *only* `scaling_wrong` fires — `FAIL_PORT_MALFORMED(scaling)` single-tag. (3) **SIGN** the outgoing $+i\,z^5/27$: `coeff`$(z^5)/i = 1/27$ ($\chi_Q = +1$), computed from $\hat{y}_2 = -3/(z\,h_2'/h_2)$ with $h_2 = j_2 + iy_2$; incoming $\to -1/27$ ($\chi_Q = -1$) $\to$ `FAIL_PORT_MALFORMED(sign)`. (4) **NONZERO** $N_{0,\mathrm{den}} \neq 0$; `zero_coupling` → `FAIL_PORT_VANISHES`. (5) **DEFERRED** the honest SIM-inconclusive branch: uncertified → `PORT_INCONCLUSIVE_SIM_DEFERRED`, proven converse → `DENSITY_PORT_HOSTED`. (6) **CLOSURE** $\bar{K}_4 - 4\bar{K}_2^2/\bar{K}_0 = 0$ and $\bar{\Gamma}_5 - 2G/(5c^5) = 0$ (both standalone), with $\bar{K}_0 = 54Gc_s^5/(5a^5c^5)$, $\bar{K}_2 = 6Gc_s^3/(5a^3c^5)$, $\bar{K}_4 = 8Gc_s/(15ac^5)$, $\bar{\Gamma}_5 = \bar{K}_0 a^5/(27c_s^5) = 2G/(5c^5)$ (Burke–Thorne). The closure is the A3 2.5PN CONSISTENCY, SHARED with Stage 028's full match-back (marked shared, not double-counted); it is a consistency over the CALIBRATED moments, not a first-principles $\bar{\Gamma}_5/G$ derivation.

The joint assembly (027 OWNS the joint) and the conditional retirement. 027 CONSUMES the ATOMIC sibling facts (not a sibling's de-counted composite predicate): 025's taint set (→ `continuity_interface` ∈ `tags` ∧ `vector_port` ∉ `tags`), ¬`vector_host_symbols`, `source_map_complete`, the `vector_free` (P2) certificate (tags are *025's*, over the baseline $N_{0,\mathrm{den}}$, not live-retainted); and 026's `moment_valid` = True, the validated I_{25}, `lineage_valid`. It assembles the 027-LOCAL subject binding `subject_binding` = (`port_moment` ∈ $N_{0,\mathrm{den}}$`.free_symbols`) ∨ ($N_{0,\mathrm{den}} = 0$ ∧ `coupling_zero`) — ★ the vanished-coupling OR-arm is 027-owned (it supersedes the source's outer `vanished_continuity_coupling`), and `port_moment` tracks the current fixture (I_{25} at baseline, $I_{\mathrm{wrong}2}$ under scaling — not 026 re-earning). Then `origin_ok` = (`continuity_interface` ∈ `tags`) ∧ (`vector_port` ∉ `tags`) ∧ (¬`vector_host_symbols`) ∧ `source_map_complete` ∧ `vector_free` ∧ (`lineage_valid`∧`moment_valid`)∧`subject_binding`, and the verdict chain yields `DENSITY_PORT_HOSTED` on all-true incl closure. With the landed joint, 027 records the EM $A_w/U, W$ vector-scaffold RETIREMENT + the pathA_43 diagnostic sliver CLOSED — ★ CONDITIONAL (R43/R44 deferred it here; a

FAIL/inconclusive does not record it; the port has not displaced the scaffold).

The able-to-fail battery (per-tooth ablation, coupling meta-test). Each tooth fires at its OWN named assert and goes vacuous when neutered (verified by the adversarial per-tooth ablation matrix on both engines, no vacuous/subsumed/tautological construct): DIM (corrupt $[N_0]$; corrupt-$[G]$ NO_FAIL a genuine scope re-run); SCALE (a-power $-3 \to$ single-tag); SIGN (incoming $\to -1/27$); NONZERO (`zero_coupling` $\to$ `FAIL_PORT_VANISHES`) + the OR-arm meta-test (disable arm 2 $\to$ flips to `FAIL_NOT_DENSITY_DERIVED`); CLOSURE (isolated post-construction $\bar{K}_4 + \delta_4$ / $\bar{\Gamma}_5 + \delta_\Gamma$, not $\bar{K}_0$; a standalone $\bar{\Gamma}_5$ assert added — the source gated it only via `closure_ok`); DEFERRED (both directions); ASSEMBLY (each of the seven atomic sibling facts corrupted separately $\to$ its own `origin_ok` assert; $\wedge$`vector_free` genuine); SUBJECT-INTEGRITY (a same-dimension/a-power foreign symbol $\to$ only the host-contract assert); POSITIVE (baseline hosted, corrupt any of six checks $\to$ flips); RETIREMENT-CONDITIONAL (force a FAIL $\to$ not recorded); and the `.wl` arity/leakage runtime scanners.

Consumed, register. *Consumed:* Stage-024's $N_{0,\mathrm{den}}$ (the subject; the checkable host contract; $\rho \equiv \rho_{\mathrm{eff}}$) + Stage-025's atomic vector-freedom facts + Stage-026's atomic continuity-lineage facts + Stage-018's DtN fingerprint (the sign gate) + Stage-021's μ-free dim machinery (cited) + Stage-020's $54/5/G$ =`GENUINE_BLOCKED` (the closure magnitude, cited) + Stage-005 $c_s + a/c/G/D_0$ — all provenance. Register: **zero** new counted knobs; new structural edge **R46** — the density-port checks + closure + the joint landing + the scaffold-retirement record (discharges nothing at the knob level, like R43/R44/R45). It upgrades the $N_{0,\mathrm{den}}$ row — the $[N_{0,\mathrm{den}}] = L^{-1}M$ dim CHECK now LANDS here (was provenance-only at 024). Part-II counted CALIB set unchanged at $\{\mu_\eta, T_w, \beta\}(013) + \{V_{p0}/\ell_c\}(015) + \{T_\Omega, \beta_2\}(017) = 6$. Exports the completed joint + the $\bar{K}$ moments $\{\bar{K}_0, \bar{K}_2, \bar{K}_4, \bar{\Gamma}_5\}$ to Stage 028.

Verification. Dual-engine, independent symbolic routes, zero file I/O — pathA_43 was already contract-clean, so the reshape "sever the bridge" step is a no-op; the work was decomposition + the joint-owning verdict framing + the `.wl` RE-AUTHOR. *Like Stages 025/026 (and unlike Stage 024's keep-native lane), Stage 027 RE-AUTHORS its `.wl`:* the source `.wl` 027 slice is a transliteration of the `.py`, so the mirror policy required genuinely independent routes — the DtN via native `SphericalHankelH1/H2` + `SeriesCoefficient`; the dimension/scaling via unit-monomial / weight-rescaling + `Exponent`; the closure via dimensionless ratio invariants with `Together/Cancel`; the verdict via an ordered `SelectFirst/Association` dispatch. Transcript-level agreement, neither engine reading the other. SymPy 55 PASS / Mathematica 55 PASS, both exit 0, CWD-independent (repo root + `/tmp`). The directive review used the Codex→Grok→Codex bookend, which caught issues at every leg: a Codex xhigh pass folded 4 BLOCKING (the certificate mis-scoping + the masked vanished-coupling OR-arm → 027-local subject binding; the SUBJECT-INTEGRITY tooth; re-authoring all verdict-bearing `.wl`; isolated $\bar{K}_4/\bar{\Gamma}_5$ mutations), a Grok-4.5 compute-verify pass caught 1 BLOCKING (the hardcoded I_{25} subject binding broke the scaling control → the current-fixture `port_moment`) + 2 nits, a Codex confirm-pass caught 3 BLOCKING consistency-sweep gaps, and a Codex final-confirm returned `DIRECTIVE_CLEAN`. Full tri-review on fresh agents: `FIDELITY_CLEAN` (independent re-derivation of $N_{0,\mathrm{den}}$ from the 2×2 inverse, the DtN $\pm 1/27$ via SymPy's own j_n/y_n, both $\bar{K}$ residuals = 0, the scaling single-tag; the `.wl` genuinely independent; the consumption matching the sibling exports) and `ADVERSARIAL_CLEAN` (the full per-tooth ablation matrix, every tooth load-bearing, no vacuous construct) — **zero remediation**.

B.23 Stage 028: The 2.5PN Burke–Thorne Match-Back (Check II-P5)

Part / anchor. Part II – Gravity (the frozen-throat $\ell = 2$ radiative-port sector, Cluster C; the CONSISTENCY cap of the pathA_43 density-port fold — the 2.5PN Burke–Thorne match-back over the port's calibrated closure moments, with the density two-port DERIVATION = Stage 024, the vector-freedom taint proof = Stage 025, the continuity-lineage token-check = Stage 026, and the port checks + closure = Stage 027)

Claim status. Exact Within Closure

Purpose. Verify — dual-engine, exact-rational, able-to-fail — that the density-mode $\ell = 2$ port's CALIBRATED closure moments $\{\bar{K}_0, \bar{K}_2, \bar{K}_4\}$ reproduce the 2.5PN Burke–Thorne normalization $\bar{\Gamma}_5 = 2G/(5c^5)$ and the moment invariant $\bar{K}_4 = 4\bar{K}_2^2/\bar{K}_0$ that the separate audited corpus `research/4d_2_5pn` left as its single open item. Five invariants INV1–INV5 (each reduced to 0) + an 11-mutation caught-by matrix, over $\bar{K}_0 = 54Gc_s^5/(5a^5c^5)$, $\bar{K}_2 = 6Gc_s^3/(5a^3c^5)$, $\bar{K}_4 = 8Gc_s/(15ac^5)$.

Status. Exact Within Closure — exact-rational residuals in moment/z-space (`expect_zero`/`expect_bool`, a no-float guard over raw and reduced residuals; no `scipy`/`numpy`/floats/tolerances). The verdict is the LOCAL consistency gate `MATCHBACK_CONSISTENT` (baseline all-zero $\wedge$ every one of the 11 mutation caught-by rows = expected $\wedge$ no-float). `MATCHBACK_CONSISTENT.` ★★ There is **no** `FAIL_*/PASS` PHYSICS token — this is a CONSISTENCY artifact. **CALIBRATED consistency, NOT a first-principles $\bar{\Gamma}_5/G$ derivation**: the $\bar{K}$ moments are hardcoded calibrated closure inputs (G =`GENUINE_BLOCKED`, 020's provenance; 54/5 and $\bar{\Gamma}_5$ =external_bridge_input, the 27 =018's derived_in_gate); the full 1PN→4PN from-throat re-derivation stays SIM-DEFERRED (Gate 6). It adds **zero** new counted knobs and caps the pathA_43 density-port fold (024∧025∧026∧027∧028).

The corpus open item and the five invariants. `research/4d_2_5pn` reduced the 2.5PN sector to one scalar normalization $\hat{m}_0^2\Gamma_5 = 2G/(5c^5)$ (equivalently $\bar{K}_4\bar{K}_0 = 4\bar{K}_2^2$ and $\bar{\Gamma}_5 = 9\bar{K}_2^{5/2}/\bar{K}_0^{3/2}$) and left it open. Stage 028 checks: **INV1** $\bar{K}_4\bar{K}_0 - 4\bar{K}_2^2 = 0$ (the corpus moment invariant; $4\bar{K}_2^2/\bar{K}_0 = (8/15)Gc_s/(ac^5) = \bar{K}_4$); **INV2** $\bar{K}_0a^5/(27c_s^5) - 2G/(5c^5) = 0$ (the pathA_43 form → Burke–Thorne; $\bar{\Gamma}_5 = 54G/(5{\cdot}27\,c^5) = 2G/(5c^5)$, the /27 = 018's earned density-Hankel fingerprint); **INV3** $9\bar{K}_2^{5/2}/\bar{K}_0^{3/2} - 2G/(5c^5) = 0$ (the corpus's OWN form → BT; $9\,(6/5)^{5/2}/(54/5)^{3/2} = 2/5$ via the $\sqrt{6}$ cancellation with $54 = 9 \cdot 6$) — ★ 028's genuinely-new tie-out that Stage 027's closure does not carry; **INV4** path_form − corpus_form = 0 (★ a REDUNDANT diagnostic ≡ INV2−INV3, not an independent tooth); **INV5** the eight INDEPENDENT literal anchors $\{54/5, 6/5, 8/15, 2/5, 27, 9, 5/2, 3/2\}$.

The A3 boundary and the coherent-rescale anti-rig. INV1 and INV2 ARE Stage-027's two port-closure residuals re-expressed (INV1 = 027-residual-1 $\times\bar{K}_0$; INV2 = 027-residual-2 exactly) — the A3 SLOT Stage 027 OWNS (R46), SHARED not double-counted; INV3/INV4/INV5 and the coherent-rescale matrix are 028's additions (R47). 028 CONSUMES 027's K-moments as PROVENANCE (restated locally, self-contained; an authoring-time A3 fidelity comparison confirms they equal 027's `closure_overlay` exports — a stale-citation guard, not a runtime tie). ★★ **The anti-rig:** the INV5 anchor literals are 028's OWN independently-restated constants, computed nowhere from the moment coefficients — they are the *only* residuals that catch the coherent-rescale mutation $\{\bar{K}_0, \bar{K}_2, \bar{K}_4, \mathrm{BT}\} \times 2$ (which passes INV1–INV4 by scale-covariance). Dedupe them and the rig passes silently (the per-gate trip-up).

The able-to-fail battery (per-tooth ablation). The GENUINE independent EXIT-1 teeth = the 11 caught-by row assertions (row 1 = coherent-rescale → the four INV5 anchors only; row 2 = the coupled mutation → {INV2,INV3,INV5_K0,INV5_K2,INV5_K4}; rows 3–11 = the nine single-parameter mutations) + the no-float guard — each ablated actual-side (the mutation fixture / a residual construction, the `EXPECTED_CAUGHT_BY` oracle held immutable) to EXIT 1 at its own named assert, with a coupling meta-test (neuter the planted drift → the expected failure disappears → the harness fails). The `.wl`-independence + honest-scope are review/acceptance gates (not EXIT-1 teeth); baseline-all-zero, the `actual`-non-empty check, and INV4 (≡ INV2−INV3) are de-counted (retained for fidelity/transcript).

Consumed, register. *Consumed:* Stage-027's $\bar{K}$ moments $\{\bar{K}_0, \bar{K}_2, \bar{K}_4, \bar{\Gamma}_5\}$ (the A3 SLOT, INV1/INV2 SHARED with R46) + Stage-018's 27 fingerprint + Stage-020's $54/5/G$ =`GENUINE_BLOCKED` + the corpus's own form $9\bar{K}_2^{5/2}/\bar{K}_0^{3/2}$ (`4d_2_5pn.tex:469`, INV3, imported) + c/a — all provenance, zero file I/O. Register: **zero** new counted knobs, zero new dims; new structural edge **R47** — the full 2.5PN Burke–Thorne match-back consistency, SHARED with R46 (the port-closure consistency counted once; 028 adds INV3/INV4/INV5 + the coherent-rescale matrix). Part-II counted CALIB set unchanged at $\{\mu_\eta, T_w, \beta\}$(013) + $\{V_{p0}/\ell_c\}$ (015) + $\{T_\Omega, \beta_2\}$(017) = 6.

Verification. Dual-engine, independent symbolic routes, zero file I/O — pathA_2_5pn_matchback was already contract-clean, so the reshape "sever the bridge" step is a no-op; the work was rename + LOCAL-ledger idioms + the A3 provenance citation + the honest-scope framing + the `.wl` RE-AUTHOR. *Like Stages 025/026/027, Stage 028 RE-AUTHORS its `.wl`:* the source `.wl` was a transliteration of the `.py`, so the mirror policy required genuinely independent routes — INV1 via a `GroebnerBasis` polynomial-identity test; INV2–INV5 via `PossibleZeroQ` over positive-domain `Refine/PowerExpand`; the mutations replace complete moment expressions directly (no coefficient/config pipeline); the caught-by rows via `Pick`; machine-real detection via `FreeQ[...,_Real]`. Transcript-level agreement, neither engine reading the other. SymPy 12 / Mathematica 12 counted EXIT-1 teeth, both exit 0, CWD-independent (repo root + `/tmp`, byte-identical transcripts). The directive review used the Codex→Grok→Codex bookend, which caught issues at every leg (Codex 2 BLOCKING + 2 nits → Grok-4.5 compute-verify `DIRECTIVE_CLEAN` → Codex confirm 2 BLOCKING + 1 nit → final-confirm 1 BLOCKING → `DIRECTIVE_CLEAN`). Full tri-review on fresh agents: `FIDELITY_CLEAN` + `ADVERSARIAL_CLEAN`.

B.24 Stage 029: The Post-Newtonian Corpus DOI-Cite (Check II-P6)

Part / anchor. Part II – Gravity (Cluster C, the CITE-ONLY provenance cap). The audited external post-Newtonian ladder (1PN→4PN + 2.5PN) — seven separately-published, Zenodo-DOI'd papers under `research/` — recorded by exact DOI so the by-DOI imports in Stages 020/027/028 have visible in-Part provenance. The LAST Part-II gravity stage; after it the sector CLOSES and the scheduled MIDWAY KNOB AUDIT runs.

Claim status. CITED (EXTERNAL CORPUS)

Purpose. Record — as visible in-Part provenance, NOT a re-derivation — the audited external PN ladder the ledger's gravity sector calibrate-predicts against, so the by-DOI imports in Stage 020 (54/5, G =`GENUINE_BLOCKED`) and Stages 027/028 (the 2.5PN Burke–Thorne match-back) are backed by an in-Part citation record (completeness standard #3: cited by DOI, not inlined).

Status. Cite-only: no `scripts/` or `mathematica/` audit, no executable audit at all. The LOCAL landing is `PN_CORPUS_CITED` — a *documentation* landing, *not* a `FAIL_*/PASS` verdict. Adds **zero** new counted knobs and **zero** new dims (the PN results are imported external facts, not ledger parameters). Coverage class: *No executable audit* (the first such stage in the rebuilt ledger). ★★ Stage 029 does **not** re-derive, re-audit, or re-verify any PN result — that audit lives in the seven corpus papers; 029 is provenance only. The ledger's own gravity earning is Stages 001–028.

Corpus provenance. The seven audited PN-ladder papers, cited by Zenodo DOI. All titles confirmed against `README.md` (the sole authoritative DOI source; see the caveat below):

- `10.5281/zenodo.19449058` — *Newtonian and 1PN Orbital Dynamics from a Superfluid Defect Toy Model* (`research/1pn_orbital_dynamics`) — 0PN Newtonian + 1PN perihelion precession; Static-Limit Theorem; $\beta = \kappa_\rho + \kappa_{\rm add} + \kappa_{\rm PV} = 3$ ($\kappa_{\rm PV} = 3/2$ fixed by 1PN GR-matching, not derived there).
- `10.5281/zenodo.19449653` — *Deriving Key Post-Newtonian Coefficients from the Unified 4D Toy Model* (`research/4d_1pn_bridge`) — 1PN coefficient bridge: $\kappa_{\rm add} = 1/2$, EOS exponent $n = 5$, $\alpha^2 = 3/4$, $K_{\rm vec} = 2/\pi^2$, universal v^4 coefficient 3/8.
- `10.5281/zenodo.19450102` — *A Controlled Derivation of the Full First Post-Newtonian Sector from the Unified 4D Toy Model* (`research/4d_1pn_full`) — full conservative 1PN EIH Lagrangian; $\beta_{\rm 1PN} = 3$, EIH cross-coefficients $C_\parallel = -7/2$, $C_L = -1/2$; matches standard EIH + perihelion shift.
- `10.5281/zenodo.19450284` — *A Controlled Derivation of the Full Conservative Second Post-Newtonian Sector from the Unified 4D Toy Model* (`research/4d_2pn`) — full conservative 2PN (order c^{-4}); Legendre-transforms exactly to the standard conservative ADM Hamiltonian through 2PN; $\lambda_\rho = 1/2$.
- `10.5281/zenodo.19492270` — *A Conditional Derivation of the Point-Particle 2.5PN Sector from the Unified 4D Toy Model* (`research/4d_2_5pn`) — *conditional* 2.5PN Burke–Thorne / Iyer–Will radiation-reaction; its open normalization $\hat m_0^2 \Gamma_5 = 2G/(5c^5)$ is met at reduced-closure by Stage 028.
- `10.5281/zenodo.19501724` — *A Full Conservative Derivation of the Two-Body 3PN Sector from the Unified 4D Toy Model* (`research/4d_3pn`) — full conservative 3PN (order c^{-6}); repair-ledger $(\mu_{\rho,3}, d_3, s_{24}) = (1/4, -45/4, -1/16)$, $\Delta H_3 = -\Delta L_3$.
- `10.5281/zenodo.19561056` — *A Conditional Full Conservative Derivation of the Two-Body 4PN Sector from the Unified 4D Toy Model* (`research/4d_4pn`) — *conditional* 4PN (local + tail); the tail coefficient $C_{\rm tail} = (GM/2c^3)\gamma^{\rm eff}_{\rm quad}$ ties to the same 2.5PN quadrupole-normalization gap.

The seven descend from the shared root *4D Toy Model — Action, Projections, and Controlled Brane Limits* (`10.5281/zenodo.19449589`), which the two 4D 1PN papers cite explicitly; it is not one of the seven PN-ladder dirs and is named here only as the derivation-chain root.

Provenance caveat. `README.md` (16 DOI-bearing bullets) is the *sole* authoritative source for these DOIs. None of the seven papers declares its own DOI in source (the two 4D 1PN papers cite only the foundational `19449589`; the other five carry no DOI). So there is no self-declared DOI to conflict with the README (zero mismatches), and the cross-checks are exact title-match (all seven agree) + both 4D 1PN papers citing the shared root. The DOIs are Zenodo-registration facts, cited from the README as authoritative, *not* represented as source-verified.

The 2.5PN line-map (backs Stage 028's INV3). In `research/4d_2_5pn/paper/4d_2_5pn.tex` (display block L467–474): L469 is the moment invariant $\bar{K}_4\bar{K}_0 = 4\bar{K}_2^2$; L471–473 the corpus form $\bar{\Gamma}_5 = 9\,\bar{K}_2^{5/2}/\bar{K}_0^{3/2}$ (Stage 028's INV3); the open normalization $\hat{m}_0^2\Gamma_5 = 2G/(5c^5)$ at L60/L822, $\gamma_{\rm GR} = 2G/(5c^5)$ at L797, the Burke–Thorne force with coefficient $-2G/(5c^5)$ at L793/L961. Stage 028's card cites a blanket `:469` for the corpus form; this stage tightens it to L471–473 (without editing the committed Stage 028).

Scope (honest). The imported $54/5$ / $\bar{\Gamma}_5$ / $2G/(5c^5)$ normalizations are `external_bridge_input` with $G =$`GENUINE_BLOCKED`: the PDE delivers the form/branch, not G or the Burke–Thorne magnitude from first principles (the full 1PN→4PN from-throat re-derivation stays SIM-DEFERRED, Gate 6). Framed strictly as a toy analog: a superfluid-defect toy model reproducing GR-like PN two-body dynamics through 4PN + 2.5PN radiation reaction, cited by DOI.

Sector close. Stage 029 closes Cluster C (Stages 024–029) and the entire Part II gravity sector; the scheduled MIDWAY KNOB AUDIT (the Parts I–II codimension dry-run) follows. The PN results here are imported external facts; the corpus's own internal coefficients $(\kappa_\rho, \kappa_{\rm add}, \kappa_{\rm PV}, \lambda_\rho, \mu_{\rho,3}, \ldots)$ are the corpus papers' calibrations, *not* ledger knobs.

Appendix C

Part III Stage Cards

C.1 Stage 003: Transverse Photon Pair and the Stray Longitudinal Mode

Part / anchor. Part III – Light

Claim status. OPEN

Purpose. Test whether the shear-surface brane carries light of the Maxwell kind. The transverse question earns a clean result – two massless photons at $c_\gamma^2 = \mu_R/\rho_{br}$; the longitudinal question earns a characterized departure – one stray second-class mode where Maxwell has a gauge-removed zero, with the tuned Maxwell locus reachable only by tuning.

Status. OPEN. FAIL-headline stage: the top-line verdict is a `FAIL_*`, but the earned content is the headline and the FAIL is the first-class characterized landing (no softening). *Earned headline:* the brane carries *light* – two massless transverse photons, with (for the first time in the program) $c_\gamma^2 = \mu_R/\rho_{br}$; verdict `PASS_TRANSVERSE_UNDISTURBED`. *Top-line departure:* `FAIL_CAUCHY_STRAY_LONGITUDINAL` – on the provenance-fixed single-medium branch the $(\nabla\cdot u) + \theta$ sector carries one stray propagating longitudinal DOF, a Dirac–Bergmann *second-class* pair rather than a Maxwell first-class Gauss gauge-removal. *Reachability:* the tuned Maxwell locus ($K_\theta = C_J^2/\rho_{br}$, $B_{\text{eff}} = 0$, $m_\theta^2 = 0$) is reachable and emits `C5_RESOLVED_MAXWELL_BY_TUNING` – so the FAIL is not hardcoded – but `BY_TUNING`, not `WITH_PROVENANCE`.

Derivation. The primitive brane Lagrangian (no pre-completed square used as input) is

$$L = \tfrac{1}{2}\rho_{br}(\partial_t u)^2 - \tfrac{1}{2}\mu_R(\nabla\times u)^2 - \tfrac{1}{2}B(\nabla\cdot u)^2 + J(\partial_t\theta)\,\delta\rho_B + \tfrac{1}{2}K_\theta(\nabla\theta)^2.$$

Transverse (earned). For a polarization $\perp$ $\mathbf{k}$, $\nabla\cdot u = 0$ and the θ couplings vanish, so $L_T = \frac{1}{2}\rho_{br}(\partial_t u_T)^2 - \frac{1}{2}\mu_R k^2 u_T^2$, giving

$$\omega^2 = \frac{\mu_R}{\rho_{br}}\,k^2, \qquad c_\gamma^2 = \frac{\mu_R}{\rho_{br}},$$

two polarizations $\Rightarrow$ physical_dof = 2, massless. The speed is able-to-fail: $\epsilon = 2\rho_{br}$ shifts it to $\mu_R/(2\rho_{br})$ (`FAIL_TRANSVERSE_DISTURBED`). **Longitudinal (departure).** Number conservation slaves $\delta\rho_B = -\rho_{B0}\nabla\cdot u$; integration by parts gives the Maxwell-form cross-term with the *derived* sign $C_J = -J\rho_{B0}$, and the finite conjugate-density stiffness the Cauchy modulus $B_{\text{eff}} = \rho_{B0}^2/\chi_c$. Since $\partial_t\theta$ enters only through the first-order cross-term, the momenta are $p_u = \rho_{br}\partial_t u_L$, $\pi_\theta = Jk\rho_{B0}u_L$, forcing the primary constraint $\Phi_1 = \pi_\theta - Jk\rho_{B0}u_L$; preservation gives $\Phi_2 = -k(Jp_u\rho_{B0} + k\kappa_{\text{phase}}\rho_{br}\theta)/\rho_{br}$, and

$$\{\Phi_1, \Phi_2\} = \frac{k^2(J^2\rho_{B0}^2 + \kappa_{\text{phase}}\rho_{br})}{\rho_{br}} \neq 0$$

$\Rightarrow$ second-class pair (0 first-class, 2 second-class), $N_{\text{phys}} = (4-2)/2 = 1$. The finite-stiffness pole $\omega^2 = k^2\kappa_{\text{phase}}\rho_{B0}^2/[\chi_c(J^2\rho_{B0}^2 + \kappa_{\text{phase}}\rho_{br})]$ has positive residue and a bounded reduced Hamiltonian: a real stray longitudinal mode, not a ghost and not a gauge-removal.

Physical use, fidelity, and the departure. The brane carries light: two transverse photons at $c_\gamma^2 = \mu_R/\rho_{br}$ from the medium's own moduli (light = in-plane brane shear). Honest scope: the order-parameter phase θ does *not*, on its frozen definitions, supply a gauge-removed Maxwell scalar – it leaves one stray second-class longitudinal mode; the obstruction is pinned to the Cauchy modulus $B_{\text{eff}} = \rho_{B0}^2/\chi_c$ plus the conventional phase-gradient sign $K_\theta = -\kappa_{\text{phase}} < 0$ (vs the electric sign $K_\theta = C_J^2/\rho_{br} > 0$). Fidelity: the decisive sign $C_J = -J\rho_{B0}$ is *derived* here by an in-script integration by parts (the earned source had asserted it), closing that stage's known `FIDELITY_ISSUES`; the conventional K_θ sign is a labeled postulate whose verdict-dependence is exercised by the reachable Maxwell-locus control. Resolution of the stray mode is calibrated / sim-deferred; μ_R, ρ_{br} are calibrated inputs.

Appendix D

Part IV Stage Cards

D.1 Stage 030: Electric Scalar Block + Localized-H Closure

Part / anchor. Part IV – Charge (IV-1; first Part-IV stage)

Claim status. EXACT WITHIN CLOSURE

Purpose. Formal ledger home of the charge sector's distinct electric scalar: a genuinely *localized* parent field H on the brane, its unique *gapless* propagating mode h, and the *stable* two-field (u_L, h) kernel it couples to. This stage is the static *substrate* the charge sector stands on — it contains no force, no sign, and no mouth source. It is a **LIGHT-with-scoping** reshape of the electric-scalar subset of the monolithic G0 closure checker (`software/em_charge_attribute/g0_closure_card_v0_checks.py / .wl`, source card `g0_closure_card_v0.md` §§2.3–2.4); surviving verdict token `ELECTRIC_SCALAR_CLOSURE_STATIC`. Every predicate is a *static prerequisite* for the assembled Green mode, not a solved assertion of it — the assembled one-/two-body/far-field claims are Part VII, and the mouth source is Stage 031.

Status. EXACT WITHIN CLOSURE. **The entire G0-v0 closure is a labeled `DRAFT v0` POSTULATED closure — the choice of Pöschl–Teller operator, the R_w-odd parity representation of $\{H, h, u_L\}$, and every numeric witness value are postulates, not derivations.** What is **EARNED** (exact within that declared closure) is that, *given* this action, the static prerequisites hold: the nonnegative parent operator $O_\perp = A^\dagger A$; the unique normalizable *zero* mode f_0 below a positive continuum gap $4/\ell^2$; the zero-mode norm, projection, and idempotence; the three reduction identities; static kernel positivity $D > 0$ with two positive squared wave speeds; reduced-h masslessness $Q_s(0, 0) = 0$; conservative Hessian symmetry; and the units-restored $[L, T, M]$ dimensional homogeneity of every electric-scalar-block term. Ledger-local earned label (not a source token): `ELECTRIC_SCALAR_STATIC_PREREQUISITES_VERIFIED`. Charge is the static $\pm w$ throat — a particle punctures the brane into $\pm w$, the bend is the geometric field $\xi_w = \ell\, h$, and h is the localized zero mode assembled here; the puncture-deflection *mechanism* (the mouth $\chi \leftrightarrow h$ source, the $1/R^2$ force, the honest `R1_REQUIRED(bc_selection)` landing) is Stage 031, which *dresses* this substrate.

The localized parent H and its Pöschl–Teller operator (EARNED spectral structure). The distinct electric scalar is a parent field $H(x, w, t)$ of dimension L^{-1} living in $y = (x, w) \in \mathbb{R}^3 \times \mathbb{R}$, with transverse operator and action

$$O_\perp = -\partial_w^2 + \frac{4 - 6\,\mathrm{sech}^2(w/\ell)}{\ell^2}, \qquad S_H = \tfrac{1}{2}\int dt\, d^4y\, \{\, 2M_4(\partial_t H)^2 - 2K_4[\, |\nabla_x H|^2 + H\, O_\perp H\,]\}.$$

With the superpotential $A = \partial_w + 2\tanh(w/\ell)/\ell$ both engines verify term-by-term that $O_\perp = A^\dagger A$ with $A^\dagger = -\partial_w + 2\tanh(w/\ell)/\ell$ (`PASS_TRANSVERSE_FACTORIZATION`), so $O_\perp \geq 0$. Direct substitution gives the zero mode $O_\perp f_0 = 0$, $f_0(w) = 1/[\ell\cosh^2(w/\ell)]$ (`PASS_PARENT_TRANSVERSE_ZERO_EIGENVALUE`). Because $O_\perp = A^\dagger A$, the zero mode is exactly $\ker A$; $Af = 0$ is a *first-order* ODE with a one-dimensional solution space, and the engines confirm the integrating-factor solution's ratio to f_0 is w-independent ($= \ell$) with finite positive L^2 norm, so $\ker A = \mathrm{span}\{f_0\}$ is the *unique* normalizable zero mode (`PASS_UNIQUE_NORMALIZABLE_KERNEL`). As $w \to \pm\infty$ the potential $\to 4/\ell^2$ (symbolic `limit` both directions), so the continuum threshold is $4/\ell^2 > 0$ and the zero mode sits a *positive gap* below it (`PASS_POSITIVE_CONTINUUM_GAP`). *Scope (no over-claim):* the engines claim a unique normalizable zero mode below a continuum starting at $4/\ell^2$; they do not enumerate the full spectrum (this $s = 2$ well additionally supports one excited bound state at $3/\ell^2$, an analytic PT aside neither computed nor relied on here).

The zero-mode reduction and its relations (EARNED; the two c_E-teeth are genuinely able-to-fail). With the parent inner weight 2 fixed by S_H, the closed-form norm is $N_0 = \int dw\, 2f_0^2 = 8/(3\ell) > 0$ (`PASS_ZERO_MODE_NORM`). Writing $H = f_0 h + H_\perp$ with $\int dw\, 2f_0 H_\perp = 0$, the reduced mode is the normalized zero-mode projection $h = P_0 H := N_0^{-1}\int dw\, 2f_0 H$, for which the engines verify $P_0 f_0 = 1$, $P_0(f_0 h) = h$, and idempotence $P_0^2 = P_0$, with $[h] = 1$ (`PASS_NORMALIZED_ZERO_MODE_PROJECTION`). The zero-mode reduction of S_H yields, each as its own dual-engine predicate, the reduced inertia $M_h = N_0 M_4$ (`PASS_REDUCED_H_INERTIA`, witness $M_h^* = 1$) and stiffness $K_h = N_0 K_4$ (`PASS_REDUCED_H_STIFFNESS`, $K_h^* = 1$), with physical positivity $M_4 N_0 > 0$ (`PASS_PHYSICAL_H_NORM`). **On the two c_E-relation teeth (not tautological):** the *value* relations $K_4 = M_4 c_E^2$ and $K_h = M_h c_E^2$ are *definitional* ($\{M_4, c_E\}$ are inputs, so K_4 is *defined* as $M_4 c_E^2$); a value self-comparison would be a vacuous $X \equiv X$. So the teeth verify their genuine production-live content instead: `PASS_PARENT_STIFFNESS_DIMENSIONAL_CONSISTENCY` checks $[M_4 c_E^2] = [K_4]$ (fails under a $[K_4]$ dimension mutation), and `PASS_REDUCED_SPEED_PRESERVATION` checks that the *separately* reduced ratio is speed-preserving, $c_E^2 = K_h/M_h = K_4/M_4$, so an inconsistent reduction (e.g. $K_h = N_0^2 K_4$) makes $K_h/M_h = 160/3 \neq 1$ and the tooth fires on a *real* error. *Dependency direction:* M_4 (a postulated `ACTION` primitive) and c_E (input) are the free data; $K_4 = M_4 c_E^2$, $M_h = N_0 M_4$, $K_h = N_0 K_4 = M_h c_E^2$ are all **derived**; the reduced normalization $M_h^* = 1$ is a `CONV` choice imposed *on* M_4.

The coupled (u_L, h) scalar block — two INDEPENDENT positivity facts (EARNED). After eliminating the brane phase θ_B (its Schur shift absorbed into $A_{\text{eff}} = \rho_{br} + C_J^2/\kappa_{\text{phase}}$), the reduced scalar action has inertia matrix $M = \mathrm{diag}(A_{\text{eff}}, M_h)$ and stiffness $K = \left[\begin{smallmatrix} B_{\text{eff}} & C_{hu} \\ C_{hu} & K_h \end{smallmatrix}\right]$. The stage splits kernel health into two teeth that are *genuinely independent* in both engines. (a) *Sylvester positivity* (`PASS_STABILITY`): $B_{\text{eff}} > 0$ and $D = B_{\text{eff}} K_h - C_{hu}^2 > 0$ (physical $[D] = M^2 T^{-4}$), confirmed by SymPy `is_positive_definite` / Wolfram `PositiveDefiniteMatrixQ+Eigenvalues`. (b) *Positive generalized wave speeds* (`PASS_POSITIVE_GENERALIZED_WAVE_SPEEDS`): the squared characteristic speeds $z = c^2/c_E^2$ solve $\det(K - zM) = 0$; in the explicitly-normalized star units ($M^* = I$, $B_{\text{eff}}^* = 2$, $K_h^* = 1$, $C_{hu}^* = 1/2$),

$$\det(K^* - z^* M^*) = z^{*2} - 3z^* + \tfrac{7}{4}, \qquad z_\pm^* = c_\pm^{*2} = \frac{3 \pm \sqrt{2}}{2} > 0,$$

so both coupled modes propagate. The two facts are independent: D depends on K only, while the speeds depend on both K and M; the (b)-tooth mutates the inertia ($A_{\text{eff}}^* : 1 \to 2$), which *keeps* the Sylvester margin $D^* = 7/4$ intact yet *changes* the roots — driving (b) to exit 1 at its own assert while (a) stays green, so the roots check is not a corollary of the margin. (The physical $\det(K - zM)$ is dimensionful and is never equated to the numeric polynomial — the determinant is stated only through the star form.)

Reduced-h masslessness and conservative Hessian symmetry (EARNED). The time-dependent quadratic operator $Q_s(\omega,k) = \left[\begin{smallmatrix} A_{\text{eff}}\omega^2 - B_{\text{eff}}k^2 & -C_{hu}k^2 \\ -C_{hu}k^2 & M_h\omega^2 - K_h k^2 \end{smallmatrix}\right]$ satisfies $Q_s(0,0) = 0$ at star values — there is *no* k^0h^2 term (no bare h mass), the static prerequisite against `FAIL_PINNED_BRANON`/`FAIL_YUKAWA` (`PASS_REDUCED_H_MASSLESSNESS`; the tooth reinstates a k^0h^2 coefficient and $Q_s(0,0) \neq 0$ fires). The gradient-energy Hessian of the (u_L, h) block equals the displayed Euler operator and is symmetric, $\left[\begin{smallmatrix} 2 & 1/2 \\ 1/2 & 1 \end{smallmatrix}\right]$ = its transpose (`PASS_CONSERVATIVE_HESSIAN_SYMMETRY`, the action-derived integrability prerequisite; the tooth desymmetrizes the cross coefficient $1/2 \to 3/4$).

Dimensional firewall (EARNED — units-restored $[L,T,M]$). Every electric-scalar-block term is checked exponent-triple homogeneous, units restored, able-to-fail: the four-dimensional parent-H terms $\{H_{\text{kin}}, H_{\text{grad}}, H_{\text{pot}}\}$ target the *bulk* density E/L^4, the three-dimensional reduced terms $\{u_{\text{kin}}, u_{\text{stiff}}, h_{\text{kin}}, h_{\text{stiff}}, \text{mix}\}$ the *brane* density E/L^3, with $[H] = L^{-1}$, $[u_L] = L$, $[h] = 1$, $[M_4] = M$, $[K_4] = E$, $[A_{\text{eff}}] = ML^{-3}$, $[B_{\text{eff}}] = ML^{-1}T^{-2}$, $[M_h] = ML^{-1}$, $[K_h] = MLT^{-2}$, $[C_{hu}] = MT^{-2}$, $[c_E] = LT^{-1}$. The mix-term tooth corrupts $[C_{hu}]$ ($MT^{-2} \to MT^{-1}$) and the homogeneity residual fires. (The mouth Robin/source-term dimensions travel with the mouth source in Stage 031, not here.)

Scope, deferrals, and first-class departures (recorded honestly). (1) The whole G0-v0 closure is a **POSTULATED `DRAFT v0` closure** — every witness $(M_4^*, c_E^*, B_{\text{eff}}^*, C_{hu}^*, \ldots)$, the Pöschl–Teller operator, and the parity representation are labeled postulates; what is earned is that, given this action, §§1–5 hold. (2) The **mouth SOURCE mechanism is DEFERRED to Stage 031**: `PASS_BARE_MOUTH_SOURCE` (the orientation-odd $\chi \leftrightarrow h$ mouth functional, $f_0(0) = 1/\ell$, the nonzero projected source) is not folded here; Stage 031 must own a *full able-to-fail* reconstruction of it. (3) **Out-of-scope, UNDISCHARGED cross-sector G0 predicates (Part VII de-dup obligation):** the source checker is monolithic across all seven G0 gates; Stage 030 folds *only* the charge-new electric-scalar subset. The remaining G0 predicates (`PASS_PLANAR_WALL_FACTORIZATION`, the bulk-witness/Fourier-Hessian group, the drain-normalization/controller group) are **not** re-folded and **not** already covered by the built Part-I/II stages (those verify physically-adjacent but *different* facts) — they are `OUT-OF-SCOPE / UNDISCHARGED CROSS-SECTOR G0 PREDICATE`, reconciled as a Part VII G0-vs-Part-I de-dup. (4) **Excluded assembled claims → Part VII:** the assembled one-body and two-body/far-field claims (constrained Hessians, the dressed $1/R$ monopole, pair-force integrability) are solved-response objects for Part VII, not discharged here. (5) **c_E has no committed cone lock:** c_E is the electric sector's first dimensional entry and is `FREE-UNREDUCED`; $c_s/c_E = 2$ was chosen expressly to avoid imposing a cone lock, and its relation to c_γ/c_s is re-adjudicated in Part VI (`pathA_40`, `PENDING`).

Verification. Dual-engine, both exit 0, genuinely independent routes: **SymPy 16 PASS / Mathematica 16 PASS**. Standalone, print-only, assert-zero (`raise SystemExit(1)` / `Exit[1]` on failure), no argparse harness, no JSON/YAML payload, zero file-I/O between engines. The `.wl` is not a transliteration: it reorders the derivation (native matrix route first) and uses native Wolfram machinery where the `.py` uses SymPy — `Integrate` for N_0, `DSolve` (counting one integration constant) for the ker A uniqueness, `Limit` for the $4/\ell^2$ threshold, and `PositiveDefiniteMatrixQ`/`CharacteristicPolynomial`/`Eigenvalues` for the coupled kernel. **16 per-tooth ablations, each `FIRED_AT_OWN_ASSERT`** (env switch `LEDGER_STAGE030_MUTATION`): a card-primitive mutation for every folded predicate — including the two *distinct* kernel teeth (C_{hu}^* fires the margin `PASS_STABILITY`; inertia A_{eff}^* fires the roots `PASS_POSITIVE_GENERALIZED_WAVE_SPEEDS`), the genuine reduction teeth (the $[K_4]$ dimension mutation and the $K_h = N_0K_4 \to N_0^2K_4$ reduction-inconsistency mutation), the norm/projection/inertia mutations, the spectral teeth (factorization coefficient, f_0 power, potential asymptote → gap), and the mundane $[C_{hu}]$ dimension tooth. Out-of-scope predicates carry no teeth (dropped, not silently green); the payload is float-free throughout (machine-`Float` atoms raise). **Tri-review CLEAN:** `FIDELITY_CLEAN` + `ADVERSARIAL_CONCERNS(2)` → remediated → fresh-agent `REVERIFY_CLEAN` — the pre-commit adversarial re-verify replaced two originally-tautological c_E-relation value self-checks with the genuine `PASS_PARENT_STIFFNESS_DIMENSIONAL_CONSISTENCY` + `PASS_REDUCED_SPEED_PRESERVATION` teeth above.

D.2 Stage 031: Puncture-Deflection Mechanism — Field Identity, Mouth Source, Exterior $1/R^2$ Form

Part / anchor. Part IV – Charge (IV-2; the EARNED-mechanism stage of the 4-stage split)

Claim status. Exact Within Closure

Purpose. The puncture-deflection *mechanism* that dresses the static charge substrate of Stage 030: the field identity $\xi_w = \ell h$; the orientation-odd mouth source that makes the charge sign the $\pm w$ $\mathbb{Z}_2$ orientation; the exterior stationary field $h(r) = h_A a/r$; the full completed-square response matrix m; and the target-blind neutral far-field shell $A = m_{gg} C$. It stops at the EARNED, target-blind structure — WHICH boundary-condition class holds, and hence the force *sign*, is Stage 032's `R1_REQUIRED(bc_selection)`. A reshape of the puncture-deflection source build (`software/em_charge_attribute/puncture_deflection_electric_sign_result.md` + `check.{py,wl}`) with several genuine new builds — the full bare-mouth reconstruction, the independent generic $I_+ > 0$ certification, the exterior derivation, and the full response matrix. Surviving verdict token `THROAT_H_SOURCE_1_OVER_R2`.

Claim status. Exact Within Closure — EARNED, target-blind, and exact-symbolic *within* the postulated G0 closure. **What is EARNED** (given the postulated G0 action and r_Σ profile class): the field identity $\xi_w = \ell h$ and the reduced coupling $g_{\chi h} = J_m/\ell$; the annulus normalization $\int \eta = 1$; $f_0(0) = 1/\ell$ evaluated (not assigned) from the actual profile; the orientation projection $Q_\chi = s$ through the odd kernel *and* the generic $I_+ > 0$ reflection-dominance certification (not N_χ/N_χ); the nonzero bare forcing $-g_{\chi h} s$; the exterior $h(r) = h_A a/r$ with positive holding curvature; the full response matrix m (κ, Z, m_{uu}, m_{ug}, m_{gg}, $\det m$, star witness) with $m_{gg} \geq 0/\det m \geq 0$ earned via z_g^2; the self-response S_{gg}; and the units-restored $[L, T, M]$ firewall over every mechanism primitive. The $1/R^2$ falloff and the $s_1 s_2$ product of the far-field shell are target-blind EARNED. **POSTULATED** (first-class): the bounded frozen $r_{\Sigma,+} \in (0, 1]$ representative with the $+w$ sleeve of frozen length L_s is a labeled postulate *completing* G0's already-postulated r_Σ class (no new knob class); strict $z_g > 0$ is a POSTULATED Robin-admissibility witness ($z_g = 1$ at G0). **DEFERRED to Stage 032** (`R1_REQUIRED(bc_selection)`): the class numerator C that fixes the force sign, the four BC-ensemble coefficients, `internal_inconsistency` =none, the sealed 23040-cell landing, and the Q_E/magnitude re-home. Ledger-local earned label: `PUNCTURE_DEFLECTION_MECHANISM_TARGET_BLIND`.

The field identity and the bare-mouth source (EARNED). The normal embedding displacement is the reduced scalar, $\xi_w = \ell h$ ($[\xi_w] = L$, $[h] = 1$; `PASS_FIELD_IDENTITY`); with $f_0(0) = 1/\ell$ (consumed from Stage 030) the live coupling reduces $-J_m Q_\chi H \to -g_{\chi h} Q_\chi h$, $g_{\chi h} = J_m/\ell$. The bare mouth source is reconstructed piece-by-piece: the actual annular kernel $\eta_i = 3 \cdot \mathbf{1}_A/[4\pi((a+\ell)^3 - a^3)]$ integrates to 1 (`PASS_ETA_ANNULUS_NORMALIZATION`); $f_0(0) = 1/\ell$ is evaluated from $f_0 = 1/[\ell \cosh^2(w/\ell)]$, the tooth also protecting the sech^2 curvature (`PASS_F0_MOUTH_VALUE_EVALUATED`); $g_{\chi h} = J_m/\ell \neq 0$ (`PASS_REDUCED_COUPLING_NONZERO`). The orientation projection $Q_\chi = s$ is proved in two parts. *(a)* With the odd kernel $o_\ell(w) = w\,e^{-w^2/(2\ell^2)}/(\sqrt{2\pi}\,\ell^3)$, the even r_{bg}, and the body reflection $r_{\Sigma,-}(w) = r_{\Sigma,+}(-w)$, one gets $I_- = -I_+$, so $Q_+ = +1$, $Q_- = -1$ as ratios (`PASS_REFLECTION_ANTISYMMETRY`, `PASS_Q_CHI_ORIENTATION`). *(b)* Oddness alone does not give $I_+ \neq 0$; with a bounded peak-normalized $r_{\Sigma,+} \in (0,1]$ carrying a one-sided sleeve L_s, the reflected denominator difference $2\sinh(2w/\ell)\sinh(L_s/\ell) > 0$ and $D(w) - D(-w) = (r_+ - r_-)(r_+ + r_-) > 0$ for $w > 0$ give

$$I_+ = \int_{w>0} o_\ell(w)\,[D(w) - D(-w)]\,dw > 0$$

parameter-generically (`PASS_I_PLUS_REFLECTION_DOMINANCE`), so $N_\chi = I_+ \neq 0$ is certified before any division (`PASS_N_CHI_NONZERO_GUARD`) and $Q_\chi = s$ follows. The one-sidedness is load-bearing: an even deformation, $L_s = 0$, or a $-w$ sleeve each drive I_+ to 0 or < 0 at their own asserts, and a non-generic family triggers the `STAGE031_STOP: I_plus_not_generic` anti-rescue guard. Varying the live mouth functional yields the h-Euler density $\eta_i(k_m h - g_{\chi h} s_i)$ ($k_m = K_m/\ell^2$) with nonzero bare component $-g_{\chi h} s$ (`PASS_BARE_FORCING_LIVE_VARIATION`).

Exterior solution and the named core-gap fact (EARNED). The exterior stationary field (3D radial Euler, decaying branch, $h(a) = h_A$) is $h(r) = h_A a/r$ with $E_{\text{ext}} = 2\pi\kappa a h_A^2$, where $\kappa > 0$ is the exterior h-gradient stiffness (a generic positive constant in the 031 exterior block, *not* identified with D/B_{eff}), and the holding curvature $4\pi\kappa a > 0$ is positive (`PASS_EXTERIOR_ONE_OVER_R`, `PASS_POSITIVE_HOLDING_CURVATURE`). A nonzero signed h_A is *not* stationary for the core-less energy (only $h_A = 0$ is): this is exported as the named neutral fact `NONZERO_HA_REQUIRES_CORE_HOLDER` (`PASS_NONZERO_HA_REQUIRES_CORE_HOLDER`), consumed by Stage 032; 031 claims no guaranteed-nonzero interaction.

The full completed-square response m (EARNED; consumes 030's D). With $b = B_{\text{eff}}$, $c = C_{hu}$, $D = bK_h - c^2$: the response Robin scale $\kappa = D/b$, $Z = \left[\begin{smallmatrix} 1 & 0 \\ -(c/b)z_b & z_g \end{smallmatrix}\right]$ (typed; lower-left $[\cdot] = L$), $m = Z^\top \operatorname{diag}(b^{-1}, \kappa^{-1}) Z$, giving

$$m_{uu} = \frac{D + c^2 z_b^2}{bD}, \quad m_{ug} = -\frac{c z_b z_g}{D}, \quad m_{gg} = \frac{b z_g^2}{D}, \quad \det m = \frac{z_g^2}{D}.$$

At the star $(b, K_h, c, z_b, z_g) = (2, 1, \frac{1}{2}, 1, 1)$, $D_* = 7/4$ (consumed from 030) and $m_* = \frac{1}{7}\left[\begin{smallmatrix} 4 & -2 \\ -2 & 8 \end{smallmatrix}\right]$ — note $m_{uu} = K_h/D = 4/7 \neq 1/b$, while $m_{ug} = -c/D = -2/7$ and $m_{gg} = b/D = 8/7$ (`PASS_RESPONSE_STAR_WITNESS`). Positivity is honest: $m_{gg} \geq 0/\det m \geq 0$ are EARNED via z_g^2 given $B_{\text{eff}} > 0$, $D > 0$ (`PASS_M_GG_NONNEGATIVE`); strict $z_g > 0$ is a POSTULATED Robin-admissibility witness (`PASS_Z_G_POSTULATED_WITNESS`). The self-response $S_{gg} = \langle \eta, L_h^{-1}\eta \rangle$ has an explicit positive-definite quadratic-form definition (`PASS_S_GG_SELF_RESPONSE`), $[S_{gg}] = E^{-1}$.

The target-blind far-field shell and the units firewall (EARNED). The two-throat interaction has the FORM $U = s_1 s_2 A/(4\pi R)$, $F = s_1 s_2 A/(4\pi R^2)$, $A = m_{gg} C$, $[A] = EL$, $[C] = E^2$ (`PASS_NEUTRAL_FAR_FIELD_FORM`): the $1/R^2$ falloff, the $s_1 s_2$ product, and $[A] = EL$ are target-blind EARNED; the class-dependent numerator C is *not* specified and *no* sign is picked — an allowed/conditional leading form, possibly with zero coefficient (`PASS_ALLOWED_ZERO_COEFFICIENT`); a nonzero amplitude needs the deferred core holder. The complete mechanism dimensional firewall (21 `PASS_UNITS_*`) checks every primitive units-restored and able-to-fail: $[\xi_w] = L$, $[h] = [h_A] = 1$, $[K_m] = ML^4T^{-2}$, $[J_m] = ML^3T^{-2}$, $[k_m] = [g_{\chi h}] = E$, $[\eta] = L^{-3}$, $[o_\ell] = L^{-2}$, $[N_\chi] = L^{-1}$, $[m_{uu}] = L^3/E$, $[m_{ug}] = L^2/E$, $[m_{gg}] = L/E$, $[\kappa] = E/L$, $[Z_{21}] = L$, $[\det m] = M^{-2}T^4$, $[S_{gg}] = E^{-1}$, $[C] = E^2$, $[A] = EL$, $[U] = E$, $[F] = E/L$, with live chains $k_m = K_m/\ell^2$, $g_{\chi h} = J_m/\ell$, $N_\chi = \int \eta \int o_\ell$, $A = m_{gg} C$, $U = A/R$, $F = A/R^2$.

Scope, deferrals, and first-class departures (recorded honestly). (1) EARNED within the postulated G0 closure; the whole G0-v0 closure stays a labeled postulate (Stage 030). (2) The bounded $r_{\Sigma,+}$ representative (with frozen sleeve L_s) is a `[POSTULATE]` completing G0's r_Σ class — no new knob class; the generic $I_+ > 0$ certification is structural to the one-sidedness, not to the magnitude. (3) Strict $z_g > 0$ is POSTULATED; $m_{gg} \geq 0/\det m \geq 0$ are earned via z_g^2. (4) The SIGN R1 (the class numerator C, the four BC ensembles, `internal_inconsistency`, the 23040-cell landing, the R1 verdict) and the magnitude re-home (Q_E; c_a, c_ξ unbounded at tier-A) are Stage 032; a guaranteed-nonzero amplitude needs the sim-deferred core holder (`NONZERO_HA_REQUIRES_CORE_HOLDER`).

Verification. Dual-engine, both exit 0, genuinely independent routes: **SymPy 50 PASS / Mathematica 50 PASS**. Standalone, print-only, assert-zero (`raise SystemExit(1)` / `Exit[1]` on failure), no argparse harness, no JSON/YAML payload, zero file-I/O between engines; float-/machine-real-free payload throughout. The `.wl` is not a transliteration: it reorders the derivation (native response matrix/`Inverse` first, then `DSolve` for $h(r) = h_A a/r$, then the native `Integrate` mouth route last) and uses native Wolfram machinery for the annulus/η normalization, the certified I_+ sleeve integral, and N_χ. **50 per-tooth ablations, each `FIRED_AT_OWN_ASSERT`** (env switch `LEDGER_STAGE031_MUTATION`): a primitive mutation for every folded predicate, including the SEPARATED norm/normalization teeth ($\int \eta$ vs the $N_\chi = 0$ pre-division guard vs the generic $I_+ > 0$ certification), the one-sidedness load-bearing teeth (even-deformation, $L_s = 0$, $-w$-sleeve), the per-entry response teeth, the exterior teeth, the neutral-shell teeth, and the complete units firewall; no vacuous or $X \equiv X$ tooth. **Tri-review CLEAN:** fidelity + adversarial fresh-agent review CLEAN. Verdict `THROAT_H_SOURCE_1_OVER_R2`, earned label `PUNCTURE_DEFLECTION_MECHANISM_TARGET_BLIND`.

D.3 Stage 032: Puncture-Deflection Charge — BC Ensembles, the Force Sign, and the Honest `R1_REQUIRED(bc_selection)` Landing

Part / anchor. Part IV – Charge (IV-3; the R1-LANDING stage of the 4-stage split)

Claim status. Open

Purpose. The R1-landing half of the puncture-deflection charge gate. Stage 031 earned the target-blind far-field STRUCTURE ($F_X = s_1 s_2 A_X / 4\pi R^2$, $A = m_{gg} C$); Stage 032 owns the four boundary-condition ensembles that fix the force *sign*, shows the committed bare model does not select among them, and lands the honest terminal `R1_REQUIRED(bc_selection)`. It selects the class numerator C per admissible mouth BC class (the four A_X), characterizes the MIXED three-regime admissibility, verifies `internal_inconsistency` =none (Q-AMEND consistency), runs the sealed 23040-cell §4 landing ladder recomputed from typed upstream facts, and re-homes Q_E/charge-magnitude as the co-blocker `R1_REQUIRED(magnitude)`. A reshape of the §4 landing table + the Q-BC/Q-AMEND/Q-COMBINE/Q-MAG blocks of the puncture-deflection source build (`software/em_charge_attribute/puncture_deflection_electric_sign_result.md` + `check.{py,wl}` + `independent_recompute.py`). Consumes Stage 031's $\{m, m_{gg}, \det m, S_{gg}, A = m_{gg} C, 1/R^2,$ `NONZERO_HA_REQUIRES_CORE_HOLDER`$\}$ and Stage 030's $\{D, D_* = 7/4\}$ by citation — no mechanism re-derived.

Claim status. Open — an honest R1 landing, *not* an exact closure. The electric SIGN is `R1_REQUIRED(bc_selection)`: **neither EARNED nor CALIBRATED**. **DECIDED** (not R1): the four coefficient formulas $A_V/A_J/A_M/A_{\mathrm{MIXED}}$ *given a class*, and the coupling $g = g_{\chi h}$ (consumed from

031). **R1** (deferred): the class SELECTION among $\{V, M, J, \mathrm{MIXED}\}$ and the ensemble data $\{\phi, q, j, \lambda\}$. The strict signs $A_V > 0$ (repel), $A_J < 0$ (attract), and the strict MIXED regimes are CONDITIONAL on a nondegenerate response ($m_{gg} > 0$ at the $z_g \neq 0$ witness, $S_{gg} > 0$, nonzero magnitudes); weak ($\geq 0/\leq 0$) in general; at $z_g = 0$ every A_X vanishes. The **magnitude** (`R1_REQUIRED(magnitude)`; c_a, c_ξ unbounded at tier-A) and the **variant / MIXED-parameter / REPLACE-vs-ADD** selections are SEPARATE downstream R1 debts — not part of the sign-class R1 and not resolved by picking a class. `internal_inconsistency` =none (charge coexists with gravity; owned + verified here). Resolver: the SIM-DEFERRED nonlinear parent-throat boundary functional / barrier / map $s \mapsto h_A$ (a sibling of gravity's R10/R30/R33).

The four DECIDED conditional ensembles (consuming 031's m_{gg}, S_{gg}). Selecting the class numerator C per admissible mouth BC class, reshaped through 031's two-mouth response *without* reconstructing the response matrix or its $1/R^2$ mechanism (the ownership firewall `PASS_SCOPE_FIREWALL` rejects any such re-derivation):

$$A_V = \frac{m_{gg}\phi^2}{S_{gg}^2}, \quad A_J = -m_{gg}(j+g)^2, \quad A_M = m_{gg}(q^2 - g^2), \quad A_{\mathrm{MIXED}} = m_{gg}\Big[(1-2\lambda)q^2 - 2\lambda qg - g^2\Big].$$

On a nondegenerate response (`PASS_A_V_FORMULA`, `PASS_A_J_FORMULA`, `PASS_A_M_FORMULA`, `PASS_A_MIXED_FORMULA`) V (Dirichlet) repels ($A_V > 0$), J (fixed-source) attracts ($A_J < 0$), M (fixed-monopole) is INDEFINITE ($\mathrm{sgn}(q^2 - g^2)$; `PASS_A_M_INDEFINITE`), and MIXED spans. The weak certificates $A_V \geq 0$, $A_J \leq 0$ hold generally (`PASS_WEAK_SIGNS_GENERAL`); the strict signs hold only at the labelled witness (`PASS_STRICT_SIGNS_NONDEGENERATE_WITNESS`); at $z_g = 0$ all $A_X = 0$ and no strict assert is raised (`PASS_DEGENERATE_Z_G_ZERO`). Typing: `coefficients` =DECIDED, `bc_selection` = `magnitude` = `mixed_parameters` = `variant` =R1 (`PASS_DECIDED_CONDITIONAL_TYPING`).

MIXED three-regime admissibility. A_{MIXED} interpolates between the endpoints A_M ($\lambda = 0$) and $-m_{gg}(q+g)^2$ ($\lambda = 1$), with the algebraic zero $\lambda_* = (q-g)/2q$ (`PASS_MIXED_ENDPOINTS`, `PASS_MIXED_INTERIOR_ZERO`). "Spans" means over the FULL combined parameter-and-magnitude domain, NOT for every fixed q/g: for $q > g$, $\lambda_* \in (0, \frac{1}{2})$ admissible so positive/null/negative all occur; for $q = g$, the zero is only at $\lambda = 0$, negative in the interior; for $q < g$, $\lambda_* < 0$ inadmissible so MIXED stays negative on $[0,1]$ (`PASS_MIXED_REGIME_Q_GT_G`, `PASS_MIXED_REGIME_Q_EQ_G`, `PASS_MIXED_REGIME_Q_LT_G`, `PASS_MIXED_FULL_DOMAIN_SPANS`).

Internal consistency, typed Q-BC, and Q-COMBINE. The two admissible amendments — REPLACE (re-types the existing h-source BC, no new rows) and ADD (keeps the source + one core h-holding row) — both scope S_{hold} to $r_B - \frac{1}{2}$ only and preserve the 13 `[POSTULATE]` zero rows, so the Q-AMEND sector map is empty $\Rightarrow$ `internal_inconsistency` =none (`PASS_AMEND_REPLACE`, `PASS_AMEND_ADD`, `PASS_ZERO_LEDGER_13`, `PASS_S_HOLD_SCOPE`, `PASS_INTERNAL_INCONSISTENCY_NONE`). The committed model's actual evidence (admissible variation, but the missing parent-throat/boundary closure `NONZERO_HA_REQUIRES_CORE_HOLDER`) classifies to `UNDETERMINED_ANALYTICALLY`, the four controls resolving to {FIXED_SOURCE, DIRICHLET_VALUE, FIXED_MONOPOLE, MIXED} (`PASS_BC_ACTUAL_CLASSIFIER`, the four `PASS_BC_*_CONTROL`, `PASS_ADMISSIBLE_CLASSES`). The REPLACE and ADD per-class totals differ, the realization is `variant_unresolved`, held-h work is not double-counted, and the class$\times$variant outcomes are not all equal $\Rightarrow$ `outcome_not_invariant` (`PASS_REPLACE_TOTALS`, `PASS_ADD_TOTALS`, `PASS_VARIANT_UNRESOLVED`, `PASS_NO_DOUBLE_COUNT`, `PASS_OUTCOME_NOT_INVARIANT`). Q_E/ magnitude is re-homed as `R1_REQUIRED(magnitude)` with all four Q-MAG hooks NO/UNDETERMINED (`PASS_MAGNITUDE_FREE_FACTOR`, the four `PASS_QMAG_*_HOOK`, `PASS_MAGNITUDE_COBLOCKER`).

The sealed 23040-cell landing $\Rightarrow$ `R1_REQUIRED(bc_selection)`. The exhaustive typed grid ($5 \times 6 \times 6 \times 4 \times 2 \times 2 \times 2^3 = 23040$ cells) is classified by a first-matching-predicate ladder, cross-checked against an independent declarative oracle on every cell (`PASS_VERDICT_TOTALITY`), and serialized to the committed digest `7627417ace0f99a17187a90efa2523ca9b68df8b64f7960d38be2dc6f553ac49` (`PASS_TRUTH_TABLE_TOTAL`, `PASS_TRUTH_TABLE_COUNTS`, `PASS_TRUTH_TABLE_DIGEST`). The class gap fires BEFORE the variant/MIXED-parameter/tier/magnitude gaps (`PASS_VERDICT_PRECEDENCE`), so the production tuple (`qbc` = `UNDETERMINED_ANALYTICALLY`, not all classes agree) lands at the FIRST match `R1_REQUIRED(bc_selection)` (`PASS_PRODUCTION_LANDING`). The landing is RECOMPUTED from typed neutral upstream facts and compared against a fresh adjudication — an injected landing token is rejected and a named alternative (`qbc` $\to$FIXED_SOURCE) re-lands at `R1_REQUIRED(sign_and_magnitude)`$\neq$ the production token (`PASS_LANDING_OWNERSHIP`, `PASS_TYPED_NEUTRAL_FACTS`, `PASS_TARGET_BLINDNESS`). Units restored and able-to-fail: $[A] = EL = ML^3T^{-2}$, $[U] = E$, $[F] = EL^{-1}$ (`PASS_UNITS_A/U/F`); range controls certify the invariance classifier (`PASS_RANGE_SIGN_FLIP/ZERO_TOUCH/SUBDOMINANT`).

Scope, deferrals, and first-class departures (recorded honestly). (1) The SIGN is `R1_REQUIRED(bc_selection)` — the committed bare model (G0 + U2-unresolved $\mathfrak{B}$ + S_{hold} scoped to $r_B - \frac{1}{2}$) does not determine the mouth boundary operator, so it selects no class and picks no sign. (2) The strict signs are CONDITIONAL on a nondegenerate response; weak in general; all A_X vanish at $z_g = 0$. (3) The magnitude (`R1_REQUIRED(magnitude)`; c_a, c_ξ unbounded at tier-A) co-blocks; the variant/MIXED-parameter/REPLACE-vs-ADD selections are further SEPARATE R1 debts. (4) The resolver is the SIM-DEFERRED nonlinear parent-throat map $s \mapsto h_A$ (sibling of R10/R30/R33); a guaranteed-nonzero amplitude needs the same core holder (`NONZERO_HA_REQUIRES_CORE_HOLDER`). (5) `internal_inconsistency` =none is a positive structural fact.

Verification. Dual-engine, both exit 0, genuinely independent routes: **SymPy 57 PASS / Mathematica 57 PASS**. Standalone, print-only, assert-zero (`raise SystemExit(1)` / `Exit[1]` on failure), no argparse harness, no JSON/YAML payload, zero file-I/O between engines; float-/machine-real-free payload throughout. Verdict token `R1_REQUIRED(bc_selection)`; co-blocker `R1_REQUIRED(magnitude)`; `INTERNAL_INCONSISTENCY` =none. The `.wl` is not a transliteration: native `Solve`/`Reduce` for the ensemble coefficients + the MIXED zero + the three-regime split, an independent native construction of the exhaustive 23040-cell landing table + SHA-256 digest, and an independent landing-ladder evaluation over the typed facts; both engines compute the truth-table digest, the 44-entry manifest partition ($\{22, 12, 10\}$, digest `e2cfd11b41cdd3d95111f25215b16e917b1ce0a9623929619338a1a83fa81014`), and the production landing at RUNTIME and agree. **58 mutations total, each `FIRED_AT_OWN_ASSERT`** (env switch `LEDGER_STAGE032_MUTATION`): the 57 teeth (every folded predicate — the four A_X, the weak/strict/degenerate sign typing, the MIXED endpoints/interior-zero/three-regime teeth, the Q-AMEND/Q-BC/Q-COMBINE/Q-MAG blocks, the range controls, the units firewall, the exhaustive total/counts/digest, the precedence + production landing, the upstream-relanding ownership tooth, target-blindness, and the 44-entry manifest) plus `MANIFEST_MISDISPOSITION`; the landing tooth mutates an UPSTREAM fact and re-lands on a NAMED DIFFERENT §4 landing, never the computed summary (the 030 $X \equiv X$ lesson); no vacuous or subsumed tooth. **Tri-review CLEAN:** fidelity + adversarial fresh-agent review CLEAN; directive Codex$\to$Grok$\to$Codex bookend + build + fresh-agent re-verify all CLEAN; manifest-hardening remediation followed by CLEAN re-verification. Verdict token `R1_REQUIRED(bc_selection)`.

D.4 Stage 033: Native P Has No Emergent Maxwell Gauss Law — the Characterized Departure `NATIVE_P_NO_EMERGENT_GAUSS`

Part / anchor. Part IV – Charge (IV-4; the LAST stage of the 4-stage split – the native-P departure)

Claim status. Open

Purpose. Record, as a first-class characterized departure, that the medium's own P-field content $\{p = P_{\|}, u, s, b, \dots\}$ cannot host exact U(1)/Maxwell electromagnetism as a native gauge structure. At quadratic order the two native families THEORY-A and THEORY-C are Dirac–Bergmann all-second-class (FC = 0; no first-class Gauss chain), and the tuned rank-drop strata that carry first-class directions have a zero Gauss descendant (`DESCENDANT_ZERO`). Six able-to-fail controls anchor the machinery. Complementary to the throat-charge mechanism of Stages 030–032 (this is the candidate *self-hosting* electric theory, not the throat); the charge-sector twin of the light-sector departure `FAIL_CAUCHY_STRAY_LONGITUDINAL` (Stage 003 / pathA_36). Consumes *nothing* numeric from 030/031/032. Reads out as: **"EM is NOT exact Maxwell."**

Claim status. Open — a **CHARACTERIZED-DEPARTURE, recorded first-class**: a load-bearing no-go, NEVER softened, NEVER rescued. It is *not* an exact closure, and equally it is *not* a knob, *not* a reduction, and *not* an R1 deferral — it discharges no knob and **does not shrink the irreducible count** (the honest departure ledger, model map §4). Verdict token `NATIVE_P_NO_EMERGENT_GAUSS` (both A and C, both engines, both exit 0). The named able-to-fail alternates the verdict would have re-derived to, had a first-class Gauss chain been found, are `FIRST_CLASS_GENERIC_EM_CANDIDATE` (from the symbolic open stratum) and `FIRST_CLASS_TUNED_INVERSE_DESIGN` (from the tuned locus). This is the quadratic Stage-1 constraint gate only: compactness, quantization, deconfinement, and native $\pm w$ current supply remain downstream and are *not* claimed.

The shared pipeline and the two native families. Each native family is supplied as a quadratic Lagrangian in the medium's own fields and reduced through the identical pipeline

$$\text{input } L \;\to\; \text{Legendre (primaries from Hessian nulls)} \;\to\; \text{Dirac closure} \;\to\; \text{PB-kernel first-class / Gauss search.}$$

THEORY-A carries $\{p_{1\text{–}3}, u_{1\text{–}3}, s, \lambda, \sigma, b\}$ with retained couplings $\{g_{tA}, g_{sA}, g_{dA}, g_{bA}\}$; THEORY-C carries the same plus a multiplier ξ with couplings $\{g_{tC}, g_{sC}, g_{dC}\}$. A first-class Gauss generator would appear as a nonzero first-class count in the weak Poisson-bracket matrix *and* a first-class primary whose Hamiltonian descendant is a $\propto k$ (Gauss-shaped) spatial action.

The quadratic FC = 0 result (both families, exact). On the regular open kinetic stratum, for all retained couplings (`PASS_THEORY_A_SIGNATURE`, `PASS_THEORY_C_SIGNATURE`; `PASS_GUARD_COUPLINGS_ENTER_A/C`):

$$\begin{aligned}
\text{THEORY-A: } & \text{Hessian } 8/2,\ \ 8 \text{ constraints stages } [0,0,1,1,2,2,3,3],\ \text{all second-class;}\\
& \text{weak PB matrix rank } 8,\ \text{FC } 0,\ \text{SC } 8,\ \text{gauss cand. } 0.\\
\text{THEORY-C: } & \text{Hessian } 7/4,\ \ 12 \text{ constraints stages } [0,0,0,0,1,1,1,1,2,2,3,3],\ \text{all second-class;}\\
& \text{weak PB matrix rank } 12,\ \text{FC } 0,\ \text{SC } 12,\ \text{gauss cand. } 0.
\end{aligned}$$

Every native coupling provably survives into the momentum-map / Hessian / constraints / PB matrix (`GUARD-COUPLINGS-ENTER` PASS). The physical kinetic-Hessian determinant factors, in both families, as $\det H_{\text{phys}} = (\rho_u - g_t^2)^3$ (per component $\rho_u - g_t^2$; `PASS_KINETIC_HESSIAN_DETERMINANT`), so the *only* additional kinetic degeneracy is the tuned locus $g_t^2 = \rho_u$. (The signature's FC $= n_{\text{constraints}} -$ rank element is a true-by-construction rank–nullity identity; the load-bearing signature elements are the counts and classes, mutated by `PASS_THEORY_{A,C}_SIGNATURE`.)

The `DESCENDANT_ZERO` hardening guard on the tuned locus. On $g_t^2 = \rho_u$ the Hessian nullity rises (A: $2 \to 5$, C: $4 \to 7$) and first-class directions can appear. The potential-null residual is solved *symbolically* (not hardcoded) for both signs $g_t = \pm 1$; within the argued boundary partition + scanned representatives the common Hessian/potential-null locus is the observed FC $= 2$ stratum. For every rejected first-class primary both engines compute the descendant $\{\text{primary}, H_2\}$ and its field action and certify it `DESCENDANT_ZERO` (identically 0) or non-gradient (`PASS_HARDENING_DESCENDANT_A/C`, `PASS_PRIMARY_FC_ACCOUNTING_A/C`): **FC-bearing strata checked = 2, rejected directions checked = 4** per family, *all* `DESCENDANT_ZERO` (`descendant_zero` =True, `secondary_action_non_gradient` =True); accounting `primary_fc` = `rejected` + `candidates` with candidates = 0. The guard fails the run if any rejected direction instead hides a nonzero $\propto k$ Gauss descendant — a tuned FC direction cannot masquerade as an emergent Gauss law.

The six able-to-fail controls. Each control is an input Lagrangian through the *same* pipeline, with its own `FIRED_AT_OWN_ASSERT` ablation (token / FC / SC / Gauss cand. / Hessian nullity):

Control	Expected class token	FC	SC	G	null
maxwell	`FIRST_CLASS_GAUSS`	2	0	1	1
gauged hard unit	`MIXED`	2	4	1	2
bare O(4) hard sigma	`SECOND_CLASS_RADIAL_NO_GAUSS`	0	4	0	1
nonconserved current	`INCONSISTENT_PRESERVATION`	2	0	1	1
Coulomb-gauge Maxwell	`SECOND_CLASS_NO_LOCAL_GAUGE`	0	8	0	3
global U(1) complex scalar	`GLOBAL_CHARGE_NO_LOCAL_GAUSS`	0	0	0	0

Maxwell and the gauged-hard-unit control yield nonzero Gauss candidates (`GUARD-SEARCH-CAPABLE` PASS — the search finds a real Gauss chain when one exists, so its absence for native A/C is meaningful). Control 4's Gauss preservation carries the explicit continuity defect $-nc_{j1} - 2\,nc_{j2} - 3\,nc_{j3} + nc_{\rho\text{-}dot}$, nonzero without a conservation rule $\Rightarrow$ `INCONSISTENT_PRESERVATION`.

Honest scope (does NOT weaken the verdict). (a) **Decisive:** the fully-symbolic open kinetic stratum — FC = 0 for A and C for *all* retained couplings. (b) **Argued + fixed-seed scanned:** the tuned $g_t^2 = \rho_u$ locus, *not* an exhaustive stratification — a 6-point semidefinite-boundary scan per family (2 signs $\times$ {generic semidefinite, first rank-drop non-common, common Hessian/potential-null locus}; the common-null representative is the FC = 2 one, all gauss cand. 0) *and* a fixed-seed randomized rank-drop sweep of 6 points per family, 12 total (seeds A = `260713`, C = `260715`; per-sign split 3+3; all FC = 0). Pinned are the seed, sign split, cardinality, and decisive per-point signatures (`FC` = 0, `gauss` = 0, `additional_G` =False), *not* the sampled coordinates (`PASS_HONEST_TUNED_SCOPE`). Any missed measure-zero first-class stratum would be a `TUNED_INVERSE_DESIGN` artifact, so the generic no-go stays decisive; the scan is not promoted to an exhaustive classification (no unqualified-exhaustiveness "only"). With FC = 0 for both families the audit records `BRANCH_2_QUADRATIC_ABSENCE` (`PASS_DECISION_ORDER_BRANCH2`): a regular nonlinear gauge identity would have a nontrivial leading linearization, so the quadratic absence is decisive; 033 records the quadratic verdict, not an all-orders proof. The verdict is re-derived from the computed searches (`PASS_VERDICT_TOTALITY`) — feeding a genuine first-class Maxwell search through the open path re-derives `FIRST_CLASS_GENERIC_EM_CANDIDATE` (the tooth reads the computed result, not the literal).

Source-to-stage manifest. No silently-dropped source claim: every source tooth / Q-claim partitions {`PRESERVED` 33, `REPLACED_BY_STRONGER` 15, `SCOPED_OUT` 3} (51 total), manifest digest (SHA-256) `6b191e77fefe24c9000445f01e4e2c6154ab1bb9b15bb40a6d1515dc463a7e9d`, computed at runtime in both engines and agreed (`PASS_SOURCE_PREDICATE_MANIFEST`; dropping one row fires it). The `argparse` harness, all JSON writing, and the cross-engine file comparator are SCOPED_OUT (print-only / zero-file-I/O / independent-tokens contract).

Verification. Dual-engine, both exit 0, genuinely independent routes: **SymPy 33 teeth / Mathematica 33 teeth**. Standalone, print-only, assert-zero (`raise SystemExit(1)` / `Exit[1]`), no argparse harness, no JSON/YAML payload, zero file-I/O between engines; float-/machine-real-free payload throughout. The `.wl` is RE-AUTHORED (not a mirror): native `Solve`/`Reduce[ForAll]` common-null derivation, native `NullSpace`/`Inverse`/`Det`/`MatrixRank`, and its own six per-tooth control ablations. Per-tooth ablation (env switch `LEDGER_STAGE033_MUTATION`): every folded predicate fires at its own assert, the six control ablations `FIRED_AT_OWN_ASSERT` in both engines, and the verdict tooth re-derives from a mutated *computed* search input to a NAMED different verdict (never the `NATIVE_P_NO_EMERGENT_GAUSS` literal). **Tri-review (falsification-first, recorded transparently): fidelity CLEAN; adversarial CONCERNS(2) + a fidelity NIT — all verified non-blocking, no coverage gap.** (i) `PASS_HONEST_TUNED_SCOPE` is a framing/summary tooth (its mutation flips a self-set flag) whose substantive content is genuinely enforced by the boundary-scan / randomized-sweep / signature teeth — redundant, not a gap. (ii) the SymPy `focused_mutation_probe` path re-derives from the same pipeline objects as the clean run — a verified-safe structural smell, no hidden vacuity. (iii) the $\mathrm{FC} = n_{\mathrm{constraints}} - \mathrm{rank}$ signature element is a true-by-construction identity; the real FC/SC/count guards are the other signature elements. Arbiter re-runs reproduce the tokens and the manifest digest. Verdict token `NATIVE_P_NO_EMERGENT_GAUSS`.

Appendix E

Part V Stage Cards

E.1 Stage 034: The Moving-Throat Transverse-Vector Action Row — `TRANSVERSE_MOVE_ACTION_ROW`

Part / anchor. Part V – Magnetism (V-1; the FIRST stage of the 6-stage split – the moving-throat action row)

Claim status. Exact Within Closure

Purpose. Record, as an EARNED action-structure result, that magnetism enters the ledger as a clean $(G0+\delta)$ transverse-vector amendment. Importing the pathA_36 transverse row into a G0 that has *no* u_T, and supplying the one finite-profile moving-defect coupling $q_T \sum_i s_i \eta_a(x - X_i)\mathbf{V}_i \cdot \mathbf{u}_T$, gives a well-formed, stable row: the transverse dispersion $\omega^2 = c_\gamma^2 k^2$, a positive-definite transverse Hessian (two polarizations, no ghost/tachyon), and *no* pre-existing G0 row changed (`internal_inconsistency` = none). The imported row (u_T kinetic/gradient, $c_\gamma^2 = \mu_R/\rho_{\rm br}$) is CITED from Stage 003 / pathA_36 — *not* earned by 034, *not* re-counted; the magnetism-NEW content is exactly $\{\text{moving coupling}, q_T, \tau_d\}$. Surviving verdict token `TRANSVERSE_MOVE_ACTION_ROW`.

Claim status. Exact Within Closure — **EARNED** (action row; *imports* pathA_36) and exact-symbolic *within* the postulated G0 closure. What is EARNED (given the postulated G0 action and the imported transverse row): the row composes into a consistent, stable, non-damaging $(G0 + \delta)$ amendment — `internal_inconsistency` = none. It is *not* a knob, *not* a reduction/codimension edge, and it does **not** shrink the irreducible count. It carries *no* electric branch/sign: the moving row inherits the electric sector's open A_E, and q_T's value is the sim-deferred nonlinear-throat R1. This is the FIRST Part-V stage — the *action-row gate only*; the native source law, the parity census, both far-field routes, the boost-consistency comparison, the sealed landing, and the b_T T-even departure are downstream (Stages 035–039) and are *not* claimed here. Ledger-local earned label `TRANSVERSE_MOVE_ACTION_ROW`.

The imported row and the transverse dispersion (`ACTION_KINETIC`). G0-v0 contains *no* transverse shear DOF u_T. The kinetic + gradient row $\frac{1}{2}\rho_{\rm br}|\dot{\mathbf{u}}_T|^2 - \frac{1}{2}\mu_R|\nabla\times\mathbf{u}_T|^2$ (with $\nabla\cdot\mathbf{u}_T = 0$) is **imported verbatim from Stage 003 / pathA_36**; the tooth checks the import is faithfully instantiated, *not* that 034 earns it. For a divergence-free field $\nabla\times\nabla\times\mathbf{u}_T = -\nabla^2\mathbf{u}_T$, so the free Euler–Lagrange equation $\rho_{\rm br}\ddot{\mathbf{u}}_T = \mu_R\nabla^2\mathbf{u}_T$ gives, for the two $\mathbf{k}\cdot\boldsymbol{\varepsilon} = 0$ transverse polarizations,

$$\rho_{\rm br}\,\omega^2 = \mu_R\,k^2 \implies \omega^2 = \frac{\mu_R}{\rho_{\rm br}}\,k^2 = c_\gamma^2 k^2, \qquad c_\gamma^2 = \frac{\mu_R}{\rho_{\rm br}}.$$

The magnetism-NEW moving coupling (`ACTION_COUPLING`). The one term 034 supplies is the finite-profile moving-defect coupling $q_T \sum_i s_i \eta_a(\mathbf{x}-\mathbf{X}_i)\, \mathbf{V}_i \cdot \mathbf{u}_T$ with $q_T = \lambda_T \tau_d$, where $s_i = \pm 1$ is the $\pm w$ orientation (charge sign), η_a the normalized mouth profile ($\int \eta_a = 1$), $\mathbf{V}_i$ the throat velocity, and τ_d the active drain's time-arrow. It is a proper current·field source (per-defect current $\mathbf{J}_{T,i} = q_T s_i \eta_a \mathbf{V}_i$), linear in $\mathbf{u}_T$, and **target-blind** — it carries only q_T, no electric A_E, no q/g knob, no downstream sign/landing token.

Stability, no G0 damage, well-formedness (`ACTION_STABILITY`, `G0_DAMAGE`, `LEDGER_READY_ROW`). Stability is a *computed* positivity object: per transverse component the kinetic Hessian is $\rho_{\rm br} I$ and the Fourier gradient Hessian is $\mu_R k^2 P^T$, so with $\rho_{\rm br}, \mu_R > 0$ both are positive-(semi)definite on the two-dimensional transverse subspace — two propagating polarizations, no ghost (kinetic PD), no tachyon ($\omega^2 \geq 0$). The amendment, computed as a diff/overlap over the parsed G0 row set, touches *only* the absent transverse DOF and the finite-profile moving coupling; every declared-zero G0 entry (scalar, drain, return $F_{\rm flux}$, wall r_B, geon, ...) is byte-for-byte unchanged $\Rightarrow$ `internal_inconsistency` = none. **The active $F_{\rm flux}$ ledger is NOT silently absorbed** into this conservative row — it is named untouched and deferred (its $O(V_1 V_2)$ contribution is R1, Stage 038 / Part VII). The row is a well-formed $(G0+\delta)$ amendment: variational, local, and linear in $\mathbf{u}_T$:

$$S_{T+\rm move} = \int dt\, d^3x \left[\tfrac{\rho_{\rm br}}{2} |\dot{\mathbf{u}}_T|^2 - \tfrac{\mu_R}{2} |\nabla \times \mathbf{u}_T|^2 + q_T \sum_i s_i \eta_a(\mathbf{x}-\mathbf{X}_i) \mathbf{V}_i \cdot \mathbf{u}_T \right], \quad \nabla \cdot \mathbf{u}_T = 0.$$

The dimensional identity (`FIELD_IDENTITY_UNITS`). Units RESTORED, able-to-fail. The preregistered field is the shear *displacement* u_T ($[u_T] = L$); the curl candidate $\mathbf{b}_T = \nabla \times \mathbf{u}_T$ is *dimensionless* (its parity/departure is Stage 039, not here). The eight dimensions $[u_T] = L$, $[\dot{\mathbf{u}}_T] = LT^{-1}$, $[\nabla \times \mathbf{u}_T] = 1$, $[\rho_{\rm br}] = ML^{-3}$, $[\mu_R] = ML^{-1}T^{-2}\,(= EL^{-3})$, $[q_T] = MT^{-1}$, $[\eta_a] = L^{-3}$, $[b_T] = 1$ make each action density term EL^{-3} (e.g. $q_T s \eta_a \mathbf{V} \cdot \mathbf{u}_T$: $MT^{-1} \cdot L^{-3} \cdot LT^{-1} \cdot L = ML^{-1}T^{-2} = EL^{-3}$), so $[S_{T+\rm move}] = [E \cdot T]$ (action).

Import-vs-earn discipline (`GUARD_IMPORT_VS_EARN`). A computed anti-overclaim accounting object enforces `earned_new_terms` = {moving coupling} and `imported_terms` $\supseteq$ $\{u_T$ kinetic, gradient, $c_\gamma\}$: the imported pathA_36 row is TAGGED provenance, not counted as a 034 earn ($\{u_T, \mu_R, c_\gamma, \rho_{\rm br}\}$ not re-added to the irreducible set; de-dup deferred to Part VII). Mutating the accounting to claim the kinetic row as a 034 earn fires the guard.

Source-to-stage manifest. No silently-dropped source claim: every source tooth of the 35-tooth `TOOTH_ORDER` (commit `53cf049f`) partitions {`PRESERVED 1`, `REPLACED_BY_STRONGER 8`, `SCOPED_OUT 26`} (35 total), manifest digest (SHA-256) `4343bd60cd974f653a0a8ac2eeced6c7aca15b1831c81be20bd358f449c454af`, computed at runtime in both engines and agreed (`SOURCE_TO_STAGE_MANIFEST`; dropping one row fires it). The 26 SCOPED_OUT teeth are the V-2... V-5 source claims, each named literally by owner stage (Stages 035–038 — no wildcard families).

Verification. Dual-engine, both exit 0, genuinely independent routes: **SymPy 12 teeth / Mathematica 12 teeth**. Standalone, print-only, assert-zero (`raise SystemExit(1)` / `Exit[1]`), no argparse harness, no JSON/YAML payload, zero file-I/O between engines. The SymPy route checks transverse stability via explicit kinetic/gradient Hessians + a generalized-eigenvalue dispersion; the `.wl` is INDEPENDENT — a `NullSpace` transverse Fourier basis, `CoefficientArrays` quadratic forms, `PositiveDefiniteMatrixQ`, and native dispersion solves; each engine decomposes its OWN action density (no dual-engine disagreement). Per-tooth ablation (env switch `LEDGER_STAGE034_MUTATION`): all 12 mutations `FIRED_AT_OWN_ASSERT` per engine; the verdict tooth mutates a *computed* object (ρ_{br} sign → non-PD Hessian) so the pipeline RE-DERIVES the named token `ROW_UNSTABLE` (never the `TRANSVERSE_MOVE_ACTION_ROW` literal — the 030 $X \equiv X$ lesson), and `GUARD_IMPORT_VS_EARN` fires when the imported row is claimed as a 034 earn. **Tri-review: fidelity found 1 NIT (a dead $X \equiv X$ clause in the `.wl` `dimensionObject`) — REMEDIATED (dimensions now computed via logarithmic derivatives of the `.wl`'s independently-rescaled density/action; its `FIELD_IDENTITY_UNITS` mutation now fires) → CLEAN; adversarial CLEAN with 2 documented non-blocking coverage notes** (verified-safe): (i) two `ACTION_KINETIC` sub-conjuncts are true-by-construction (orthogonal basis / `NullSpace`) but ride a genuinely able-to-fail coefficient/dispersion assert; (ii) compound teeth carry one mutation covering their sub-checks, within the directive's permissive per-tooth scope. Arbiter re-runs reproduce the tokens and the manifest digest. Verdict `TRANSVERSE_MOVE_ACTION_ROW`, `internal_inconsistency` = none.

E.2 Stage 035: The Native Moving-Throat Source Law and Parity Census — `CONVECTION_LIKE_CONDITIONAL`

Part / anchor. Part V – Magnetism (V-2; the SECOND stage of the 6-stage split – the native source law + parity census)

Claim status. Exact Within Closure

Purpose. Record, as an EARNED source-law + parity result, what current the moving $\pm w$ throat sources. Translating the actual signed dent $\sigma_i = s_i\eta_a(\mathbf{x} - \mathbf{X}_i(t))$ under local continuity $\partial_t\sigma_i + \nabla\cdot(\sigma_i\mathbf{V}_i) = 0$ fixes the unique isotropic local flux coefficient $\alpha = 1$, yielding the native source $\mathbf{I}_i = s_i\eta_a\mathbf{V}_i$, $\mathbf{J}_{T,i} = q_T\mathbf{I}_i = q_T s_i\eta_a\mathbf{V}_i$ with $q_T = \lambda_T\tau_d$. The source is DERIVED from continuity, *not* imported from the barred $j \propto sV$/pathA_39 current (`SOURCE_NOT_IMPORTED` enforces free-symbol disjointness from the barred amplitudes $\{N_u, a_T, a'_T, a_L, q^T_A, q_L\}$). Because q_T is unfixed (R1), the compact limit is `CONVECTION_LIKE_CONDITIONAL`. The stage also derives the 24-cell parity census of $\{s, \mathbf{V}, \tau_d/q_T, \mathbf{J}_T, \mathbf{u}_T, \mathbf{b}_T\}$ under R_w/P_w/rotation/time-reversal. The imported q_T, τ_d, u_T, η_a are CITED from Stage 034 / Stage 003 — *not* earned by 035, *not* re-counted; the magnetism-NEW content is exactly {the continuity-derived source law, the parity census}. Surviving verdict token `CONVECTION_LIKE_CONDITIONAL`.

Claim status. Exact Within Closure — **EARNED** (source law + parity census, target-blind) and exact-symbolic. What is EARNED: signed-dent continuity fixes the unique isotropic flux coefficient $\alpha = 1$ and yields the native convective source, and the parity census is a target-blind structural fact. This is a **CurrentIdentity CLASSIFICATION** (the source-identity token), *not* a sealed landing — the sealed 1152-cell first-match landing (`R1_REQUIRED(electric_bc_selection)` + co-blockers) is Stage 038 (V-5). It is *not* a knob, *not* a reduction/codimension edge, and it does **not** shrink the irreducible count. It carries *no* electric branch/sign; the source's MAGNITUDE rides the unfixed q_T (R67), which is exactly why the classification is *conditional*. Ledger-local earned label `CONVECTION_LIKE_CONDITIONAL`.

The signed-dent continuity translation (`SOURCE_TRANSLATION_CONTINUITY`). The $\pm w$ throat is a finite signed dent $\sigma_i = s_i\eta_a(\mathbf{x} - \mathbf{X}_i(t))$ rigidly translating with $\mathbf{X}_i(t) = \mathbf{X}_i(0) + \mathbf{V}_i t$ ($\int \eta_a = 1$). The translation-derivative identity is $\partial_t\sigma_i = -\mathbf{V}_i\cdot\nabla\sigma_i$, so imposing continuity with a candidate isotropic local flux $\alpha\sigma_i\mathbf{V}_i$ leaves

$$\partial_t\sigma_i + \nabla\cdot(\alpha\sigma_i\mathbf{V}_i) = (\alpha-1)\sum_k V_{i,k}\,\partial_{x_k}\sigma_i,$$

which vanishes for arbitrary translating profiles *only* at $\boxed{\alpha = 1}$ (the unique root of the continuity-residual object). The unique local flux is the convective flux $\sigma_i\mathbf{V}_i$ — no free flux coefficient.

The native source law (`SOURCE_BASIS`, `SOURCE_NOT_IMPORTED`). With $\alpha = 1$ fixed,

$$\mathbf{I}_i = s_i\eta_a\mathbf{V}_i, \qquad \mathbf{J}_{T,i} = q_T\mathbf{I}_i = q_T s_i\eta_a\mathbf{V}_i, \qquad q_T = \lambda_T\tau_d,$$

the exact current appearing in Stage 034's coupling $q_T\sum_i s_i\eta_a\mathbf{V}_i\cdot\mathbf{u}_T$, here *derived* from continuity rather than posited. It reduces component-wise to precisely this continuity-flux basis (not a doubled/rescaled/ansatz basis). The native source **REPLACES** the barred $j \propto sV$/pathA_39 current, enforced at runtime as a computed **free-symbol-disjointness** object: the free-symbol set of the computed source vector is disjoint from the barred markers $\{N_u, a_T, a'_T, a_L, q_A^T, q_L\}$ — a real free-symbol intersection, *not* a source grep (no string-concat/`chr()` dodge defeats it). This is the surviving-solution discipline: the ledger shows the native source clean; the barred pathA_39 scope goes to the failures-paper backlog.

The 24-cell parity census (`PARITY_RW`, `PARITY_PW`, `PARITY_ROTATION`, `PARITY_TIME_REVERSAL`). A target-blind structural fact — six objects × four discrete columns, computed as invariants of the derived source/field:

object	R_w	P_w	rotation	T
s $(\pm w)$	odd	odd	scalar	even
$\mathbf{V}$	even	even	polar	odd
τ_d, q_T	even	even	scalar	odd
$\mathbf{J}_T = q_T s\eta_a\mathbf{V}$	odd	odd	polar	even
$\mathbf{u}_T$	odd	odd	polar	even
$\mathbf{b}_T = \nabla\times\mathbf{u}_T$	odd	odd	axial	even

so $\mathbf{J}_T, \mathbf{u}_T$ are R_w- and P_w-odd, $\mathbf{u}_T$ is polar while $\mathbf{b}_T$ is axial (curl of polar), and $\tau_d, \mathbf{V}$ are T-odd while $\mathbf{J}_T, \mathbf{u}_T$ are T-even. P_w **denotes a w-type reflection of the transverse coordinate**, NOT ordinary spatial parity $\mathbf{x} \to -\mathbf{x}$; under that reading the s, ∇, and $\mathbf{b}_T$ assignments are self-consistent. **$\mathbf{b}_T$'s axial + T-even parity is recorded RAW here**; its interpretation as the "not exact Maxwell" DEPARTURE (`B_TIME_REVERSAL_EVEN`; a physical Maxwell $\mathbf{B}$ is T-odd) is Stage 039 (V-6), *not* here.

Why `CONVECTION_LIKE_CONDITIONAL`. The source is fully DERIVED in tensor form — a profile-smeared $s\mathbf{V}$ — but its magnitude is conditional on $q_T = \lambda_T\tau_d$, which rides the R1 nonlinear moving-throat solve and is *not* fixed. Hence `CONVECTION_LIKE_CONDITIONAL`: convection-like tensor form, magnitude conditional on the unfixed q_T. An unresolved magnitude is *not* an unresolved source basis — it does not demote the source-identity token.

Cite-not-count discipline, dimensions (`UNITS_RESTORED`). Units RESTORED, able-to-fail, free-carrier-independent: $[\sigma] = [\eta_a] = L^{-3}$, $[\mathbf{V}] = LT^{-1}$, $[q_T] = MT^{-1}$, $[\mathbf{I}] = L^{-2}T^{-1}$, $[\mathbf{J}_T] = ML^{-2}T^{-2}$, $[u_T] = L$, $[b_T] = 1$, and $J_T = q_T s\eta_a V$ closes ($MT^{-1}\cdot L^{-3}\cdot LT^{-1} = ML^{-2}T^{-2}$); each parity assignment is dimensionless/structural. $q_T = \lambda_T\tau_d$, τ_d, u_T, η_a are CITED from Stage 034 (R67/R68) / Stage 003 — re-instantiated for the dual-engine check but *not* re-derived, *not* re-counted (de-dup deferred to Part VII); the source law introduces NO new irreducible knob (its only free magnitude is R67 q_T), and the census does not shrink the irreducible count.

Source-to-stage manifest. No silently-dropped source claim: every source tooth of the 35-tooth `TOOTH_ORDER` (commit `53cf049f`) partitions {`PRESERVED 8`, `REPLACED_BY_STRONGER 2`, `SCOPED_OUT 25`} (35 total), manifest digest (SHA-256) `d85de3d8f6b7d615900c8ead24f1589eed3c681e74cf546adb999f2cc7fa5b36`, computed at runtime in both engines and agreed (`SOURCE_TO_STAGE_MANIFEST`; dropping one row fires it). The 25 SCOPED_OUT teeth are the V-1 (done) and V-3. . . V-5 source claims, each named literally by owner stage (Stages 034/036–038 — no wildcard families); V-6 (039) owns no source tooth (it reuses 035's parity results).

Verification. Dual-engine, both exit 0, genuinely independent routes: **SymPy 12 teeth / Mathematica 12 teeth**. Standalone, print-only, assert-zero (`raise SystemExit(1)` / `Exit[1]`), no argparse harness, no JSON/YAML payload, zero file-I/O between engines. The SymPy route builds the continuity residual via the abstract translation identity $\partial_t\sigma = -\mathbf{V}\cdot\nabla\sigma$ and `sp.solves` for α; the `.wl` is INDEPENDENT — an explicit normalized Gaussian mouth profile with time-dependent centers, native `D[,t]`/`Div` + `Reduce` for α, operator-generated parity transforms, and `Det`/`Cross` polar–axial classification (no dual-engine disagreement). Per-tooth ablation (env switch `LEDGER_STAGE035_MUTATION`): all 12 mutations `FIRED_AT_OWN_ASSERT` per engine; `SOURCE_NOT_IMPORTED` fires when a code-native barred symbol (`Nu`) is injected into the computed source's free-symbol set (a computed intersection, not a grep). The verdict tooth mutates a *computed* object (`flux_coefficient`=2 → nonzero continuity residual) so the pipeline RE-DERIVES the build-native token `R1_SOURCE_BASIS` (a census flip re-derives `PARITY_INCONSISTENT`; an unresolved magnitude still yields `CONVECTION_LIKE_CONDITIONAL`, never the literal — the 030 $X \equiv X$ lesson). **Tri-review: fidelity found 2 NITs — both can't-fail code (a guaranteed-true `structural_cells_ok`/`ParityStructural` conjunct ANDed into `UNITS_RESTORED`; a vestigial `deriveVerdict` dead branch in the `.wl`) — both REMEDIATED (removed; `UNITS_RESTORED` now asserts only the able-to-fail dimension-equality and its mutation still fires; manifest digest unchanged) → CLEAN; adversarial CLEAN with 3 documented non-blocking notes** (verified-safe): (i) the census INPUT rows $s/\mathbf{V}/\tau_d$ are fixed generators while DERIVED rows $\mathbf{J}_T/\mathbf{u}_T/\mathbf{b}_T$ are propagated + mutation-tested — the u_T-polar sub-conjunct is true-by-construction (the Stage-034 transversality class); (ii) compound teeth carry one mutation covering sub-checks (the directive's permissive scope); (iii) a cosmetic ablation-string label in the manifest tooth. Arbiter re-runs reproduce the tokens and the manifest digest. Verdict `CONVECTION_LIKE_CONDITIONAL`.

E.3 Stage 036: The Route-A Maxwell–Darwin Reference Kernel (Boost of the Electric Interaction) — `MAXWELL_DARWIN_REFERENCE`

Part / anchor. Part V – Magnetism (V-3; the THIRD stage of the 6-stage split – the Route-A Maxwell–Darwin reference kernel)

Claim status. Exact Within Closure

Purpose. Record, as an EARNED (tier-A conditional) reference-kernel result, what far field the moving $\pm w$ throat produces along the boost route. Route A boosts the electric interaction $U_E = s_1 s_2 A_E/4\pi R$: inverse-Fourier-transforming the Coulomb-gauge transverse projector $P^T_{ij} = \delta_{ij} - k_i k_j/k^2$ over k^2 reconstructs the Maxwell–Darwin transverse Darwin kernel $\mathcal{F}^{-1}[P^T_{ij}/k^2] = \delta_{ij}/4\pi R - \mathcal{F}^{-1}[k_i k_j/k^4] = (\delta_{ij} + n_i n_j)/8\pi R \equiv I_{ij}(R)$ (from twice-differentiating the radial seed $\mathcal{F}^{-1}[k^{-4}] = -R/8\pi$), and Lorentz-completing the electric anchor gives the general independent-velocity interaction $U_A = (s_1 s_2 A_E/4\pi R)[1-(D_V + A_V)/2c_\gamma^2] + O(v^4/c_\gamma^4)$, with $D_V = \mathbf{V}_1\cdot\mathbf{V}_2$, $A_V = (\mathbf{V}_1\cdot\mathbf{n})(\mathbf{V}_2\cdot\mathbf{n})$, $\mathbf{n} = (\mathbf{X}_2 - \mathbf{X}_1)/R$. Route A is a *boost of the electric interaction*, so the kernel and anchor INHERIT the electric A_E `R1_REQUIRED(bc_selection)` (sign + normalization unselected) — the scope is EARNED (reference kernel) but `tier_A_conditional`. The imported A_E (Part IV, R63) and c_γ (Part III / Stage 003) are CITED — *not* earned by 036, *not* re-counted; the magnetism-NEW content is exactly {the transverse Darwin reference kernel, the Lorentz-completed $O(v^2/c_\gamma^2)$ anchor}. Surviving verdict token `MAXWELL_DARWIN_REFERENCE`.

Claim status. Exact Within Closure — **EARNED** (reference kernel, target-blind) and exact-symbolic, at `tier_A_conditional`. What is EARNED: the inverse-FT reconstruction is an exact target-blind derivation of the Maxwell-consistent transverse Darwin kernel, and the Lorentz-completion is exact through $O(v^2/c_\gamma^2)$. It is *conditional* because Route A boosts the electric interaction, so the kernel and anchor INHERIT the electric A_E `R1_REQUIRED(bc_selection)` (sign + normalization unselected; Stage 032, R63) — the structure is exact, the overall coefficient rides the open A_E. Hence EARNED-but-conditional, *not* GAPS-CLOSED. It is *not* a knob, *not* a reduction/codimension edge, and it does **not** shrink the irreducible count (it rides $A_E + c_\gamma$, introducing no new knob). **036 produces the reference ONLY** — the direct route (Route B), the structural comparison, and the boost-consistency / emergent-Lorentz result (the two routes matching) are Stage 037 (V-4), stated so a reader never reads 036 as claiming emergent Lorentz. Ledger-local earned label `MAXWELL_DARWIN_REFERENCE` at `tier_A_conditional`.

The transverse Darwin kernel (`BOOST_PROJECTOR`). The Coulomb-gauge transverse projector $P^T_{ij}(k) = \delta_{ij} - k_i k_j / k^2$ divided by k^2 inverse-Fourier-transforms to the transverse Darwin kernel:

$$\mathcal{F}^{-1}\left[\frac{P^T_{ij}}{k^2}\right] = \frac{\delta_{ij}}{4\pi R} - \mathcal{F}^{-1}\left[\frac{k_i k_j}{k^4}\right] = \frac{\delta_{ij} + n_i n_j}{8\pi R} \equiv I_{ij}(R),$$

obtained from the radial seed $\mathcal{F}^{-1}[k^{-4}] = -R/8\pi$ via $\mathcal{F}^{-1}[k_i k_j/k^4] = -\partial_i\partial_j(-R/8\pi) = (\delta_{ij} - n_i n_j)/8\pi R$, with $\mathbf{n} = (\mathbf{X}_2 - \mathbf{X}_1)/R$, $R = |\mathbf{X}_2 - \mathbf{X}_1|$. This is the Maxwell-consistent (Coulomb-gauge / Darwin) reference kernel. It is certified as a component-wise zero-difference object (reconstructed `transverse`$[i, j]$ minus $(\delta_{ij} + n_i n_j)/8\pi R$ symbolically zero, with $n_z^2 = 1 - n_x^2 - n_y^2$); the coefficient $1/8\pi$ and the transverse $(\delta_{ij} + n_i n_j)$ structure are both load-bearing.

The Lorentz-completed anchor (`BOOST_GENERAL_VELOCITIES`). Contracting the kernel with the independent velocities gives $\mathbf{V}_1^\top I\, \mathbf{V}_2 = (D_V + A_V)/8\pi R$, so the $O(v^2/c_\gamma^2)$ interaction piece is $U_{A,2} = -s_1 s_2 A_E (D_V + A_V)/(8\pi c_\gamma^2 R)$ (the build's `EXPECTED_A2`), and

$$U_A = \frac{s_1 s_2 A_E}{4\pi R}\left[1 - \frac{D_V + A_V}{2c_\gamma^2}\right] + O(v^4/c_\gamma^4),$$

$$\mathbf{F}_{A,2} = \frac{s_1 s_2 A_E}{8\pi c_\gamma^2 R^2}\left[(\mathbf{V}_2\cdot\mathbf{n})\mathbf{V}_1 + (\mathbf{V}_1\cdot\mathbf{n})\mathbf{V}_2 - (D_V + 3A_V)\mathbf{n}\right], \quad F_{A,2,r} = -\frac{s_1 s_2 A_E (D_V + A_V)}{8\pi c_\gamma^2 R^2}.$$

Certified as a symbolic zero-difference `route_a_u` $-$ `EXPECTED_A2` $= 0$; the anchor, force, and radial expressions are reproduced and dimension-checked.

The order structure (`BOOST_NEXT_ORDER`). The $O(1)$ and $O(v^2/c_\gamma^2)$ orders are explicit; the next term $O(v^4/c_\gamma^4)$ is *named but NOT computed* — the anchor's velocity order is exactly 2, the next order an honest R1-adjacent remainder (owned downstream at 038's higher-order hook). Certified as a runtime order object `velocity_order` $== 2$; the mutation raises the claimed order to 4 to prove the tooth pins the honest truncation.

Why `MAXWELL_DARWIN_REFERENCE`, tier-A conditional. The kernel is fully reconstructed in tensor form and the Lorentz-completion is exact through $O(v^2/c_\gamma^2)$, but the overall coefficient rides the electric A_E, which carries `R1_REQUIRED(bc_selection)`. Hence `MAXWELL_DARWIN_REFERENCE` at `tier_A_conditional`. **The open electric A_E R1 is a scope tag, not a certification failure** — production is `MAXWELL_DARWIN_REFERENCE` REGARDLESS of the open A_E; an unresolved coefficient is not a broken kernel, so it must *not* re-derive to the `MAXWELL_DARWIN_REFERENCE_UNCERTIFIED` alternate.

Cite-not-count discipline, dimensions (`UNITS_RESTORED`). Units RESTORED, able-to-fail, free-carrier-independent: $[I_{ij}] = L^{-1}$, $[A_E] = EL = ML^3T^{-2}$, $[c_\gamma^2] = L^2T^{-2}$, $[\mathbf{V}] = LT^{-1}$, $[D_V] = [A_V] = L^2T^{-2}$, $[R] = L$, $[U_A] = E = ML^2T^{-2}$, $[\mathbf{F}_{A,2}] = EL^{-1} = MLT^{-2}$, $[s] = 1$, and each velocity order is dimensionless bookkeeping. A_E (Part IV, R63) and c_γ (Stage 003) are CITED — re-instantiated for the dual-engine check but *not* re-derived, *not* re-counted (de-dup deferred to Part VII); the reference kernel introduces NO new irreducible knob (it rides the electric A_E R1 + the shared c_γ), and the $O(v^4/c_\gamma^4)$ remainder is not a new knob.

Source-to-stage manifest. No silently-dropped source claim: every source tooth of the 35-tooth `TOOTH_ORDER` (commit `53cf049f`) partitions {`PRESERVED` 2, `REPLACED_BY_STRONGER` 4, `SCOPED_OUT` 29} (35 total), manifest digest (SHA-256) `f8b1569834c1c5cfd404fee4fba7bc49d3d7263f332e551270ad499fd3585cac`, computed at runtime in both engines and agreed (`SOURCE_TO_STAGE_MANIFEST`; dropping one row fires it). The 29 SCOPED_OUT teeth are the V-1/V-2 (done) and V-4/V-5 source claims, each named literally by owner stage (Stages 034/035 done; 037/038 — no wildcard families, `BOOST_COMMON_VELOCITY` placed in 037); V-6 (039) owns no source tooth.

Verification. Dual-engine, both exit 0, genuinely independent routes: **SymPy 8 teeth / Mathematica 8 teeth**. Standalone, print-only, assert-zero (`raise SystemExit(1) / Exit[1]`), no argparse harness, no JSON/YAML payload, zero file-I/O between engines. The SymPy route is a Poisson-seed Hessian reconstruction (the seed $-R/8\pi$ fixed by $\nabla^2(\text{seed} \cdot r) = -1/4\pi R$, then the kernel by the Cartesian Hessian $-\partial_i\partial_j$); the `.wl` is materially INDEPENDENT of *both* the stage `.py` and the source `check.wl` (no seeded Hessian) — an analytically-continued Riesz transform $\mathcal{F}^{-1}[k^{-2\alpha}] = \Gamma(3/2-\alpha)R^{2\alpha-3}/(4^\alpha\pi^{3/2}\Gamma(\alpha))$ at $\alpha = 2$, genuine Schwinger/Gaussian momentum integrals, and a parallel/perpendicular velocity decomposition (no dual-engine disagreement). Per-tooth ablation (env switch `LEDGER_STAGE036_MUTATION`): all 8 teeth's mutations `FIRED_AT_OWN_ASSERT` per engine (16 across both engines). The verdict tooth mutates a *computed* object — it DOUBLES the computed kernel (`verdict_kernel` $= 2 \cdot$ `KERNEL_AT_R`), recomputes the projector residuals, and RE-DERIVES the authored stage-local token `MAXWELL_DARWIN_REFERENCE_UNCERTIFIED` (three negative controls — projector / anchor / order — test the else-branch; the open electric A_E R1 still yields `MAXWELL_DARWIN_REFERENCE` at `tier_A_conditional`, never the alternate — the 030 $X \equiv X$ lesson). **Tri-review: fidelity and adversarial CONVERGED on 2 NITs — both can't-fail conjuncts (a `NEXT_UNCOMPUTED_ORDER==4` literal $4 == 4$ ANDed into `BOOST_NEXT_ORDER`; the `dimension_firewall` inventory `len==len` guard, $13 == 13/29 == 29$) — both REMEDIATED (removed; `BOOST_NEXT_ORDER` still fires via the $(0,2)$/`claimed_order`/ε^4 order objects, `UNITS_RESTORED` still fires via the dimension residuals; manifest digest unchanged) → CLEAN**, with one documented non-blocking note (verified-safe): `BOOST_GENERAL_VELOCITIES` bundles the force-derivation / component / radial residuals under one compound mutation (the directive's permissive scope). Arbiter re-runs reproduce the tokens and the manifest digest. Verdict `MAXWELL_DARWIN_REFERENCE` at `tier_A_conditional`.

Appendix F

Reproducibility Map

F.1 Purpose of This Appendix

Placeholder for the rebuilt ledger's reproducibility rules.

F.2 Reproducibility Layers

Layer	What it controls	Minimum evidence required
No reproducibility layers populated yet.		

F.3 Canonical Source Registry

Placeholder for source artifacts.

F.4 Primary Symbolic-Audit Registry

Placeholder for symbolic audit artifacts.

F.5 Repository Freeze Protocol

Placeholder for future release-freeze requirements.

Appendix G

Source File Index

G.1 Purpose of This Appendix

Placeholder for the rebuilt ledger's source-file index.

G.2 Source Classes

Placeholder for source classes.

G.3 Generated Ledger Files

File family	Role
`paper/frontmatter/*.tex`	Placeholder governance chapters.
`paper/parts/*.tex`	Future theorem-block chapters.
`paper/stages/stage_NNN.tex`	Future per-stage cards.
`scripts/ledger_stageNNN_*.py`	Future SymPy audit scripts.
`mathematica/ledger_stageNNN_*.wl`	Future Mathematica audit scripts.

G.4 Precedence and Conflict-Resolution Rules

Placeholder for source precedence and conflict-resolution rules.

Appendix H

Layer-2 Fit-Insertion Index

H.1 Generated Coverage Summary

Item	Count
Phase-A candidates	0
Audited candidates	0

H.2 Phase-A Candidate Rows

No candidate rows exist in the empty ledger skeleton.

H.3 Visible Grouping and Exclusion Accounting

No grouping or exclusion accounting exists yet.

Appendix I

Layer-2 Parameter-Provenance Audit

I.1 Generated Coverage Summary

Item	Count
Top-level provenance parameter records	0
Canonical candidates	0

I.2 Published-Target Benchmark Sources

No benchmark rows exist in the empty ledger skeleton.

I.3 Parameter Genealogy Rows

No parameter genealogy rows exist yet.

I.4 Visible Alias Accounting

No alias accounting exists yet.

Appendix J

Fill Workflow and Editorial Rules

J.1 Purpose of This Appendix

Placeholder for the rebuilt ledger fill workflow.

J.2 What Filled Means

Placeholder for completion criteria.

J.3 Governing Maps Used During Filling

Placeholder for status, notation, anchor, dependency, source, and reproducibility controls.

J.4 Stage Appendix Fill Protocol

Placeholder for future stage-card authoring rules.

J.5 Theorem-Block Fill Protocol

Placeholder for future Part authoring rules.

J.6 Minimal Build and Static-Check Workflow

1. Confirm every retained `\input` target exists.
2. Run the SymPy and Mathematica audit runners.
3. Build `paper/pde_ledger.tex`.